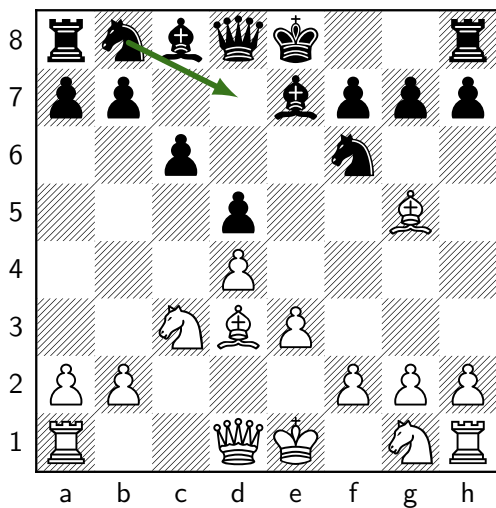


Melanie-BOTS (CCRL engines) — QGD, Exchange

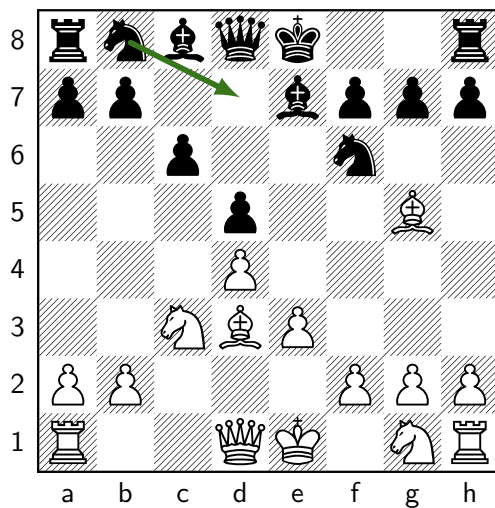
Melanie (Deep_Vision)

Abstract

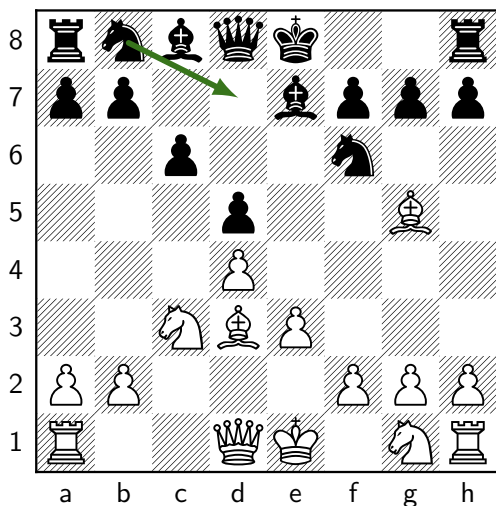
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



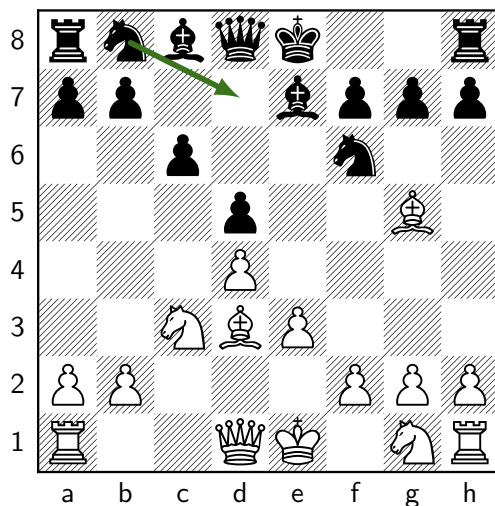
(a)



(b)

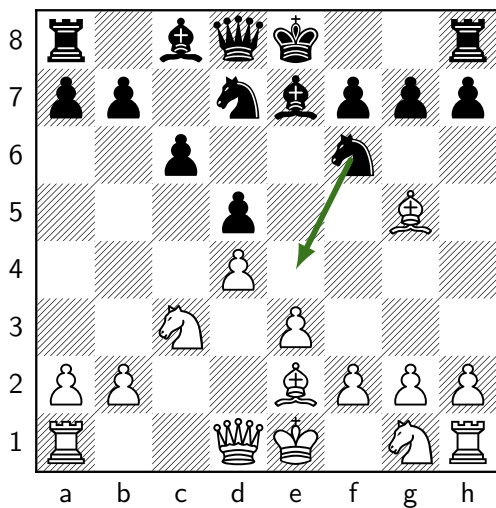


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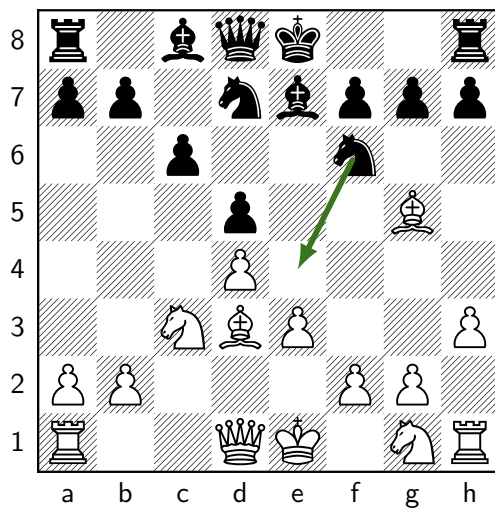


(d)

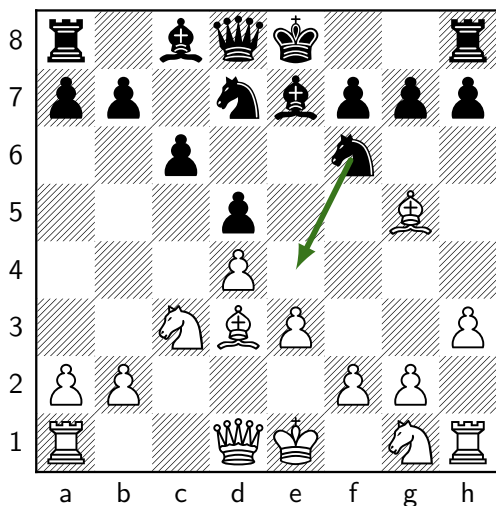
Figure 1: **Move 7, Black: Nbd7.** (a) Our position; Melanie plays Nbd7 (b8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2618` (ply 13) [structural distance $d=0$, QAP obj=49]: the master played Nbd7; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Pf7* as their *Be7*, our *Be7* as their *pa2*; transferred move b8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2619` (ply 13) [structural distance $d=0$, QAP obj=49]: the master played Nbd7; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Pf7* as their *Be7*, our *Be7* as their *pa2*; transferred move b8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1394` (ply 13) [structural distance $d=0$, QAP obj=49]: the master played Nbd7; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Pf7* as their *Be7*, our *Be7* as their *pa2*; transferred move b8d7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Nbd7 (summed QAP objective 389, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



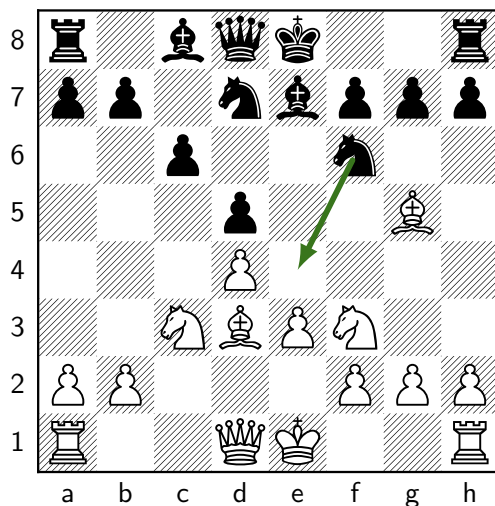
(a)



(b)

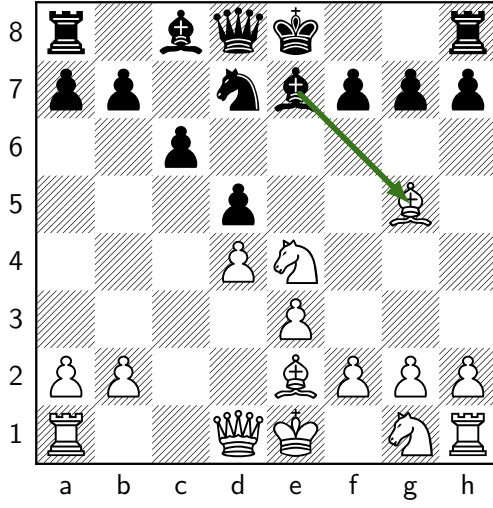


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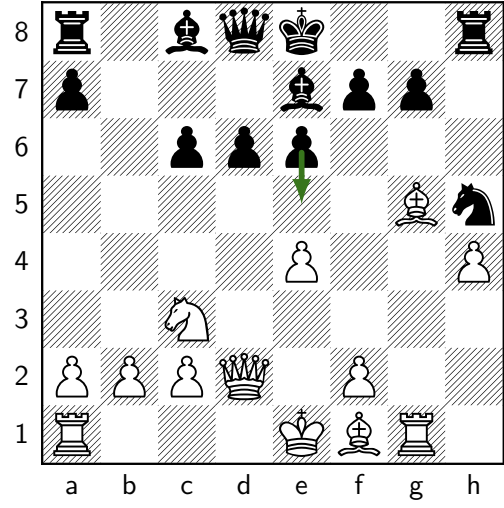


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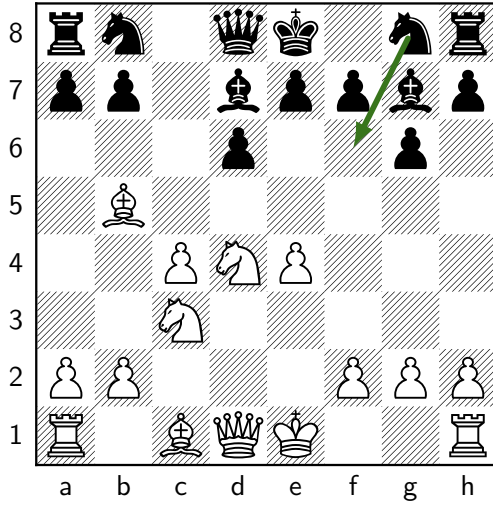
Figure 2: **Move 8, Black: Ne4.** (a) Our position; Melanie plays Ne4 (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2594` (ply 15) [structural distance $d=38$, QAP obj=35]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa7 as their Ra8, our Pd5 as their Bc8; transferred move f6e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2595` (ply 15) [structural distance $d=38$, QAP obj=35]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa7 as their Ra8, our Pd5 as their Bc8; transferred move f6e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3461` (ply 15) [structural distance $d=45$, QAP obj=34]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pd5 as their Ra8, our Pg7 as their Rh8; transferred move f6e4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ne4 (summed QAP objective 340, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



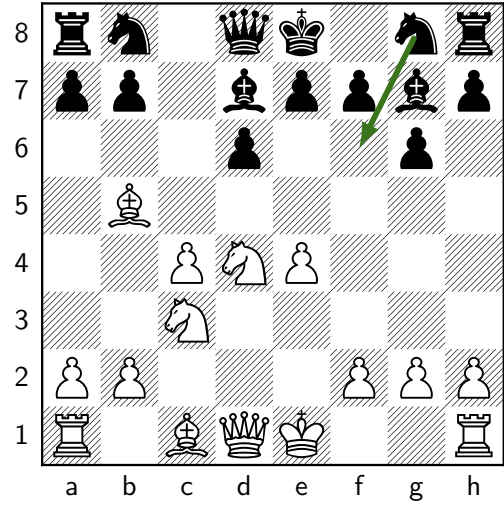
(a)



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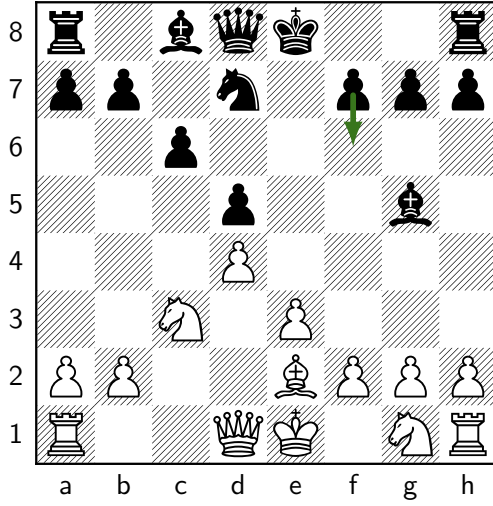


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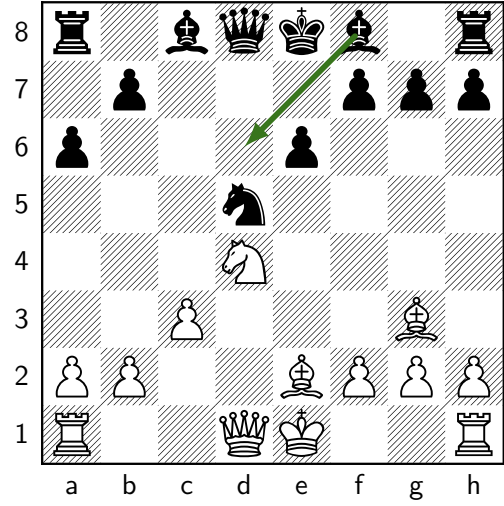


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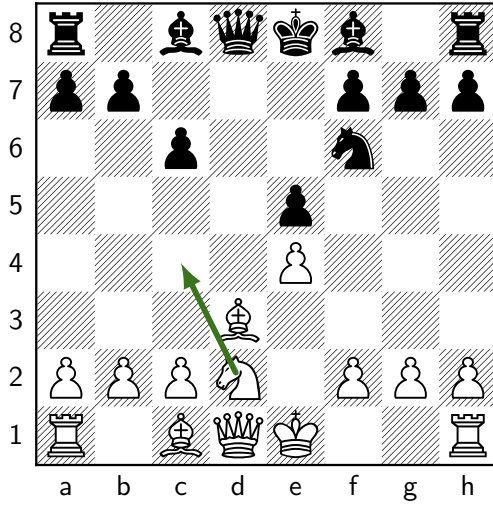
Figure 3: **Move 9, Black: Bxg5.** (a) Our position; Melanie plays Bxg5 (e7g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#334` (ply 23) [structural distance $d=55$, QAP obj=65]: the master played e5; under μ our *Pb2* plays the role of that game’s *Pe3*; cross-type roles: our *ng1* as their *Ra8*, our *Ra8* as their *Bc8*; transferred move b7b5. (c) Retrieved from `ccr1_top_p0003#667` (ply 13) [structural distance $d=64$, QAP obj=65]: the master played Nf6; under μ our *Pb2* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Ra8*, our *Pg7* as their *Nb8*; transferred move b7b6. (d) Retrieved from `ccr1_top_p0003#668` (ply 13) [structural distance $d=64$, QAP obj=65]: the master played Nf6; under μ our *Pb2* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Ra8*, our *Pg7* as their *Nb8*; transferred move b7b6. Synthesis: no retrieved analog transferred to Bxg5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



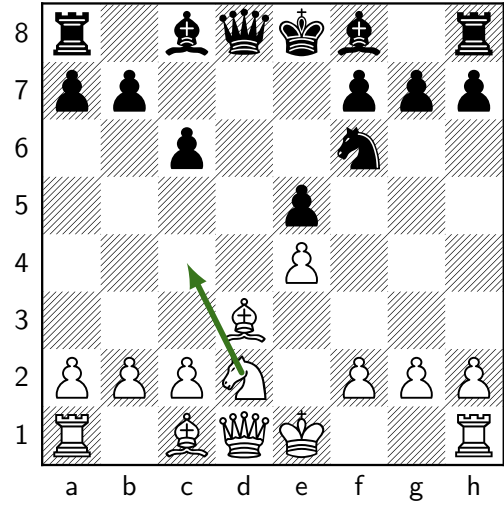
(a)



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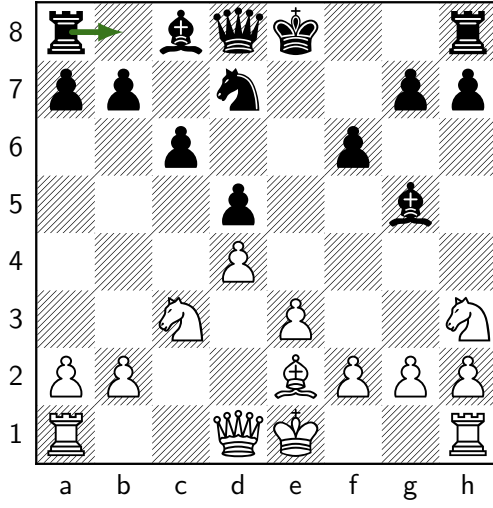


(c)

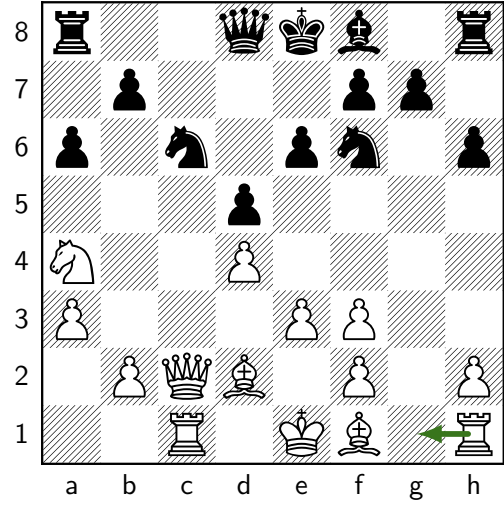


(d)

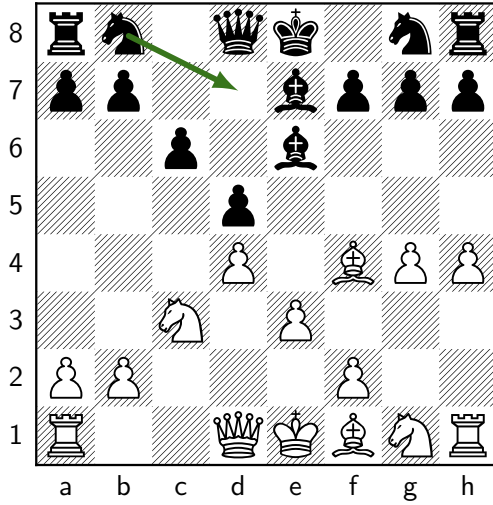
Figure 4: **Move 10, Black: f6.** (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#907` (ply 21) [structural distance $d=78$, QAP obj=48]: the master played Bd6; under μ our Pf2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our pb2 as their Bc8, our Nd7 as their Pf7; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1460` (ply 14) [structural distance $d=80$, QAP obj=37]: the master played Nc4; under μ our Pf2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra1, our ng1 as their Pa2; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1461` (ply 14) [structural distance $d=80$, QAP obj=37]: the master played Nc4; under μ our Pf2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra1, our ng1 as their Pa2; transferred move f7f6 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to f6 (summed QAP objective 414, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



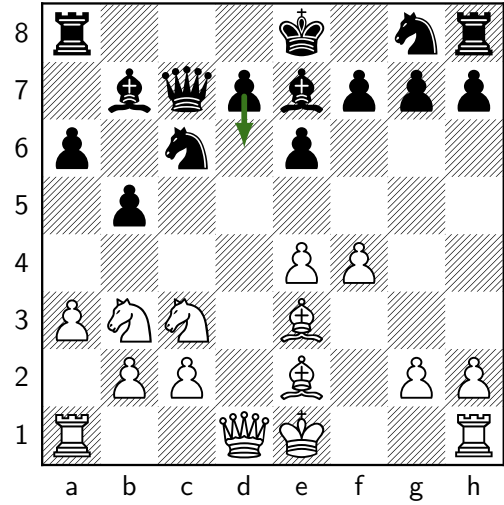
(a)



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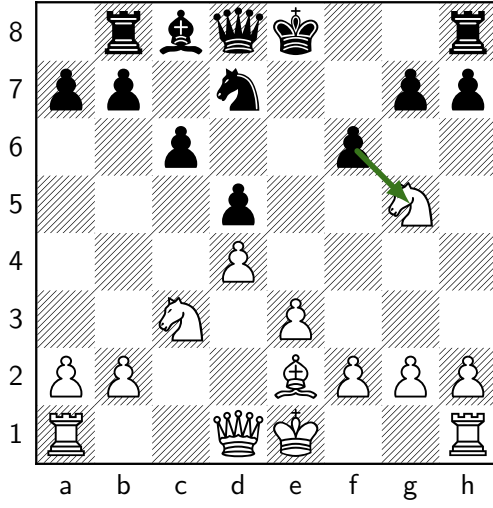


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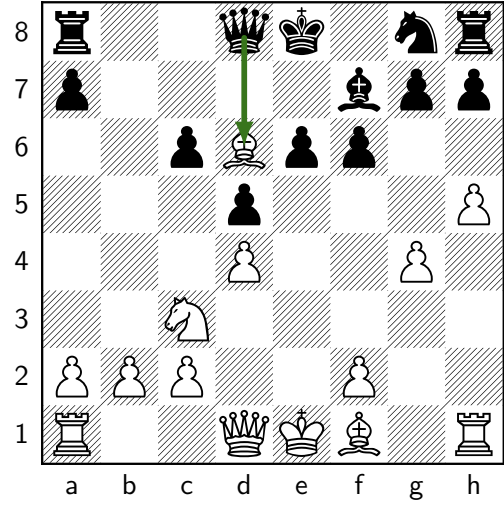


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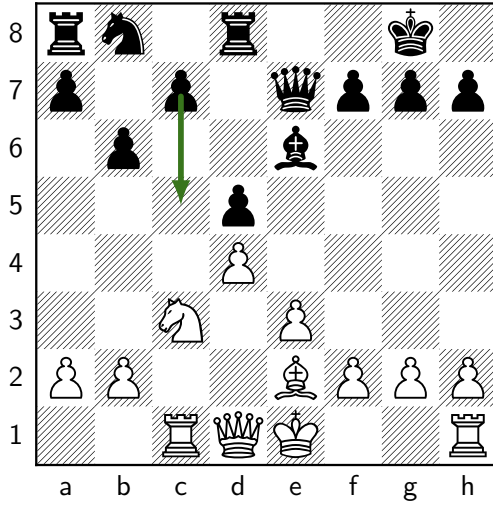
Figure 5: **Move 11, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#949` (ply 28) [structural distance $d=64$, QAP obj=32]: the master played Rg1; under μ our Ra1 plays the role of that game’s Rh1; cross-type roles: our Qd8 as their Rc1, our nh3 as their Qc2; transferred move a8b8 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#267` (ply 15) [structural distance $d=59$, QAP obj=22]: the master played Nd7; under μ our Bg4 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ra8, our Pf6 as their Ng8; transferred move g5h6. (d) Retrieved from `cctl_top_p0003#2422` (ply 19) [structural distance $d=61$, QAP obj=12]: the master played d6; under μ our Pb2 plays the role of that game’s Pd2; cross-type roles: our Pg7 as their Ra8, our qd1 as their Rh8; transferred move b7b6. Synthesis: 1 retrieved analog transferred to Rb8 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



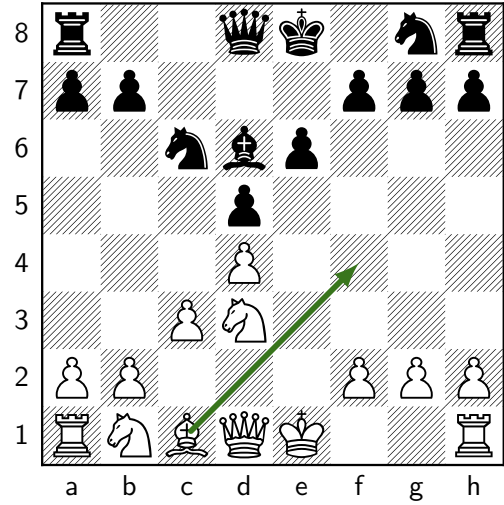
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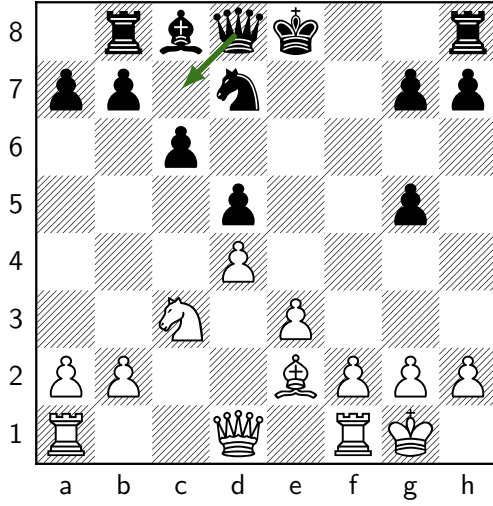


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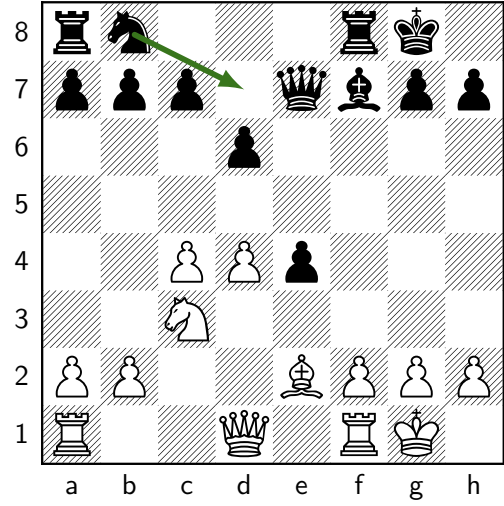


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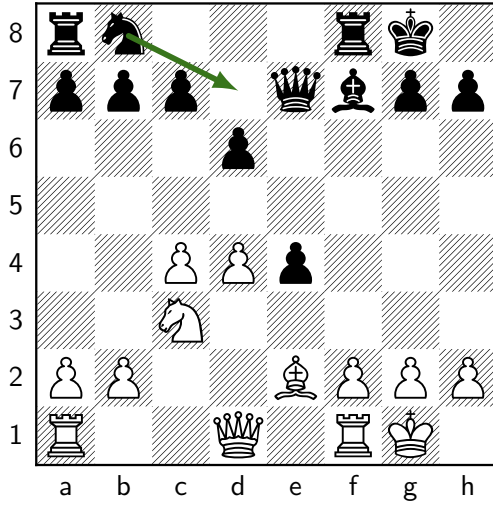
Figure 6: **Move 12, Black: f6g5.** (a) Our position; Melanie plays f6g5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#307` (ply 25) [structural distance $d=63$, QAP obj=15]: the master played Qxd6; under μ our Pf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Ra8, our Pd5 as their Ng8; transferred move f6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3322` (ply 23) [structural distance $d=71$, QAP obj=11]: the master played c5; under μ our Rb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Pa7 as their Nb8; transferred move b8a8. (d) Retrieved from `ccr1_top_p0001#1110` (ply 20) [structural distance $d=75$, QAP obj=6]: the master played Bf4; under μ our Nd2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Ra1, our Rb8 as their Nb1; transferred move d7e5. Synthesis: 1 retrieved analog transferred to f6g5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



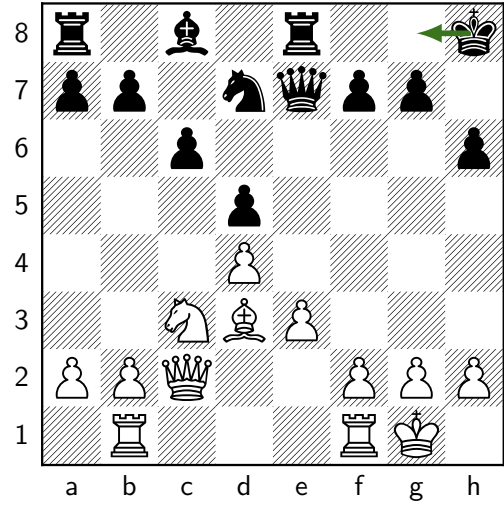
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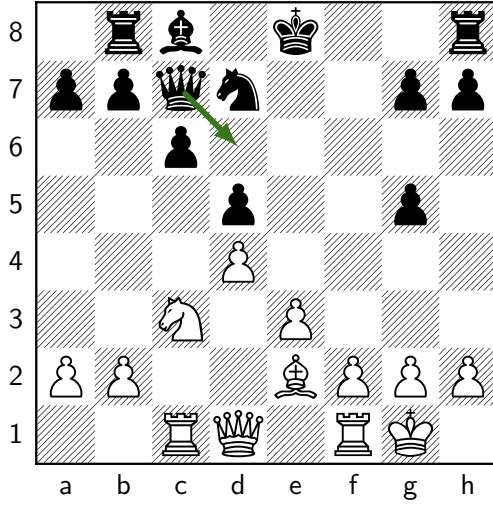


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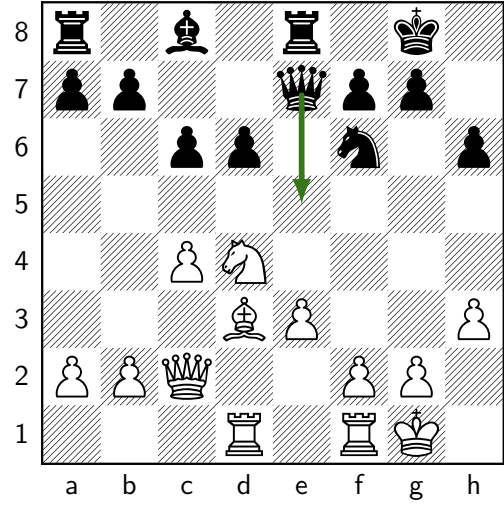


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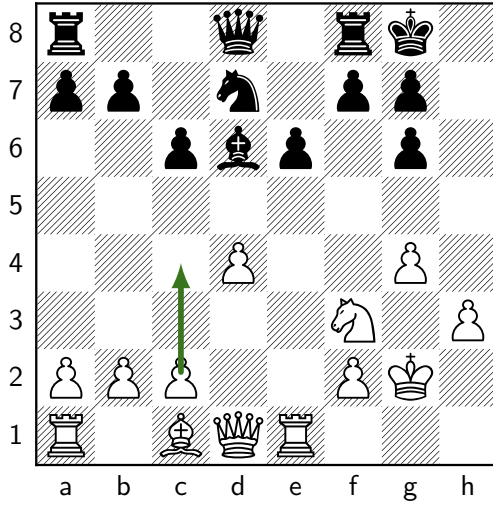
Figure 7: **Move 13, Black: Qc7.** (a) Our position; Melanie plays Qc7 (d8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3716` (ply 25) [structural distance $d=48$, QAP obj=26]: the master played Nd7; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pe3 as their Rf8, our Nd7 as their Pc7; transferred move d8c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3717` (ply 25) [structural distance $d=48$, QAP obj=26]: the master played Nd7; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pe3 as their Rf8, our Nd7 as their Pc7; transferred move d8c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2394` (ply 31) [structural distance $d=66$, QAP obj=25]: the master played Kg8; under μ our Ke1 plays the role of that game’s Kh1; cross-type roles: our Bc8 as their Ra8, our Nd7 as their Bc8; transferred move e8e7. Synthesis: 2 retrieved analogs transferred to Qc7 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



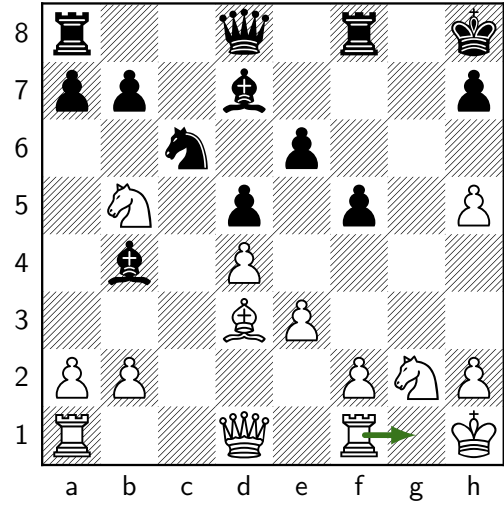
(a)



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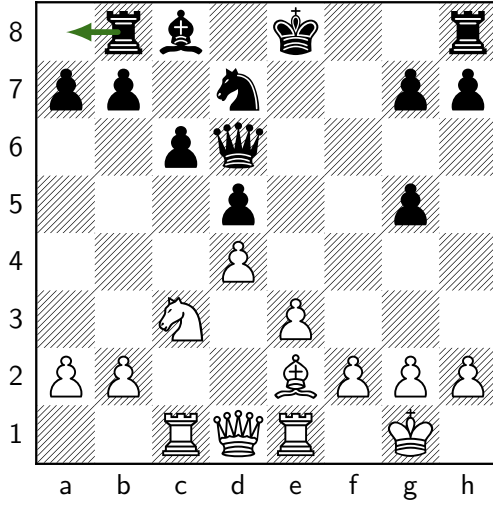


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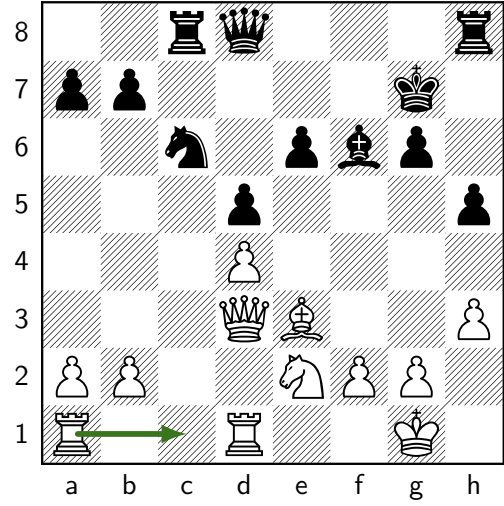


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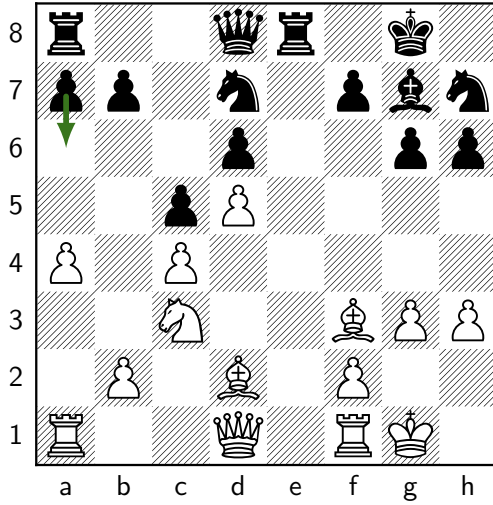
Figure 8: **Move 14, Black: Qd6.** (a) Our position; Melanie plays Qd6 (c7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#2831` (ply 31) [structural distance $d=60$, QAP obj=32]: the master played Qe5; under μ our Qc2 plays the role of that game’s Qe2; cross-type roles: our Pa7 as their Ra8, our pb2 as their Re8; transferred move c7d6 (**endorses this move**). (c) Retrieved from `cctl_top_p0002#3812` (ply 26) [structural distance $d=70$, QAP obj=13]: the master played c4; under μ our Qc2 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our pb2 as their Ra1, our Nd7 as their Bc1; transferred move c7d6 (**endorses this move**). (d) Retrieved from `cctl_top_p0001#2888` (ply 34) [structural distance $d=68$, QAP obj=43]: the master played Rg1; under μ our Rh1 plays the role of that game’s Rf1; cross-type roles: our Ph7 as their Qd1, our Bc8 as their Pa2; transferred move h8f8. Synthesis: 2 retrieved analogs transferred to Qd6 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



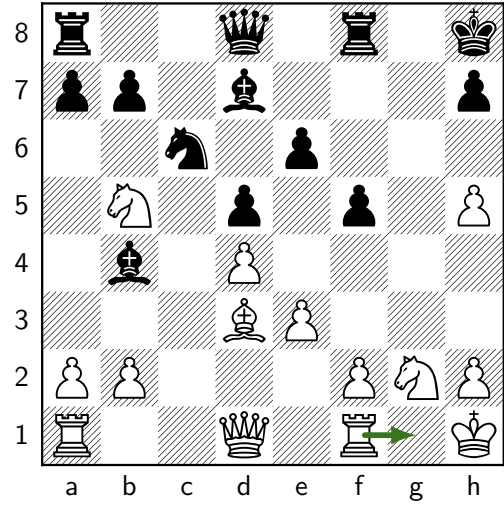
(a)



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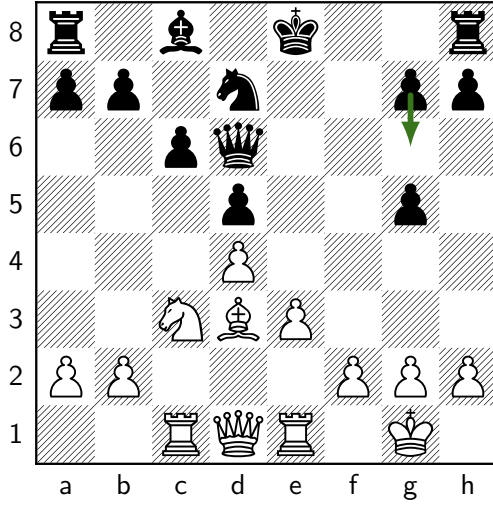


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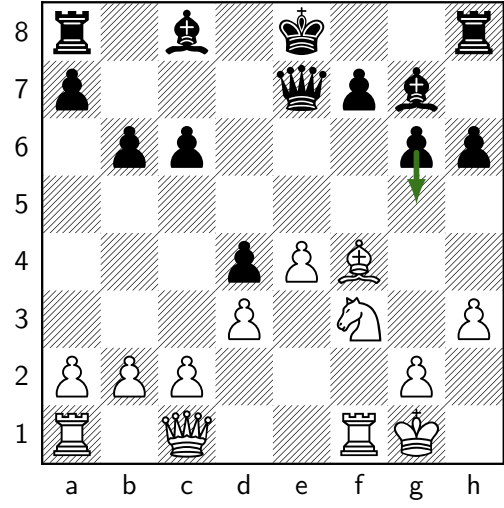


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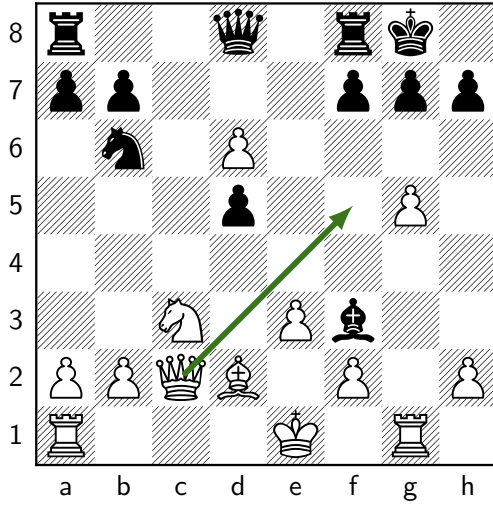
Figure 9: **Move 15, Black: Ra8.** (a) Our position; Melanie plays Ra8 (b8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#175` (ply 44) [structural distance $d=56$, QAP obj=54]: the master played Rac1; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Bc8 as their Pa2, our Pc6 as their Ne2; transferred move b8a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1610` (ply 29) [structural distance $d=82$, QAP obj=20]: the master played a6; under μ our Rb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our pd4 as their Qd8; transferred move b8a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2888` (ply 34) [structural distance $d=74$, QAP obj=36]: the master played Rg1; under μ our Rh1 plays the role of that game’s Rf1; cross-type roles: our be2 as their Qd1, our Bc8 as their Pa2; transferred move h8f8. Synthesis: 2 retrieved analogs transferred to Ra8 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



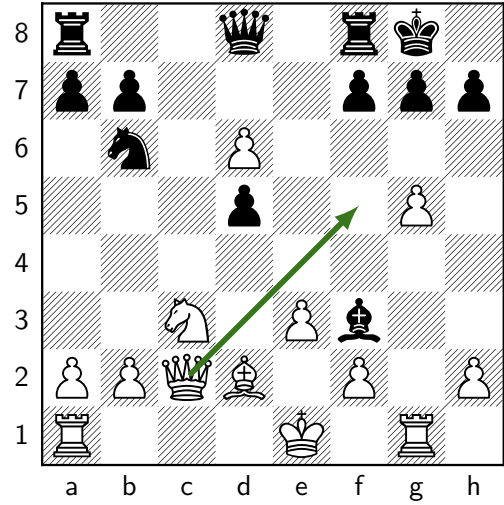
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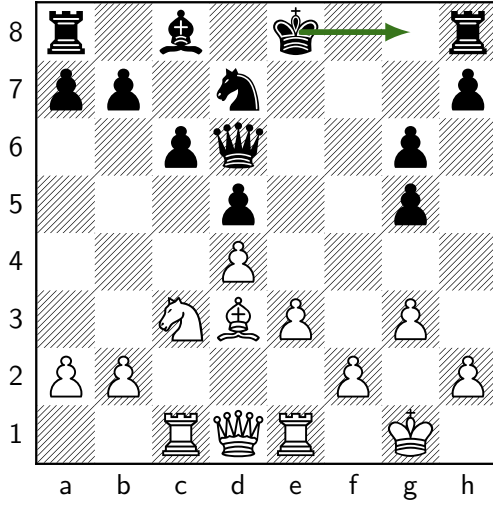


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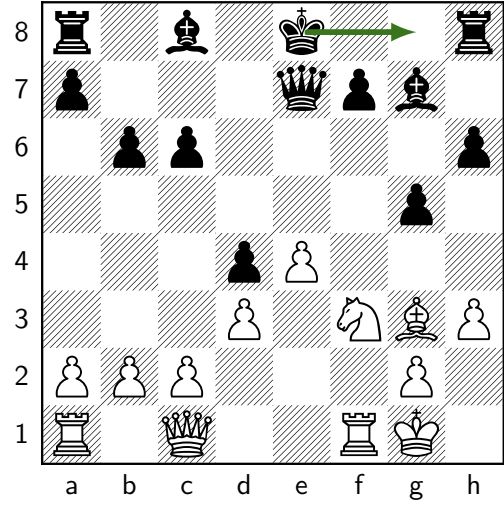


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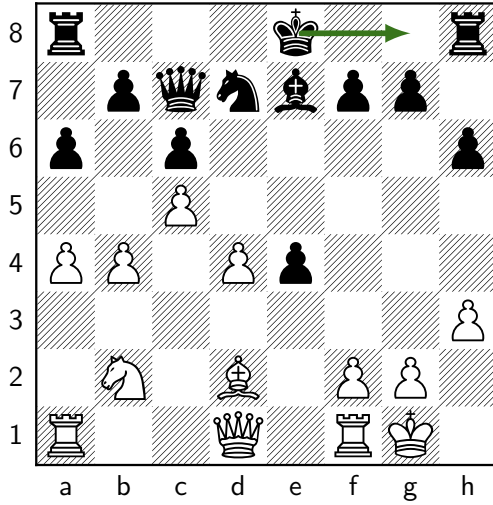
Figure 10: **Move 16, Black: g6** (g7g6). (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1335` (ply 35) [structural distance $d=71$, QAP obj=43]: the master played g5; under μ our *Pg2* plays the role of that game’s *Pg3*; cross-type roles: our *Pb7* as their *Bc8*, our *Nd7* as their *Pf7*; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1148` (ply 28) [structural distance $d=68$, QAP obj=49]: the master played Qf5; under μ our *Qd3* plays the role of that game’s *Qc2*; cross-type roles: our *Bc8* as their *Rg1*, our *Ra8* as their *Pa2*; transferred move d6f4. (d) Retrieved from `ccrl_top_p0002#2624` (ply 28) [structural distance $d=68$, QAP obj=49]: the master played Qf5; under μ our *Qd3* plays the role of that game’s *Qc2*; cross-type roles: our *Bc8* as their *Rg1*, our *Ra8* as their *Pa2*; transferred move d6f4. Synthesis: 1 retrieved analog transferred to g6 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



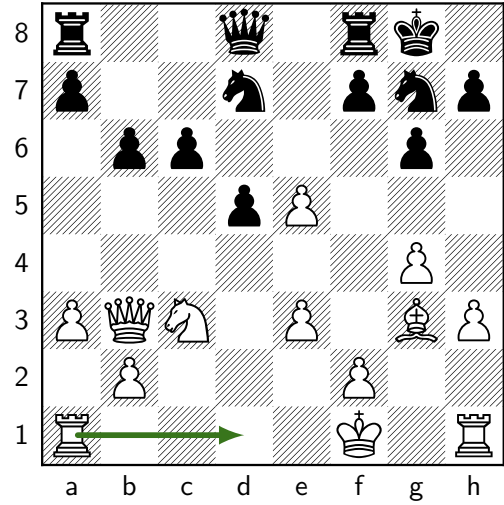
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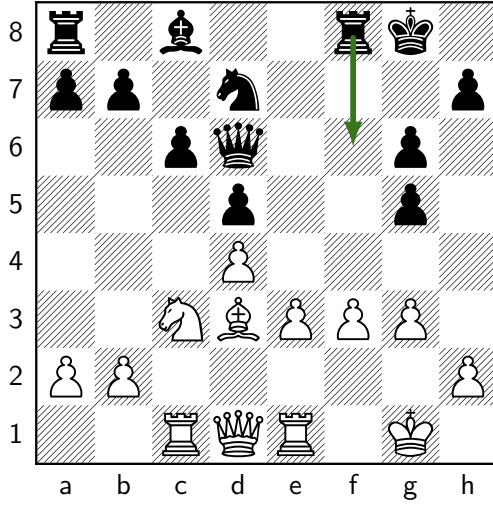


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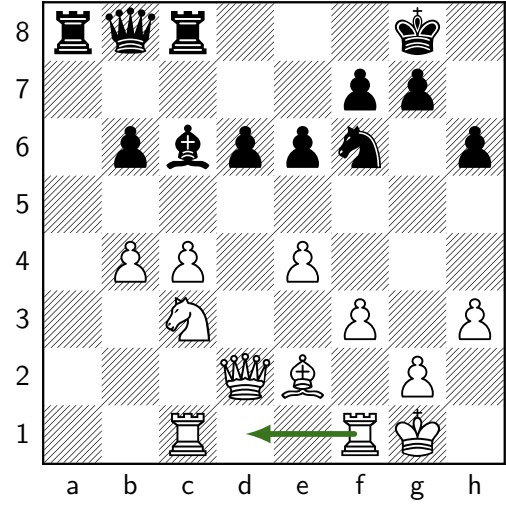


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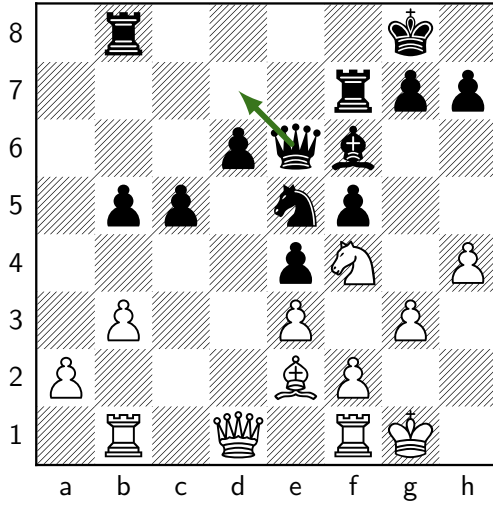
Figure 11: **Move 17, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1335` (ply 37) [structural distance $d=80$, QAP obj=67]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *Bc8* as their *Rh8*, our *Nd7* as their *Pa7*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1847` (ply 35) [structural distance $d=84$, QAP obj=47]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *Qd6* as their *Rh8*, our *rc1* as their *Qc7*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#287` (ply 32) [structural distance $d=81$, QAP obj=61]: the master played Rd1; under μ our *Ra1* plays the role of that game's *Ra1*; cross-type roles: our *rc1* as their *Pb2*, our *Nd7* as their *Pf2*; transferred move a8b8. Synthesis: 2 retrieved analogs transferred to 0-0 (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



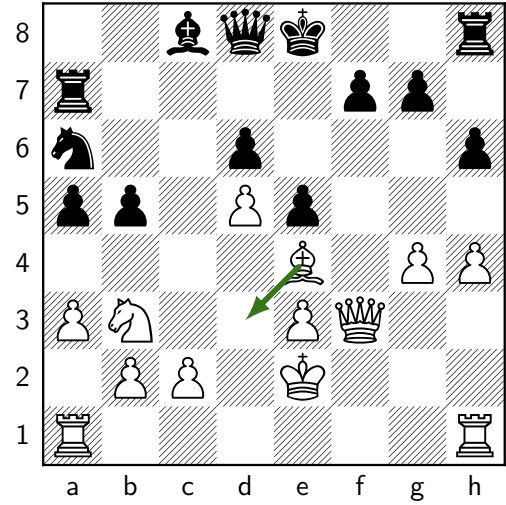
(a)



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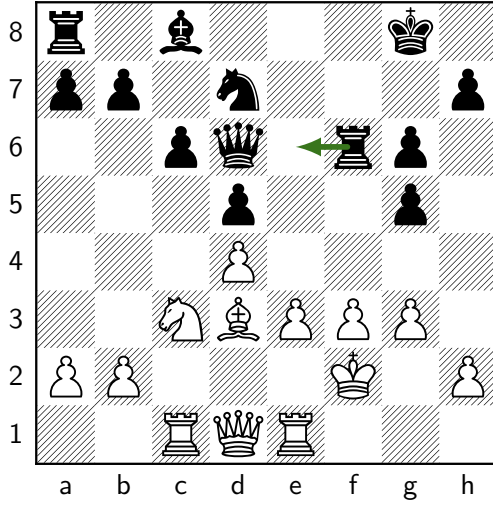


(c)

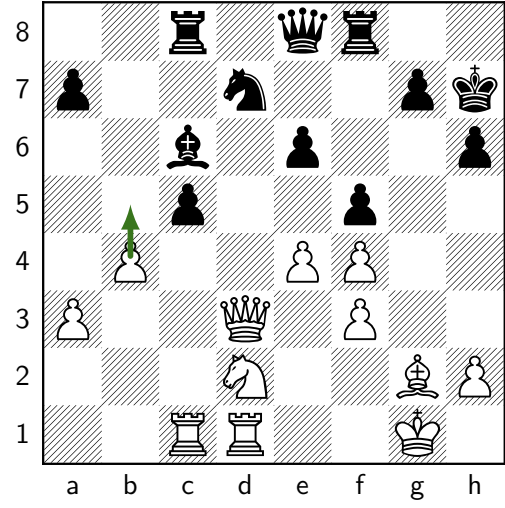


(d)

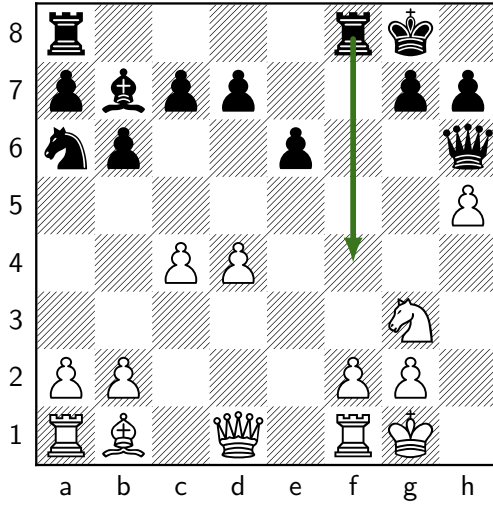
Figure 12: **Move 18, Black: Rf6.** (a) Our position; Melanie plays Rf6 (f8f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1250` (ply 48) [structural distance $d=68$, QAP obj=56]: the master played Rfd1; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Nd7 as their Rc1, our Pg6 as their Be2; transferred move f8f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#489` (ply 45) [structural distance $d=68$, QAP obj=5]: the master played Qd7; under μ our Rf1 plays the role of that game’s Qe3 (a **cross-type** correspondence); cross-type roles: our pe3 as their Rf7, our Pg6 as their Bf6; transferred move f8f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1069` (ply 36) [structural distance $d=76$, QAP obj=53]: the master played Bd3; under μ our Nd2 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our pa2 as their Ra1, our qd1 as their Rh1; transferred move d7f6. Synthesis: 2 retrieved analogs transferred to Rf6 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



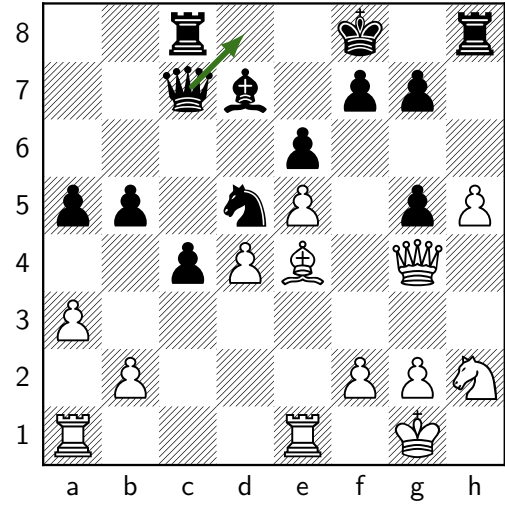
(a)



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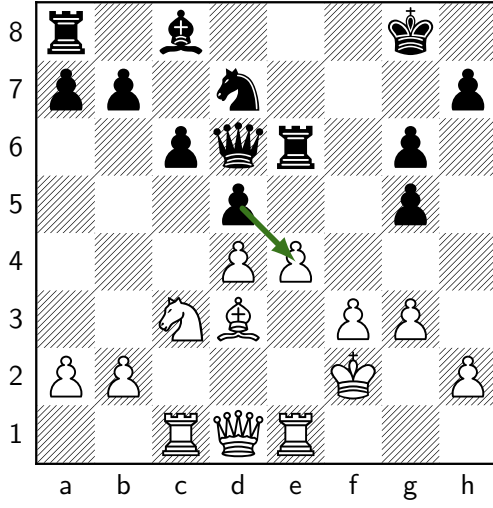


(c)

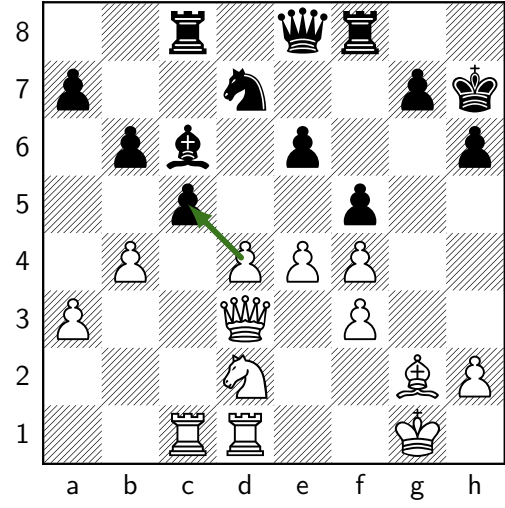


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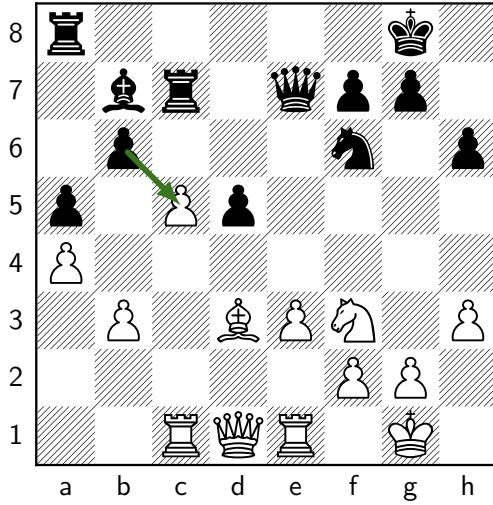
Figure 13: **Move 19, Black: Re6.** (a) Our position; Melanie plays Re6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#656 (ply 42) [structural distance $d=60$, QAP obj=49]: the master played b5; under μ our Rf3 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Rd1, our Pc6 as their Nd2; transferred move f6e6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#685 (ply 33) [structural distance $d=76$, QAP obj=26]: the master played Rf4; under μ our Rf3 plays the role of that game’s Rf1; cross-type roles: our Nd7 as their Ra8, our Qd6 as their Bb7; transferred move f6e6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3736 (ply 47) [structural distance $d=80$, QAP obj=24]: the master played Qd8; under μ our Rf3 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Rc8, our bd3 as their Rh8; transferred move f6e6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re6 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



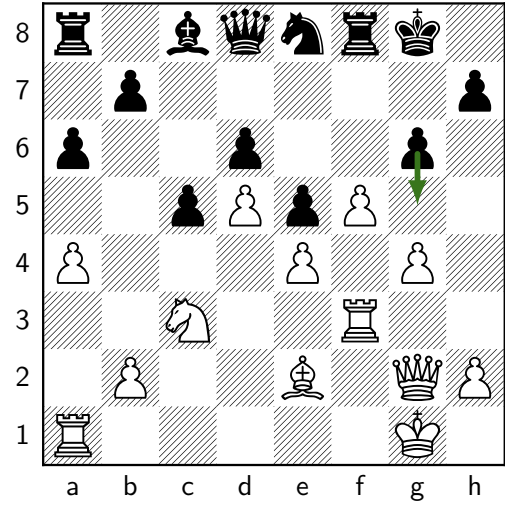
(a)



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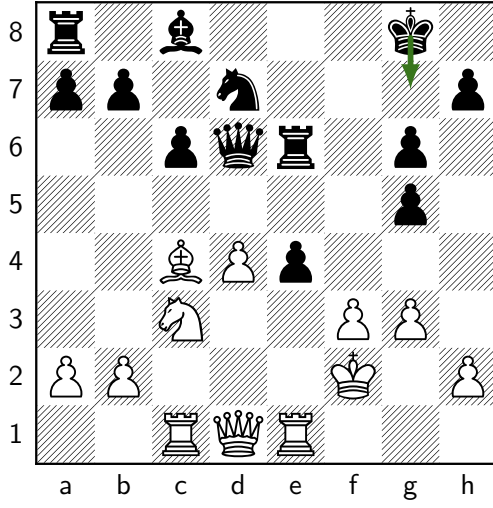


(c)

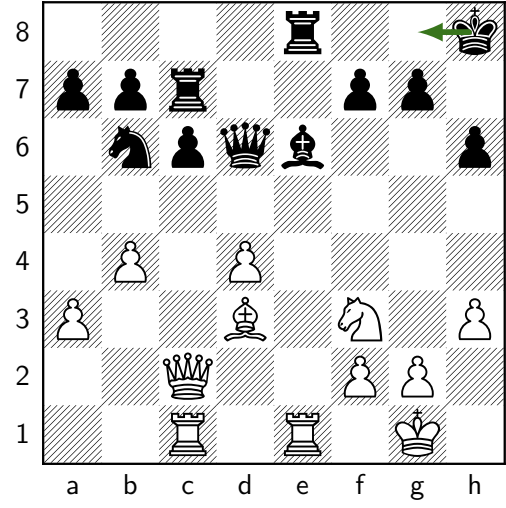


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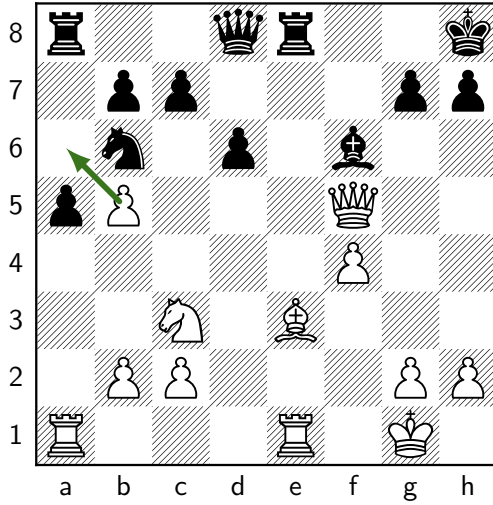
Figure 14: **Move 20, Black: dxe4.** (a) Our position; Melanie plays **dxe4** (**d5e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#656` (ply 40) [structural distance $d=81$, QAP obj=176]: the master played **dx c5**; under μ our *Pd4* plays the role of that game’s *Pd4*; cross-type roles: our *Pg6* as their *Rd1*, our *pb2* as their *Bg2*; transferred move **d5e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#860` (ply 37) [structural distance $d=80$, QAP obj=140]: the master played **bxc5**; under μ our *Pd4* plays the role of that game’s *Pb3*; cross-type roles: our *Pb7* as their *Ra8*, our *Nd7* as their *Bb7*; transferred move **d5e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1309` (ply 43) [structural distance $d=72$, QAP obj=102]: the master played **g5**; under μ our *Pd4* plays the role of that game’s *Pg3*; cross-type roles: our *Ph7* as their *Ra8*, our *Re6* as their *Bc8*; transferred move **d5e4** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **dxe4** (summed QAP objective 713, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



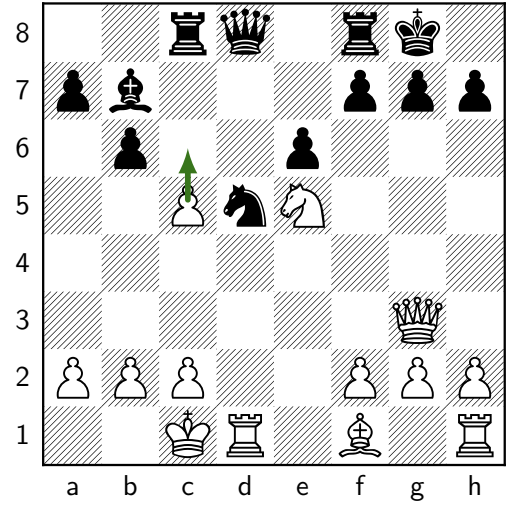
(a)



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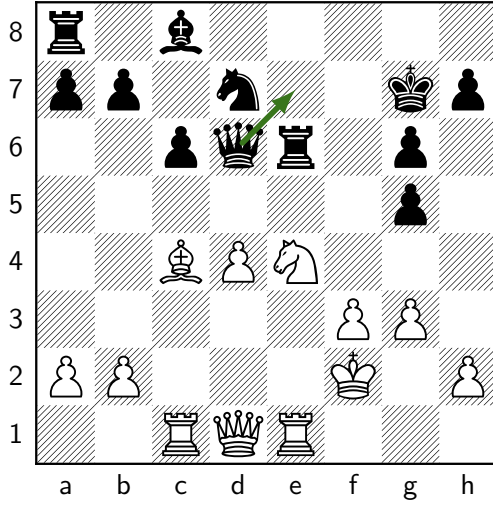


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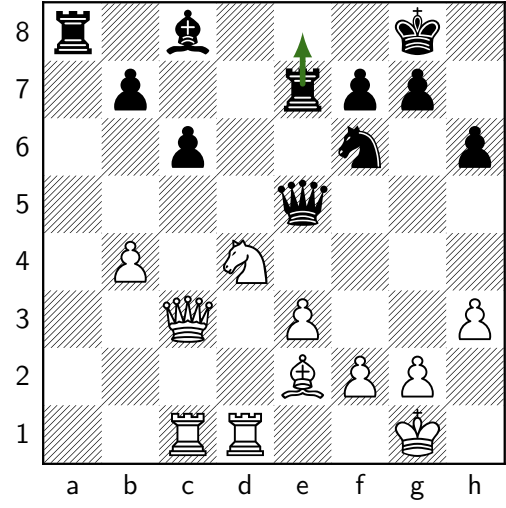


(d)

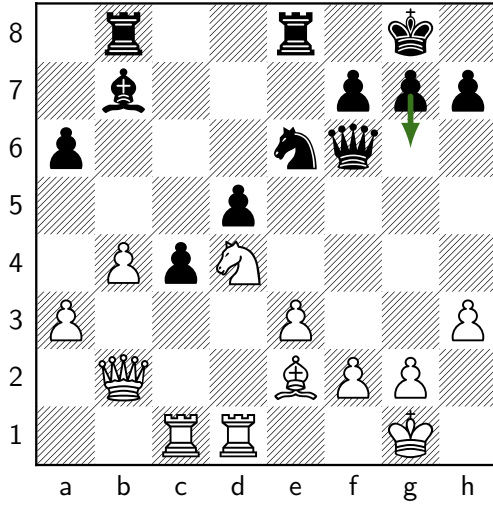
Figure 15: **Move 21, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1429` (ply 45) [structural distance $d=80$, QAP obj=21]: the master played Kg8; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our Pg6 as their Rc7, our Re6 as their Pf7; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#133` (ply 36) [structural distance $d=87$, QAP obj=43]: the master played bxa6; under μ our Pg4 plays the role of that game’s Pb5; cross-type roles: our Bc8 as their Ra1, our nc3 as their Pc2; transferred move g5g4. (d) Retrieved from `ccrl_top_p0001#517` (ply 28) [structural distance $d=77$, QAP obj=40]: the master played c6; under μ our Pe5 plays the role of that game’s Pc5; cross-type roles: our bc4 as their Rd1, our pa2 as their Bf1; transferred move e4e3. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



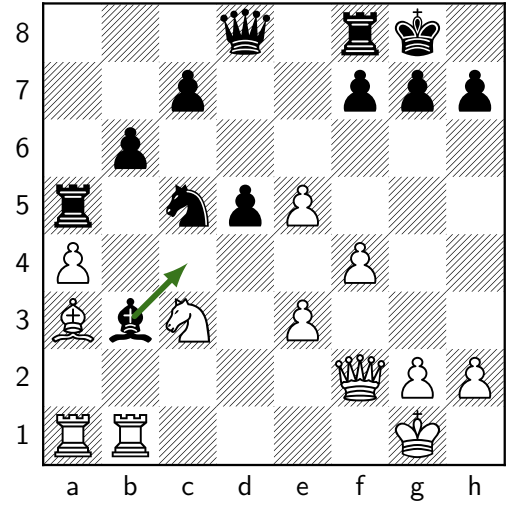
(a)



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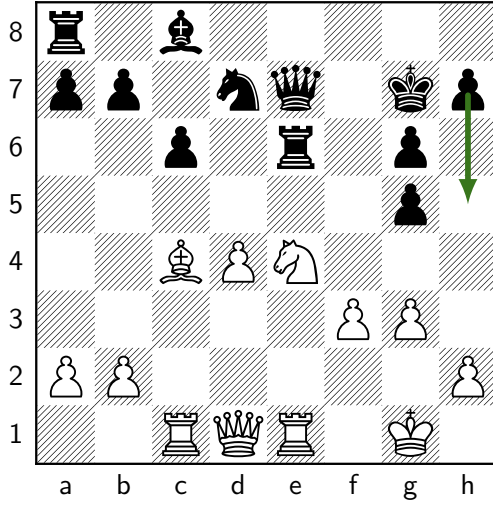


(c)

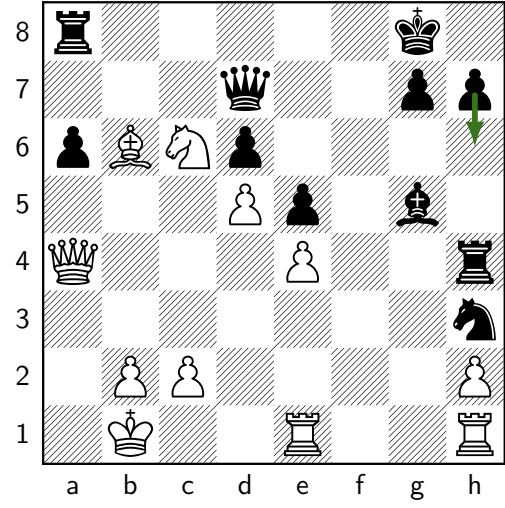


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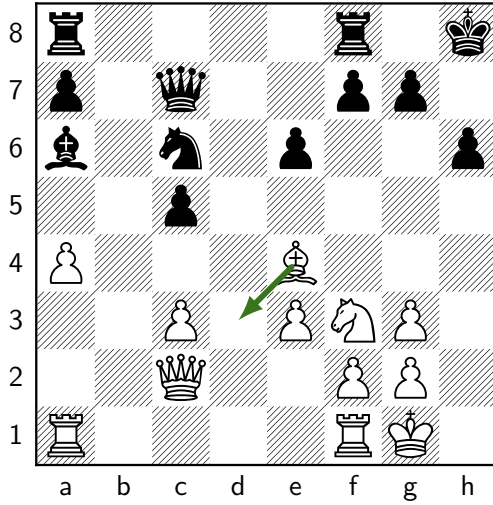
Figure 16: **Move 22, Black: Qe7**. (a) Our position; Melanie plays **Qe7** (d6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#317` (ply 43) [structural distance $d=53$, QAP obj=23]: the master played **Re8**; under μ our *Qd3* plays the role of that game’s *Re2* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Bc8*, our *rc1* as their *Pf7*; transferred move **d6e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2929` (ply 43) [structural distance $d=68$, QAP obj=50]: the master played **g6**; under μ our *Pa2* plays the role of that game’s *Pg2*; cross-type roles: our *pb2* as their *Rb8*, our *Ra8* as their *Pf7*; transferred move **a7a6**. (d) Retrieved from `ccr1_top_p0002#3190` (ply 43) [structural distance $d=68$, QAP obj=43]: the master played **Bc4**; under μ our *Re3* plays the role of that game’s *Bb6* (a **cross-type** correspondence); cross-type roles: our *pd4* as their *Qd8*, our *Ph7* as their *Rf8*; transferred move **e6e5**. Synthesis: 1 retrieved analog transferred to **Qe7** (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



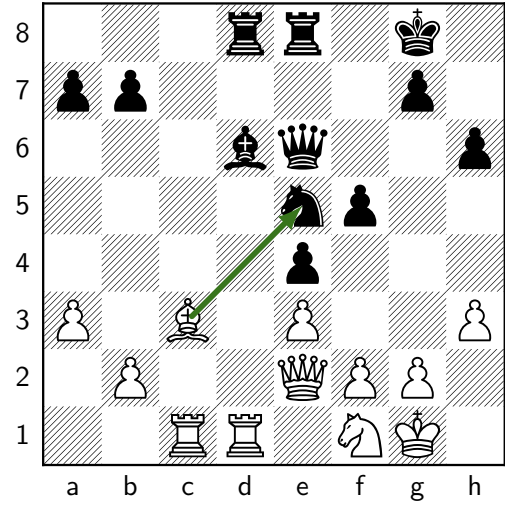
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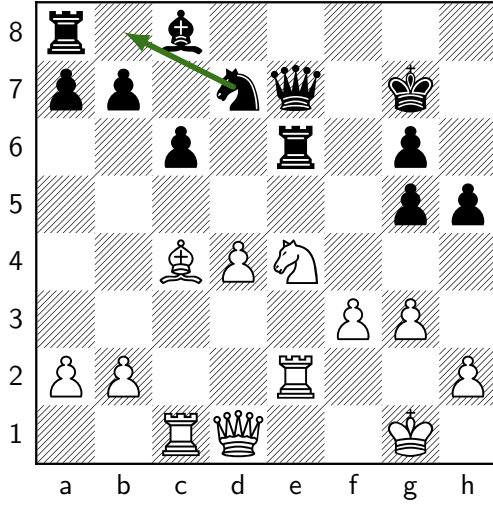


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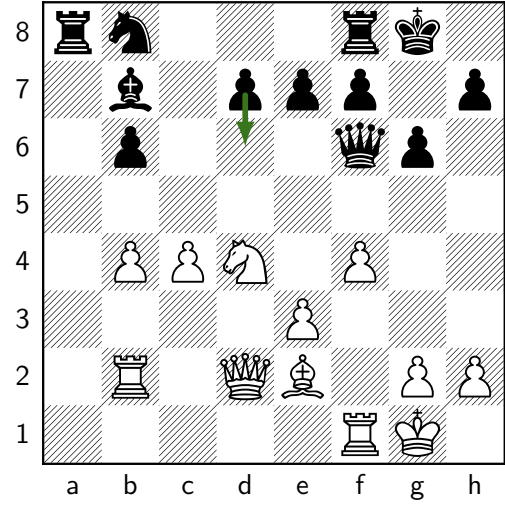


(d)

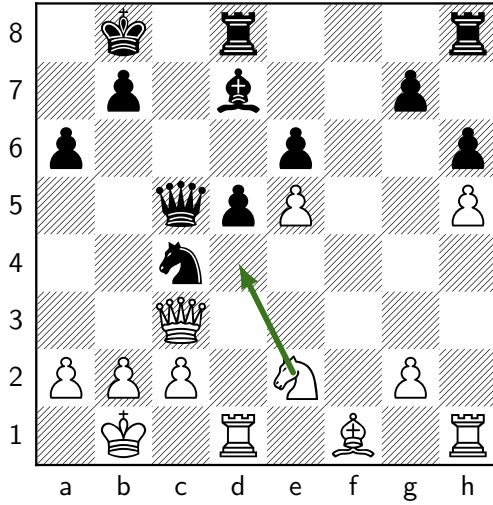
Figure 17: **Move 23, Black: h5**. (a) Our position; Melanie plays h5 (h7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1935` (ply 53) [structural distance $d=69$, QAP obj=50]: the master played h6; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Qe7* as their *Ra8*, our *Re6* as their *Qd7*; transferred move h7h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#359` (ply 36) [structural distance $d=63$, QAP obj=46]: the master played Bd3; under μ our *Re3* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Pg6* as their *Ra1*, our *Nd7* as their *Pf2*; transferred move e6e5. (d) Retrieved from `ccr1_top_p0001#1502` (ply 58) [structural distance $d=70$, QAP obj=44]: the master played Bxe5; under μ our *Re3* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Qe7* as their *Rc1*, our *Pc6* as their *Nf1*; transferred move e6e4. Synthesis: 1 retrieved analog transferred to h5 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



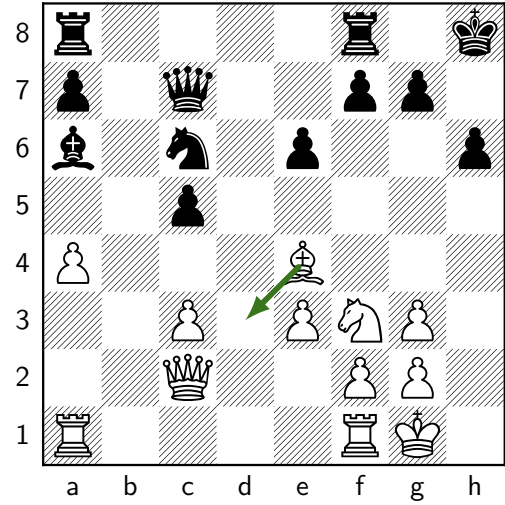
(a)



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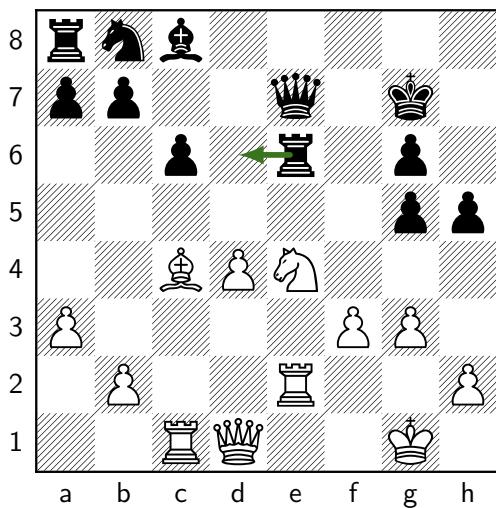


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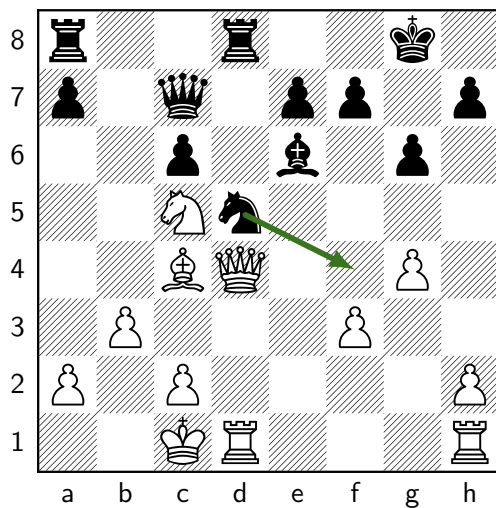


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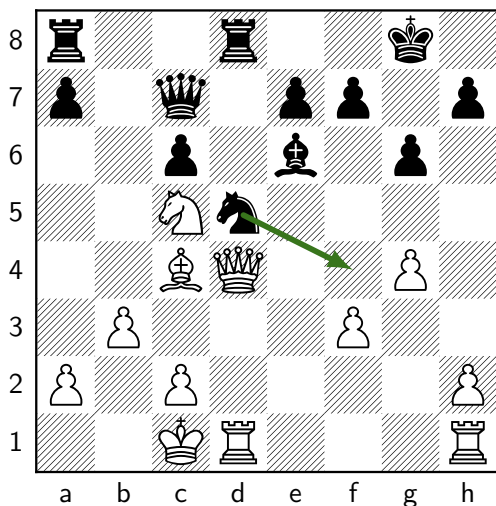
Figure 18: **Move 24, Black: Nb8**. (a) Our position; Melanie plays Nb8 (d7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3264` (ply 37) [structural distance $d=67$, QAP obj=41]: the master played d6; under μ our *Nd2* plays the role of that game’s *Pd2* (a **cross-type** correspondence); cross-type roles: our *pd4* as their *Nb8*, our *Qe7* as their *Rf8*; transferred move d7b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1585` (ply 48) [structural distance $d=66$, QAP obj=54]: the master played Nd4; under μ our *Pb2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Qe7* as their *Rd1*, our *pa2* as their *Bf1*; transferred move b7b6. (d) Retrieved from `ccr1_top_p0001#359` (ply 36) [structural distance $d=66$, QAP obj=47]: the master played Bd3; under μ our *Re3* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Pc6* as their *Ra1*, our *Nd7* as their *Pe3*; transferred move e6e5. Synthesis: 1 retrieved analog transferred to Nb8 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



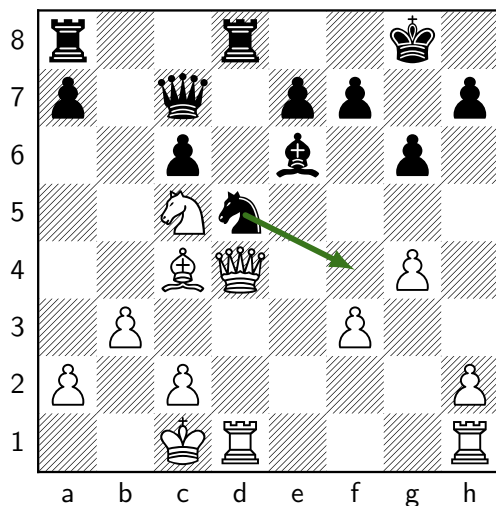
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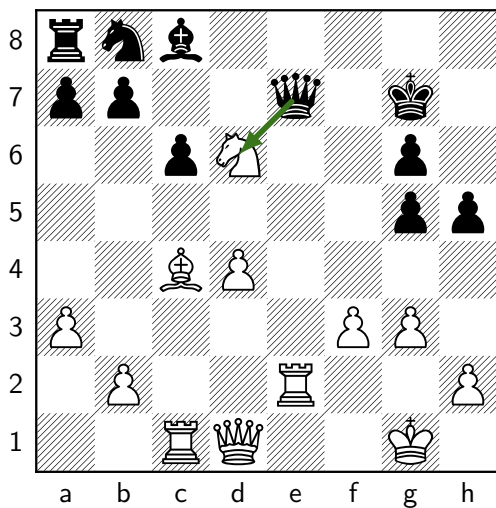


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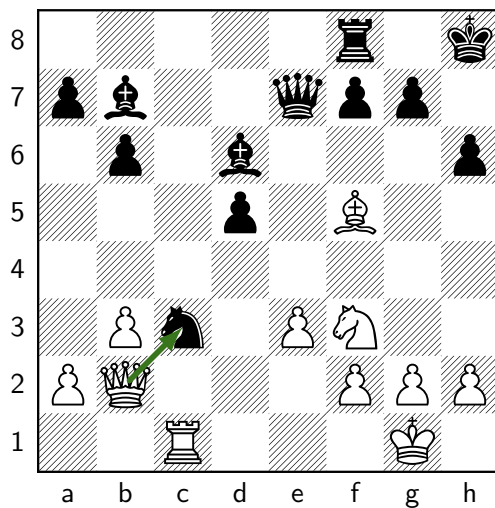


(d)

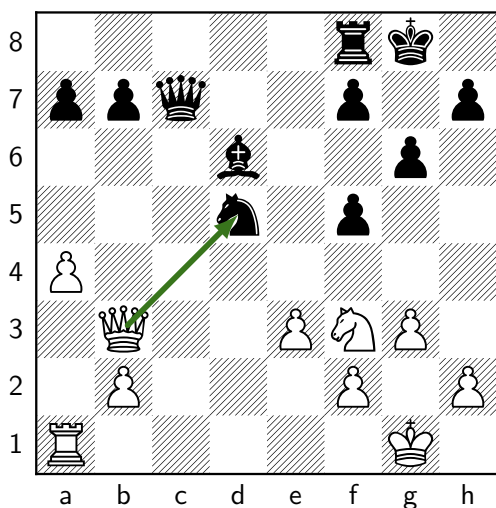
Figure 19: **Move 25, Black: Rd6.** (a) Our position; Melanie plays Rd6 (e6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1668` (ply 35) [structural distance $d=67$, QAP obj=39]: the master played Nf4; under μ our *Re3* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Rd8*, our *Nb8* as their *Pa7*; transferred move e6d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1164` (ply 35) [structural distance $d=67$, QAP obj=39]: the master played Nf4; under μ our *Re3* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Rd8*, our *Nb8* as their *Pa7*; transferred move e6d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1595` (ply 35) [structural distance $d=67$, QAP obj=39]: the master played Nf4; under μ our *Re3* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Rd8*, our *Nb8* as their *Pa7*; transferred move e6d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rd6 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



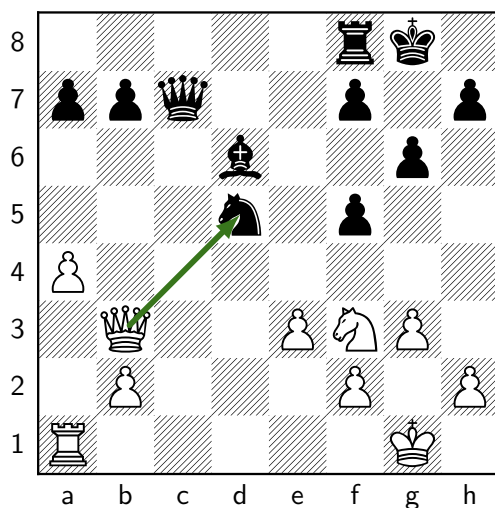
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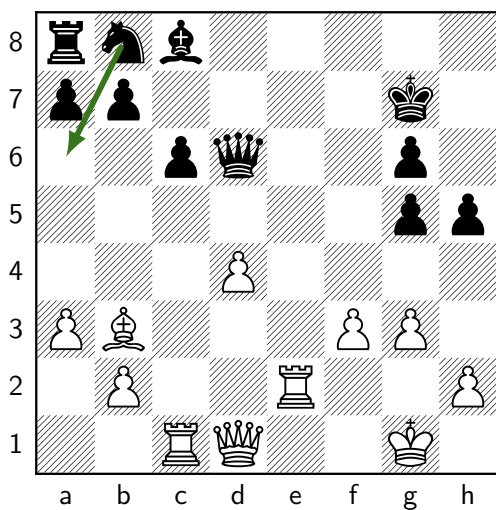


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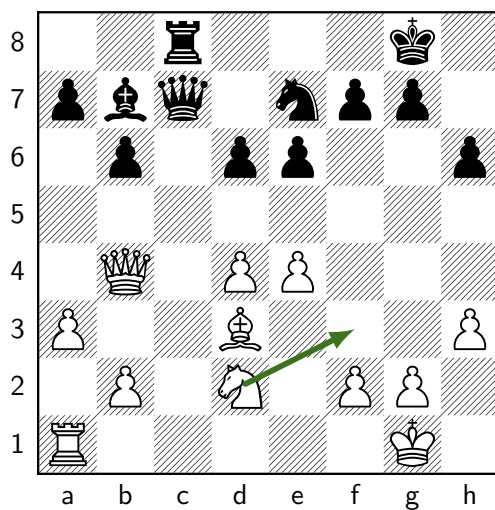


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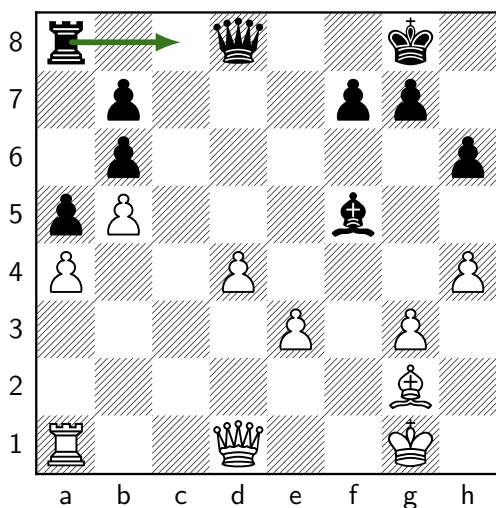
Figure 20: **Move 26, Black: Qxd6.** (a) Our position; Melanie plays Qxd6 (e7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#1392` (ply 42) [structural distance $d=70$, QAP obj=32]: the master played Qxc3; under μ our Qe2 plays the role of that game’s Qb2; cross-type roles: our Qe7 as their Rc1, our Bc8 as their Ph2; transferred move e7d6 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#798` (ply 44) [structural distance $d=81$, QAP obj=16]: the master played Qxd5; under μ our Qe2 plays the role of that game’s Qb3; cross-type roles: our Nb8 as their Ra1, our Bc8 as their Pb2; transferred move e7d6 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#1517` (ply 44) [structural distance $d=81$, QAP obj=16]: the master played Qxd5; under μ our Qe2 plays the role of that game’s Qb3; cross-type roles: our Nb8 as their Ra1, our Bc8 as their Pb2; transferred move e7d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qxd6 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



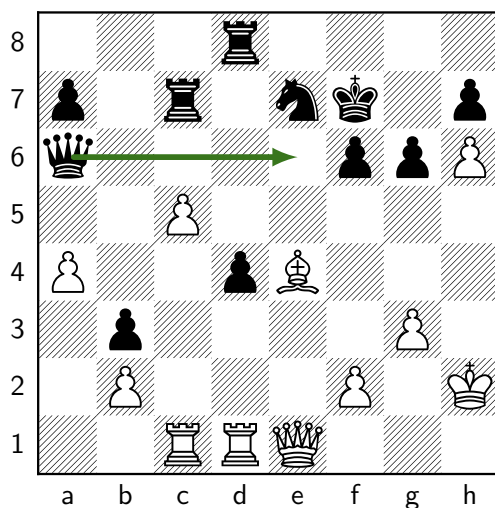
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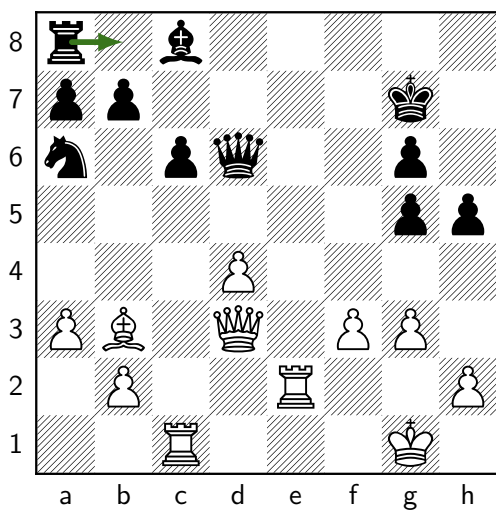


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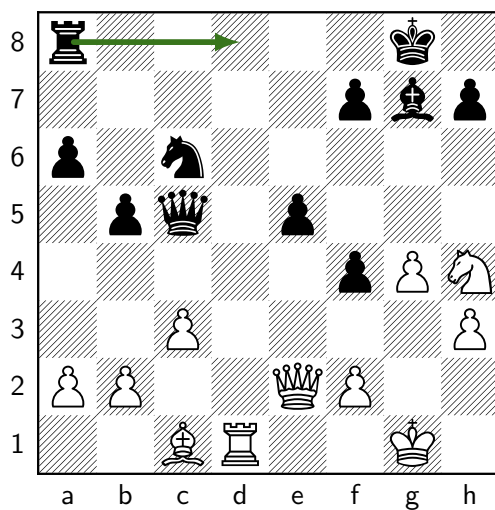


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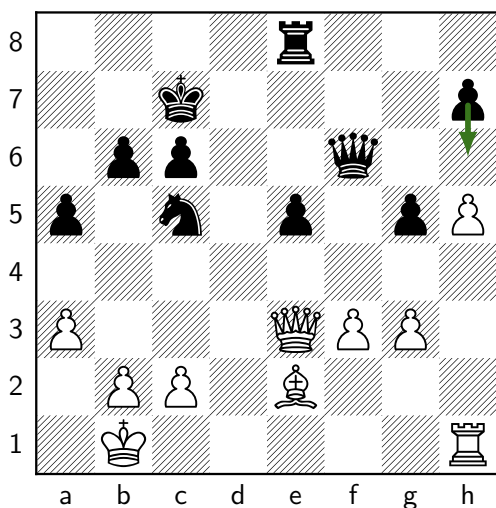
Figure 21: **Move 27, Black: Na6.** (a) Our position; Melanie plays Na6 (b8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#3350` (ply 44) [structural distance $d=68$, QAP obj=93]: the master played Nf3; under μ our Nb1 plays the role of that game’s Nd2; cross-type roles: our Bc8 as their Pf2, our Ph5 as their Bd3; transferred move b8a6 (**endorses this move**). (c) Retrieved from `ccl1_top_p0002#935` (ply 43) [structural distance $d=67$, QAP obj=62]: the master played Rc8; under μ our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pf7, our rc1 as their bg2; transferred move b8a6 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#2657` (ply 63) [structural distance $d=76$, QAP obj=82]: the master played Qe6; under μ our Qd3 plays the role of that game’s Qa3; cross-type roles: our Nb8 as their Rd8, our Qd6 as their Rc7; transferred move d6f6. Synthesis: 3 retrieved analogs transferred to Na6 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



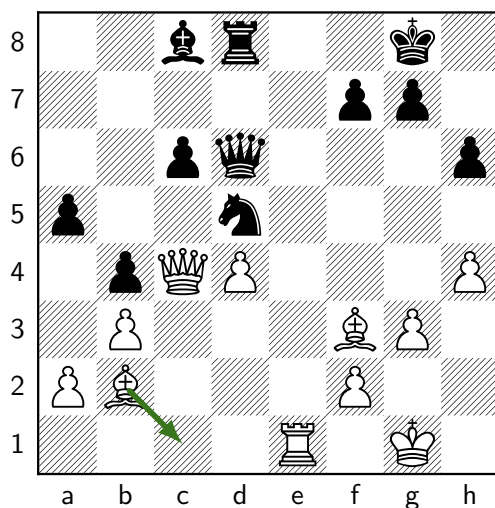
(a)



(b)



(c)



(d)

Figure 22: **Move 28, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1373` (ply 55) [structural distance $d=67$, QAP obj=72]: the master played Rd8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Pa7* as their *Bg7*, our *Pc6* as their *Nc6*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1676` (ply 59) [structural distance $d=67$, QAP obj=61]: the master played h6; under μ our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *Re8*, our *Bc8* as their *Pa5*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1812` (ply 56) [structural distance $d=67$, QAP obj=33]: the master played Bc1; under μ our *Ra1* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Pb7* as their *Re1*, our *qd3* as their *Bf3*; transferred move a8b8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb8 (summed QAP objective 166, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

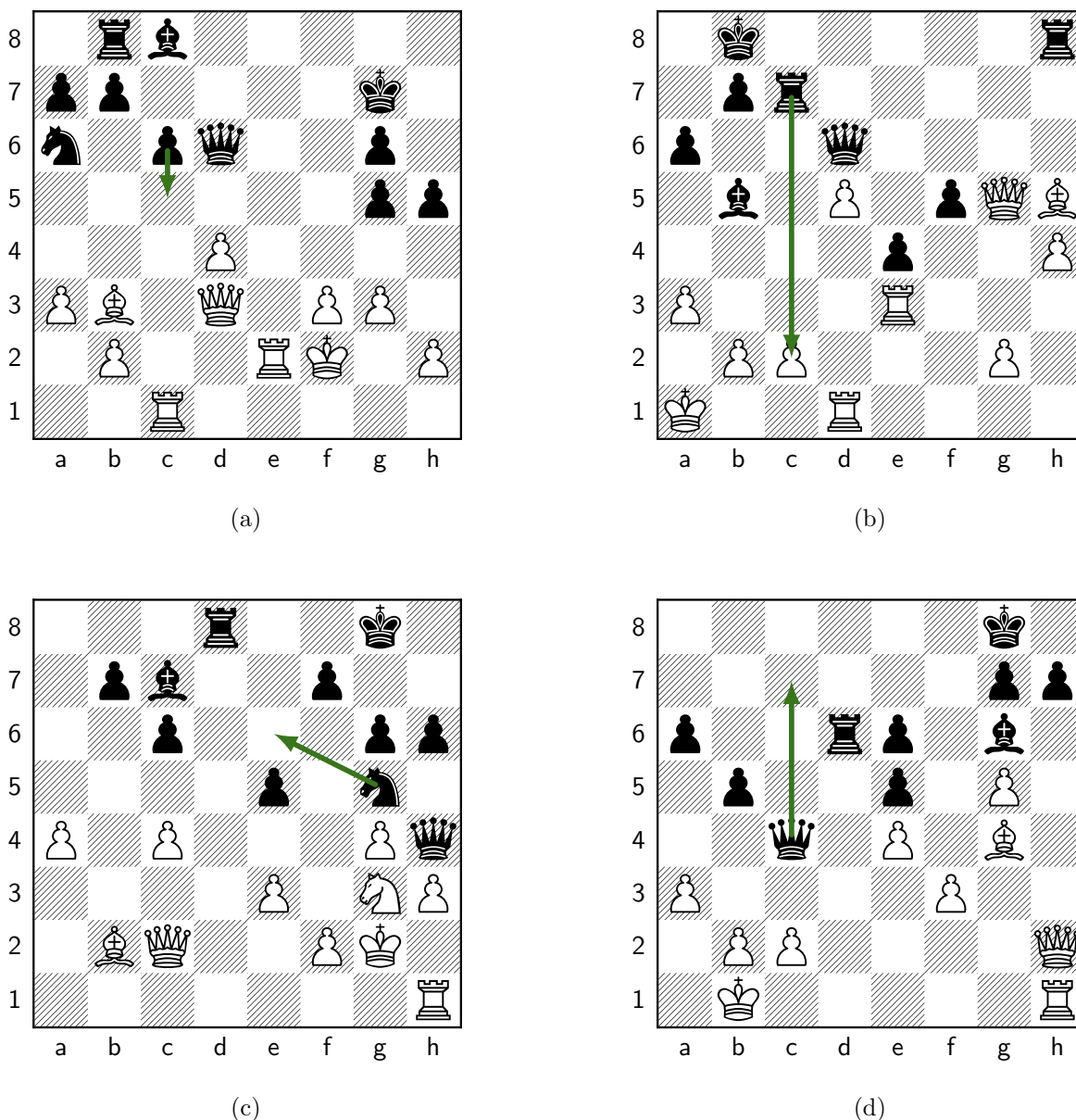
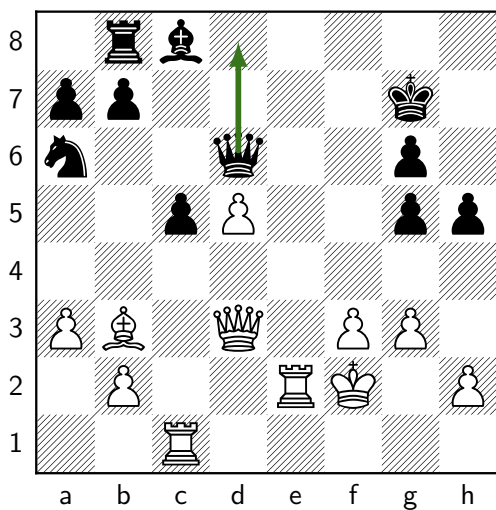
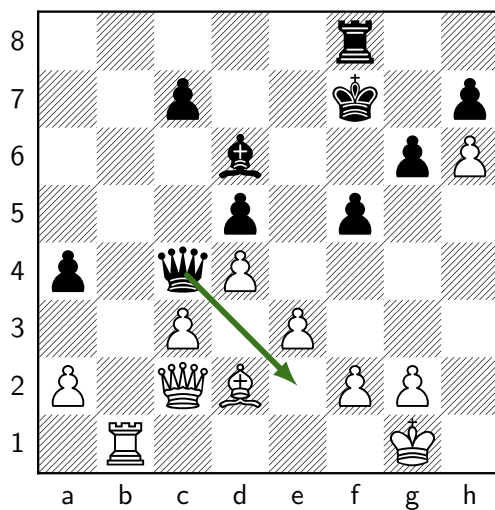


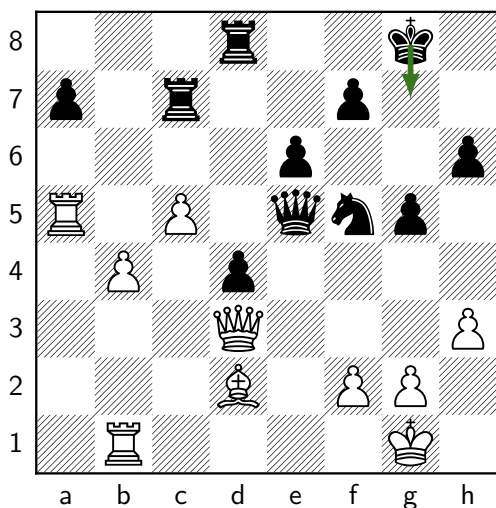
Figure 23: **Move 29, Black: c5.** (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2796` (ply 61) [structural distance $d=71$, QAP obj=70]: the master played `Rxc2`; under μ our `Pc3` plays the role of that game’s `Rc2` (a **cross-type** correspondence); cross-type roles: our `qd3` as their `Rh8`, our `Ph5` as their `Bb5`; transferred move `c6c5` (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#307` (ply 71) [structural distance $d=59$, QAP obj=19]: the master played `Ne6`; under μ our `Pc3` plays the role of that game’s `Ng4` (a **cross-type** correspondence); cross-type roles: our `Bc8` as their `Pb7`, our `Rb8` as their `Bc7`; transferred move `c6c5` (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1314` (ply 57) [structural distance $d=72$, QAP obj=81]: the master played `Qc7`; under μ our `Qd3` plays the role of that game’s `Qc5`; cross-type roles: our `Pa7` as their `Rd6`, our `Na6` as their `Pe6`; transferred move `d6e7`. Synthesis: 2 retrieved analogs transferred to c5 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



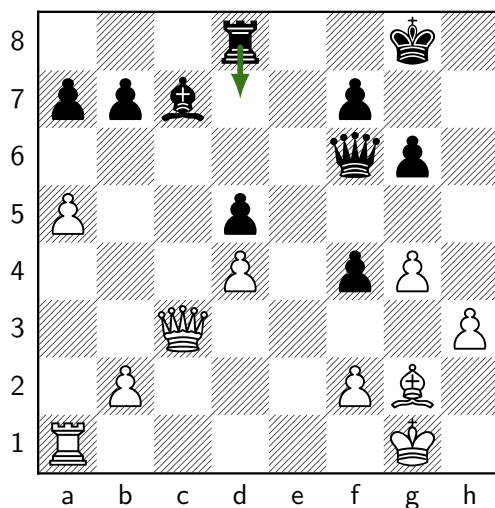
(a)



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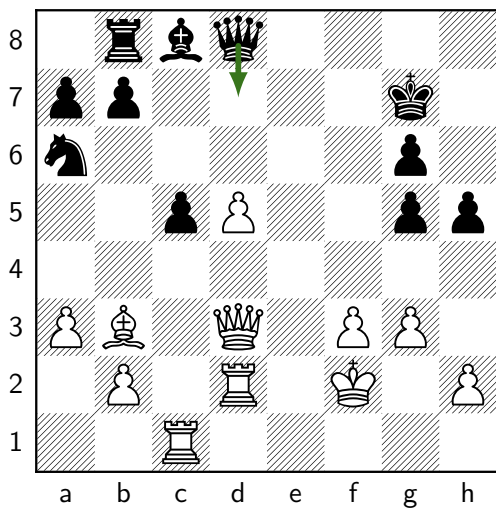


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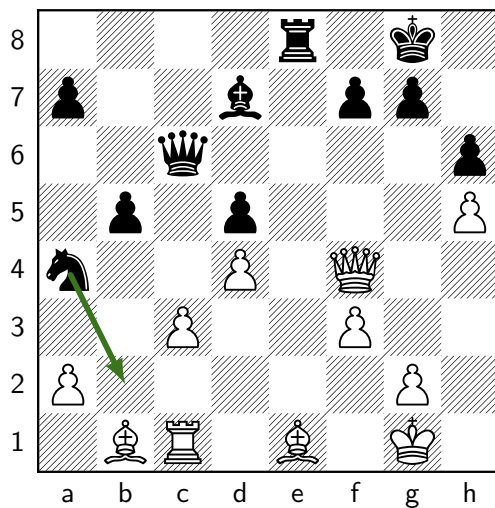


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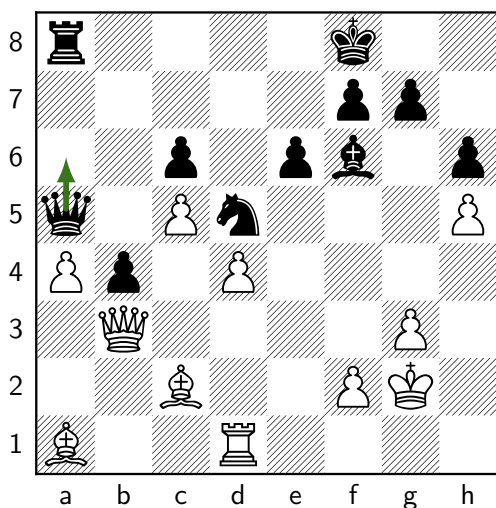
Figure 24: **Move 30, Black: Qd8.** (a) Our position; Melanie plays Qd8 (d6d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3485` (ply 59) [structural distance $d=72$, QAP obj=93]: the master played Qe2; under μ our Qd3 plays the role of that game’s Qc5; cross-type roles: our Na6 as their Pa4; transferred move d6d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1911` (ply 67) [structural distance $d=68$, QAP obj=65]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Bc8 as their Rd8, our Na6 as their Pa7; transferred move g7f8. (d) Retrieved from `ccr1_top_p0000#3916` (ply 57) [structural distance $d=68$, QAP obj=49]: the master played Rd7; under μ our Rb1 plays the role of that game’s Rd1; cross-type roles: our Bc8 as their Pb7, our Na6 as their Bc7; transferred move b8a8. Synthesis: 2 retrieved analogs transferred to Qd8 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



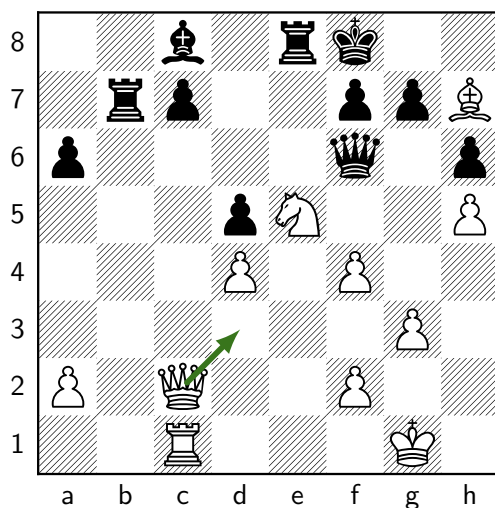
(a)



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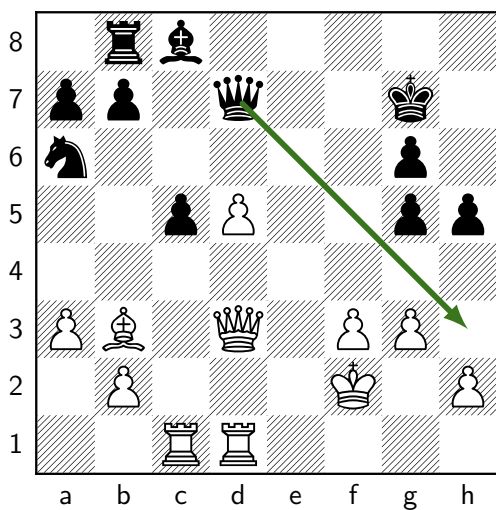


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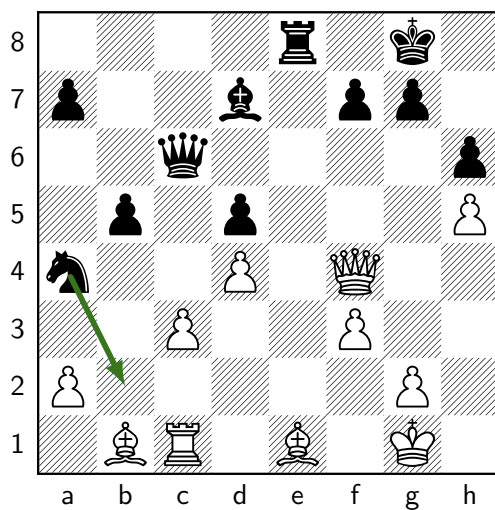


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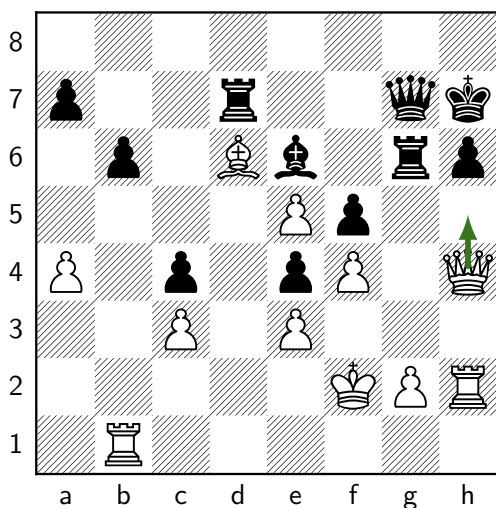
Figure 25: **Move 31, Black: Qd7.** (a) Our position; Melanie plays Qd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2734` (ply 61) [structural distance $d=66$, QAP obj=54]: the master played Nb2; under μ our Qd1 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa7, our Pg5 as their Bd7; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1225` (ply 77) [structural distance $d=67$, QAP obj=48]: the master played Qa6; under μ our Qd1 plays the role of that game’s Qa4; cross-type roles: our Bc8 as their Ra8, our rc1 as their Bf6; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3474` (ply 58) [structural distance $d=63$, QAP obj=27]: the master played Qd3; under μ our Qd1 plays the role of that game’s Qc2; cross-type roles: our bb3 as their rb7, our rd2 as their pf7; transferred move d8d7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd7 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



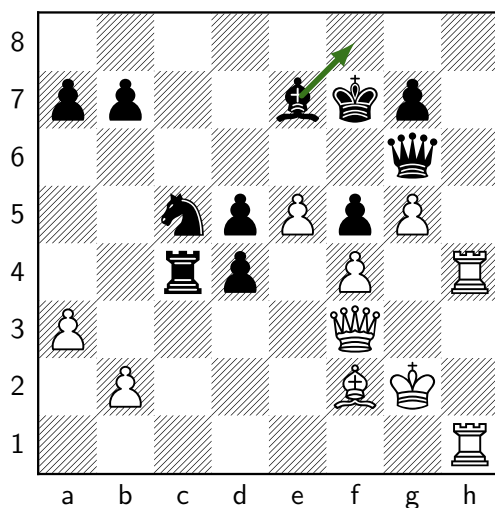
(a)



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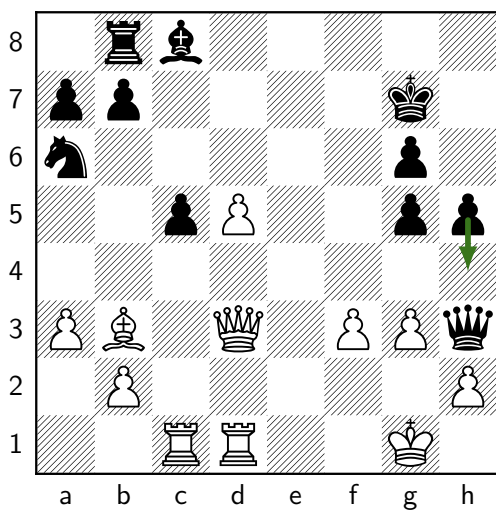


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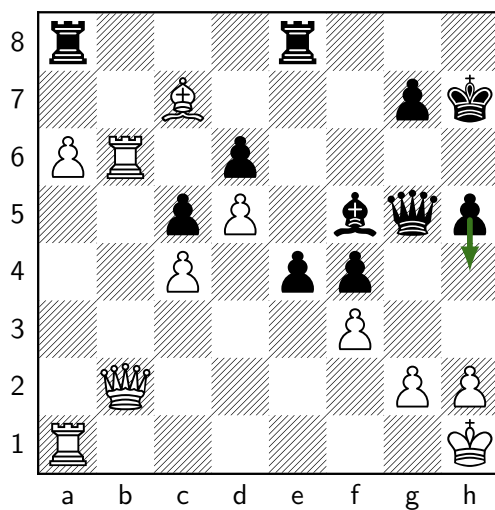


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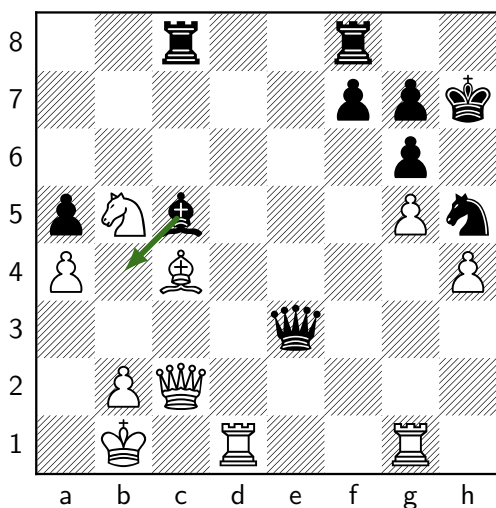
Figure 26: **Move 32, Black: Qh3.** (a) Our position; Melanie plays Qh3 (d7h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2734` (ply 61) [structural distance $d=65$, QAP obj=52]: the master played Nb2; under μ our Qd2 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our Qd7 as their Re8, our Rb8 as their Pa7; transferred move d7h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2211` (ply 74) [structural distance $d=78$, QAP obj=36]: the master played Qh5; under μ our Qd2 plays the role of that game’s Qh4; cross-type roles: our Bc8 as their Rh2, our rc1 as their Pe3; transferred move d7h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3808` (ply 71) [structural distance $d=76$, QAP obj=63]: the master played Bf8; under μ our Qd2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pb7, our Bc8 as their Pg7; transferred move d7f7. Synthesis: 2 retrieved analogs transferred to Qh3 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



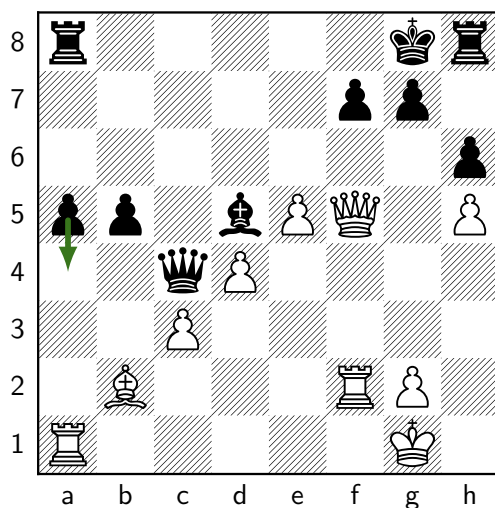
(a)



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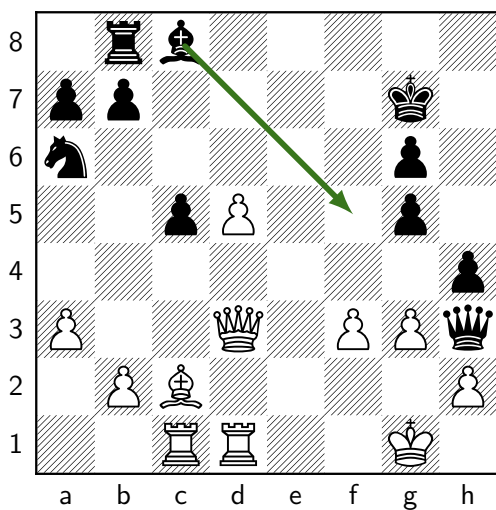


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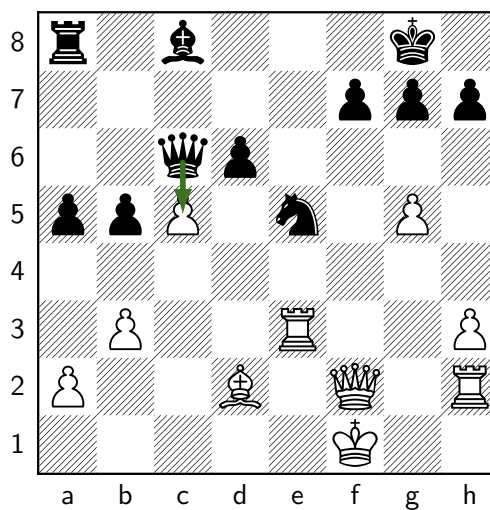


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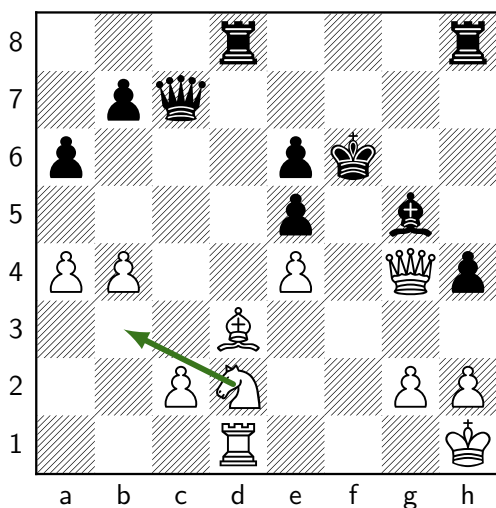
Figure 27: **Move 33, Black: h4**. (a) Our position; Melanie plays h4 (h5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1580` (ply 73) [structural distance $d=85$, QAP obj=66]: the master played h4; under μ our Ph_4 plays the role of that game’s Ph_4 ; cross-type roles: our $qd3$ as their $bc7$, our $qd3$ as their $rb6$; transferred move h5h4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2617` (ply 59) [structural distance $d=84$, QAP obj=33]: the master played Bb4; under μ our Ph_4 plays the role of that game’s Bc_4 (a **cross-type** correspondence); cross-type roles: our $Pb7$ as their $Rc8$, our $Na6$ as their $Pg7$; transferred move h5h4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#990` (ply 67) [structural distance $d=80$, QAP obj=84]: the master played a4; under μ our Pc_4 plays the role of that game’s Pa_4 ; cross-type roles: our $Pb7$ as their $Ra8$, our $Bc8$ as their $Rh8$; transferred move c5c4. Synthesis: 2 retrieved analogs transferred to h4 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



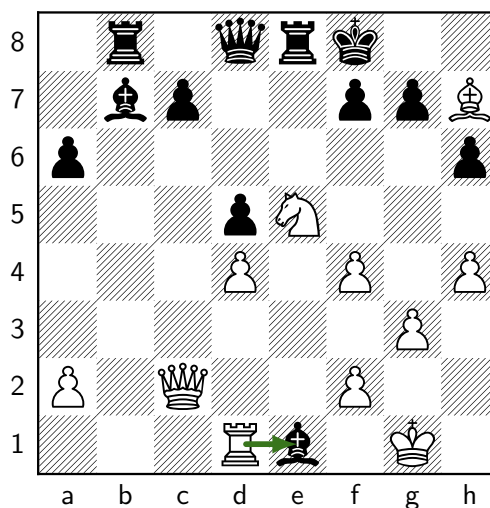
(a)



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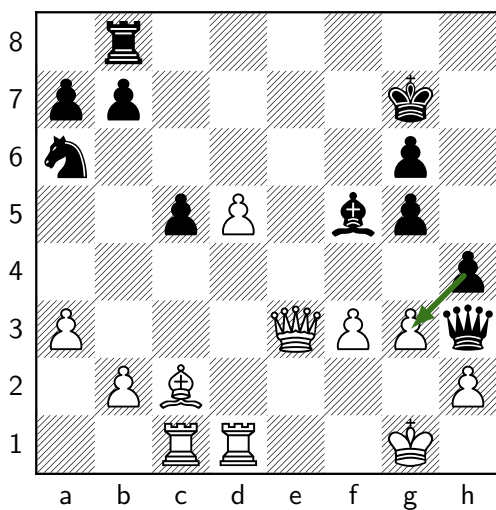


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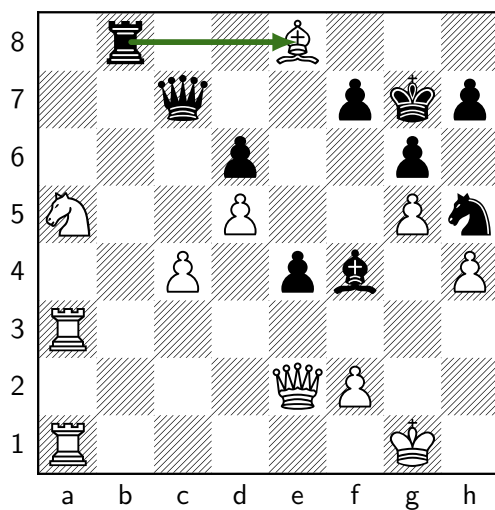


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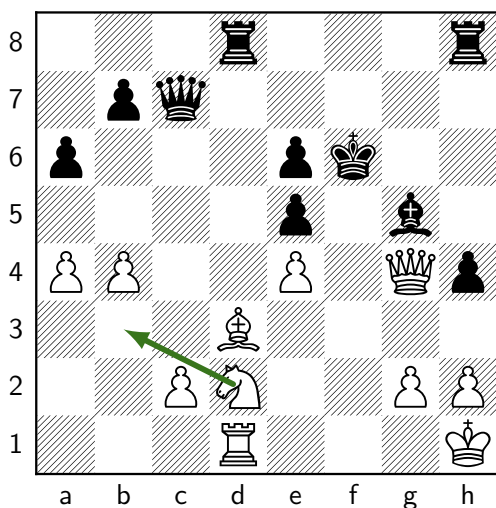
Figure 28: **Move 34, Black: Bf5**. (a) Our position; Melanie plays Bf5 (c8f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3846` (ply 53) [structural distance $d=79$, QAP obj=81]: the master played Qxc5; under μ our Qh6 plays the role of that game’s Qc3; cross-type roles: our Bc8 as their Ra8, our pf3 as their Bc8; transferred move h3e6. (c) Retrieved from `ccr1_top_p0001#1320` (ply 56) [structural distance $d=96$, QAP obj=46]: the master played Nb3; under μ our Ph5 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pg5 as their Rd1, our Rb8 as their Ph2; transferred move h4g3. (d) Retrieved from `ccr1_top_p0001#3474` (ply 52) [structural distance $d=91$, QAP obj=32]: the master played Rxe1; under μ our Qh6 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pg3 as their be1, our Rb8 as their Pa2; transferred move h3d7. Synthesis: no retrieved analog transferred to Bf5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



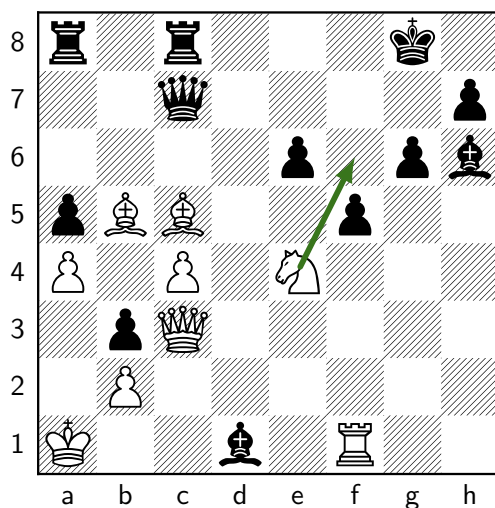
(a)



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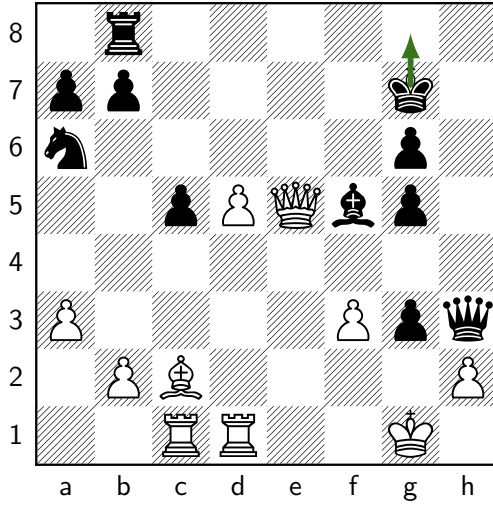


(c)

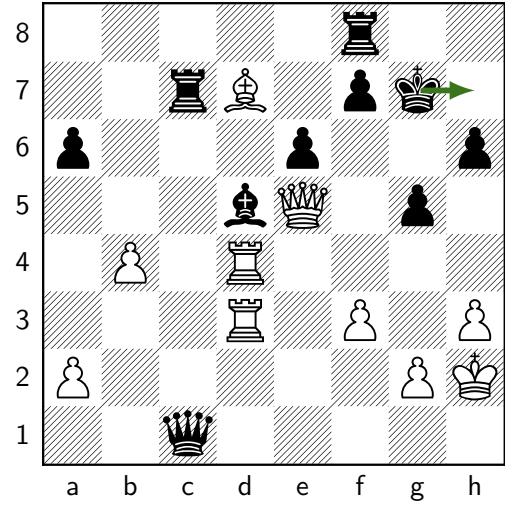


(d)

Figure 29: **Move 35, Black: hxg3**. (a) Our position; Melanie plays hxg3 (h4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1684` (ply 67) [structural distance $d=82$, QAP obj=66]: the master played Rxe8; under μ our Ph5 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our pg3 as their be8, our Na6 as their Pg6; transferred move h4g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1320` (ply 56) [structural distance $d=78$, QAP obj=34]: the master played Nb3; under μ our Ph5 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Rd1, our Bf5 as their Pg2; transferred move h4g3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1384` (ply 66) [structural distance $d=90$, QAP obj=30]: the master played Nf6+; under μ our Ph5 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our ph2 as their bd1, our Qh3 as their Rf1; transferred move h4g3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to hxg3 (summed QAP objective 140, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

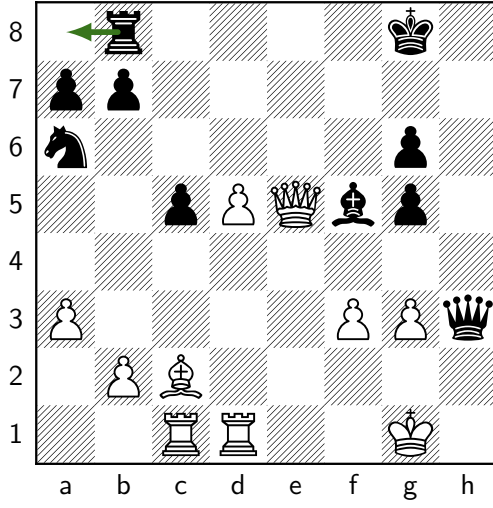


(a)

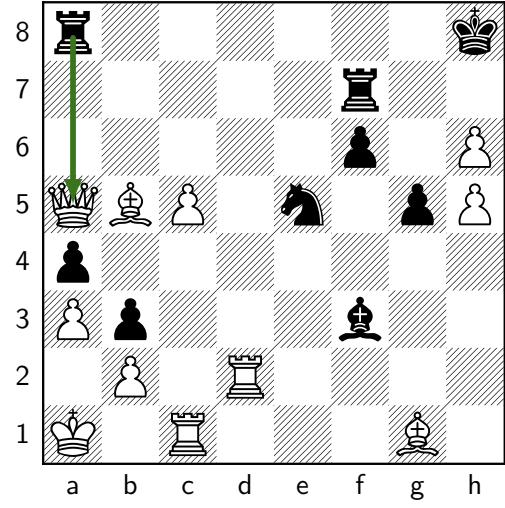


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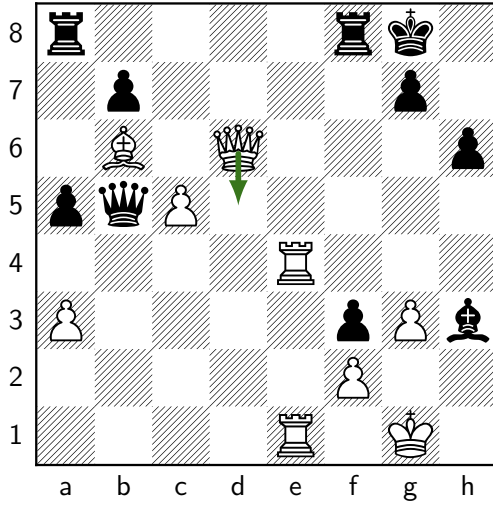
Figure 30: **Move 36, Black: Kg8.** (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2885` (ply 57) [structural distance $d=78$, QAP obj=80]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Qh3 as their Rf8, our pa3 as their Rc7; transferred move g7g8 (**endorses this move**). Synthesis: 1 retrieved analog transferred to Kg8 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



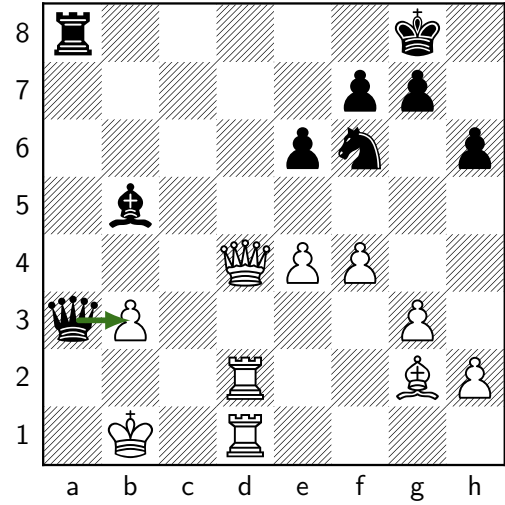
(a)



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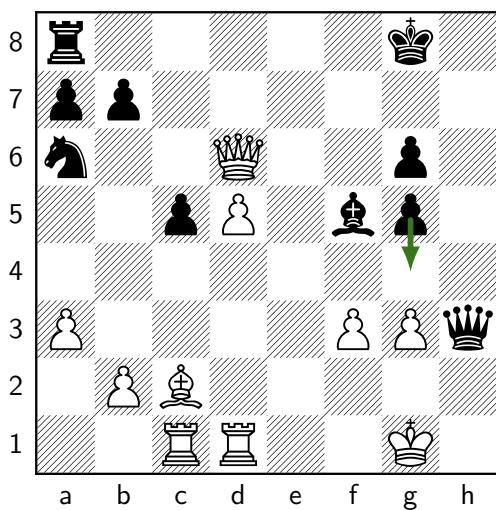


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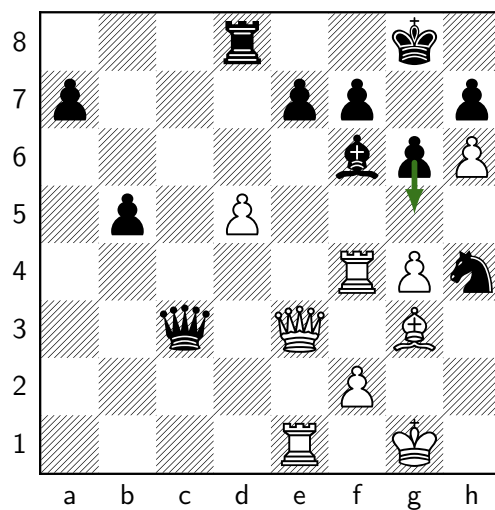


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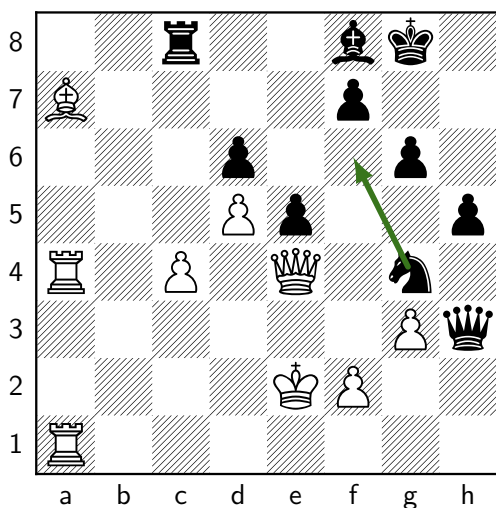
Figure 31: **Move 37, Black: Ra8.** (a) Our position; Melanie plays Ra8 (b8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#889` (ply 85) [structural distance $d=80$, QAP obj=30]: the master played Rxa5; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Pb7 as their Rf7, our Na6 as their ph6; transferred move b8a8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#290` (ply 64) [structural distance $d=86$, QAP obj=45]: the master played Qd5+; under μ our Qh6 plays the role of that game’s Qd6; cross-type roles: our Na6 as their Re1, our rd1 as their Pg3; transferred move h3g3. (d) Retrieved from `ccrl_top_p0001#3033` (ply 73) [structural distance $d=64$, QAP obj=32]: the master played Qxb3+; under μ our Qh6 plays the role of that game’s Qa6; cross-type roles: our Na6 as their Pe6, our Bf5 as their Nf6; transferred move h3g3. Synthesis: 1 retrieved analog transferred to Ra8 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



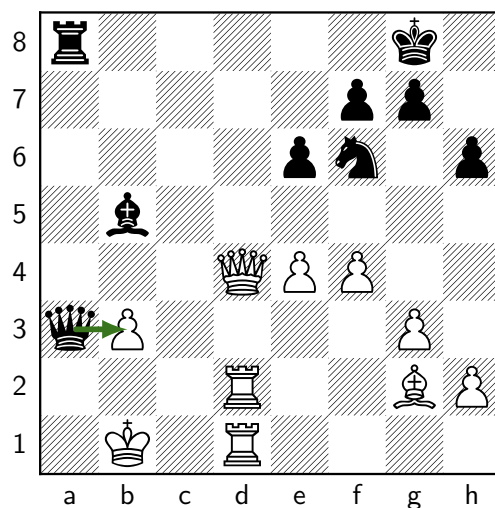
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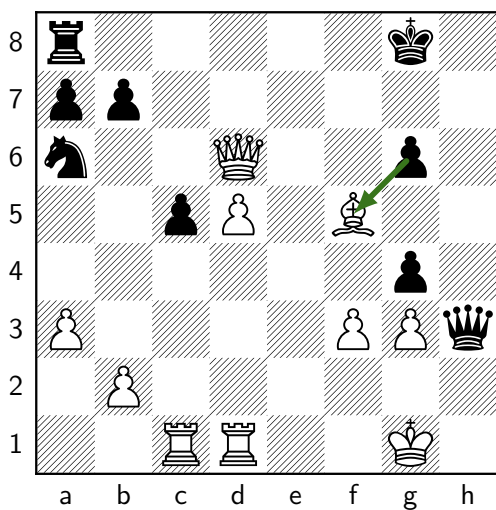


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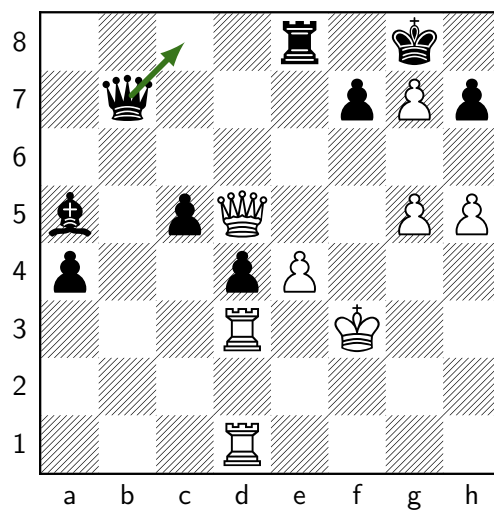


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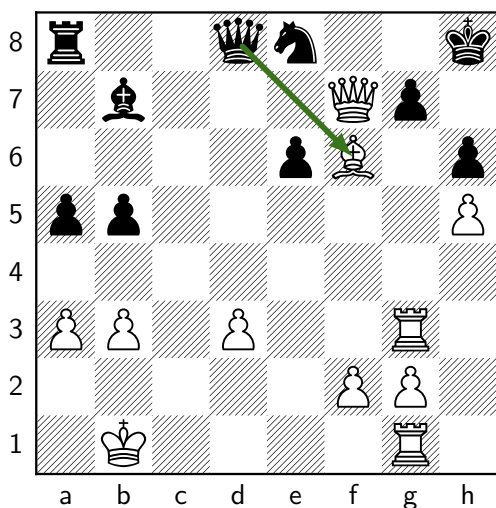
Figure 32: **Move 38, Black: g4**. (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3043` (ply 63) [structural distance $d=77$, QAP obj=17]: the master played g5; under μ our Pg_4 plays the role of that game's Pg_3 ; cross-type roles: our Na_6 as their Pe_7 , our Pc_5 as their Bf_6 ; transferred move g5g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1864` (ply 67) [structural distance $d=85$, QAP obj=53]: the master played Nf6; under μ our Na_3 plays the role of that game's Ng_5 ; cross-type roles: our Bf_5 as their Rc_8 , our Pg_6 as their Bf_8 ; transferred move a6c7. (d) Retrieved from `ccrl_top_p0001#3033` (ply 73) [structural distance $d=70$, QAP obj=33]: the master played Qxb3+; under μ our Qh_6 plays the role of that game's Qa_6 ; cross-type roles: our rc_1 as their Pf_7 , our bc_2 as their Nf_6 ; transferred move h3g3. Synthesis: 1 retrieved analog transferred to g4 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



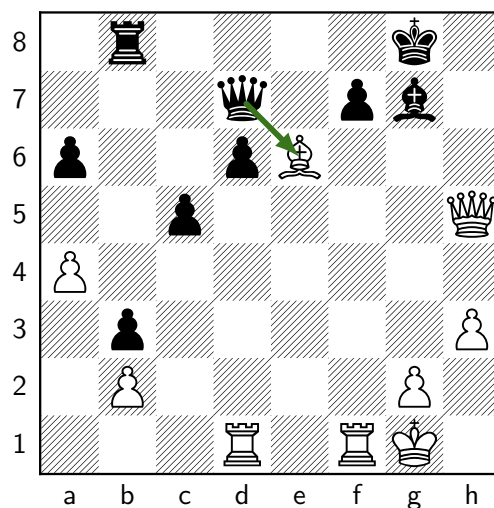
(a)



(b)



(c)



(d)

Figure 33: **Move 39, Black: gxf5.** (a) Our position; Melanie plays **gxf5** (**g6f5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3527` (ply 91) [structural distance $d=86$, QAP obj=87]: the master played **Qc8**; under μ our **Pg3** plays the role of that game’s **Qb2** (a **cross-type** correspondence); cross-type roles: our **Qh3** as their **Re8**, our **Na6** as their **Ph7**; transferred move **g6f5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#885` (ply 61) [structural distance $d=89$, QAP obj=67]: the master played **Qxf6**; under μ our **Pg3** plays the role of that game’s **Qd1** (a **cross-type** correspondence); cross-type roles: our **Pg6** as their **Ne8**, our **Qh3** as their **Bb7**; transferred move **g6f5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3642` (ply 61) [structural distance $d=78$, QAP obj=45]: the master played **Qxe6**; under μ our **Pg3** plays the role of that game’s **Qd2** (a **cross-type** correspondence); cross-type roles: our **Qh3** as their **Rb8**, our **Na6** as their **Bg7**; transferred move **g6f5** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **gxf5** (summed QAP objective 199, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

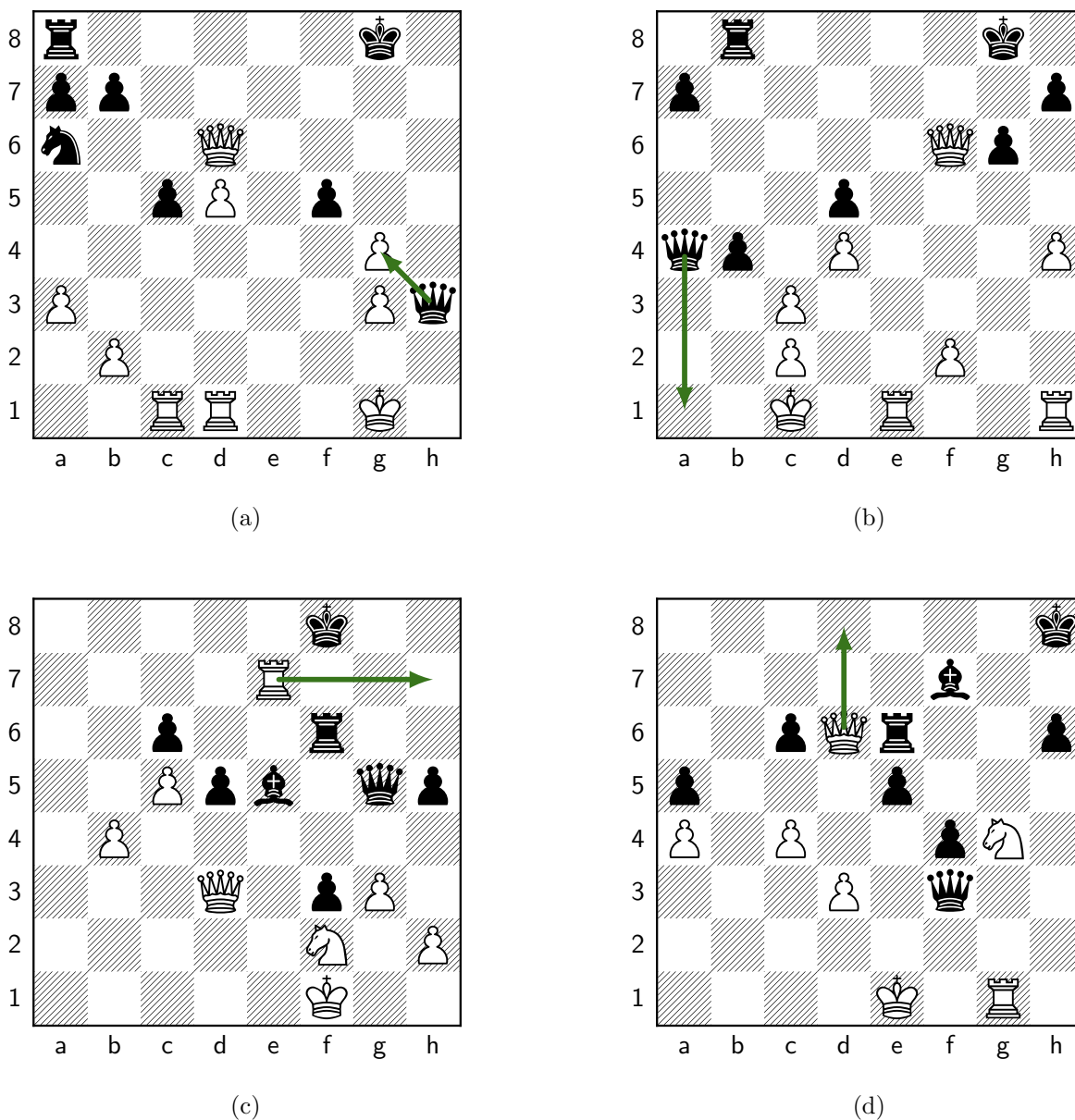
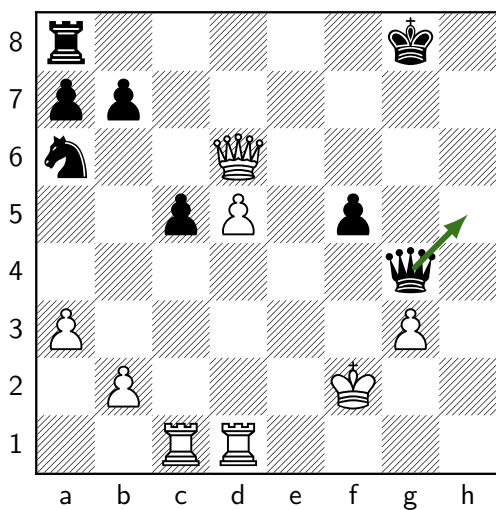
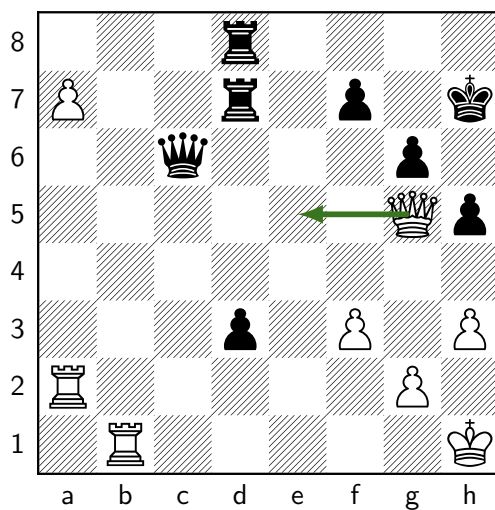


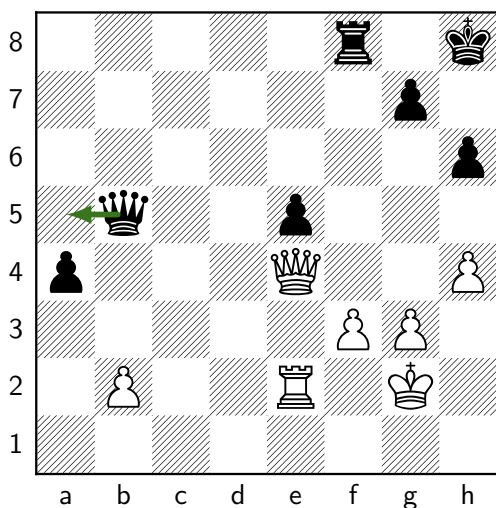
Figure 34: **Move 40, Black: Qxg4.** (a) Our position; Melanie plays Qxg4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#80 (ply 63) [structural distance $d=64$, QAP obj=70]: the master played Qa1+; under μ our Qh6 plays the role of that game’s Qa5; cross-type roles: our Na6 as their Pg6; transferred move h3g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1013 (ply 82) [structural distance $d=65$, QAP obj=8]: the master played Rh7; under μ our Qh6 plays the role of that game’s Re7 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Nf2, our Na6 as their Ph2; transferred move h3g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2518 (ply 84) [structural distance $d=51$, QAP obj=8]: the master played Qd8+; under μ our Qh6 plays the role of that game’s Qd6; cross-type roles: our Pf5 as their Ng4, our Na6 as their pc6; transferred move h3g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxg4 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



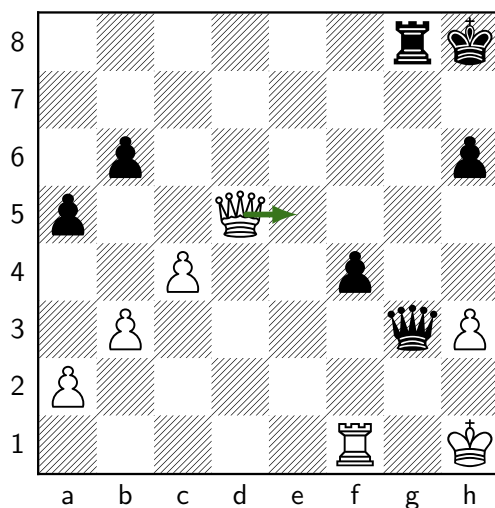
(a)



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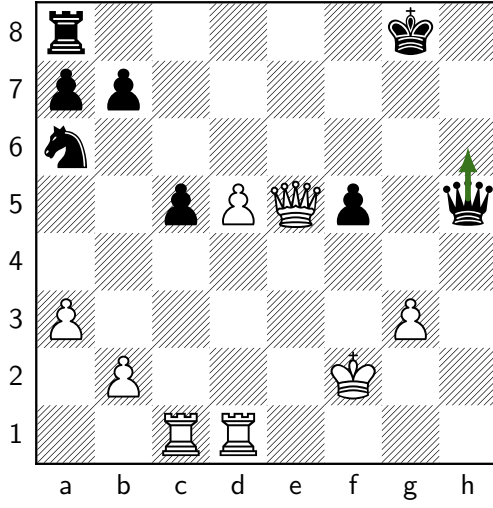


(c)

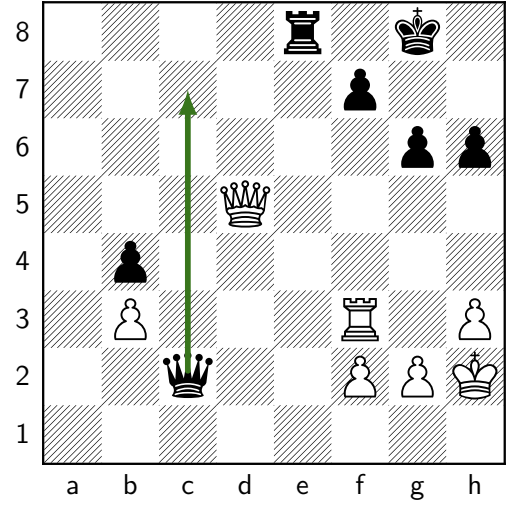


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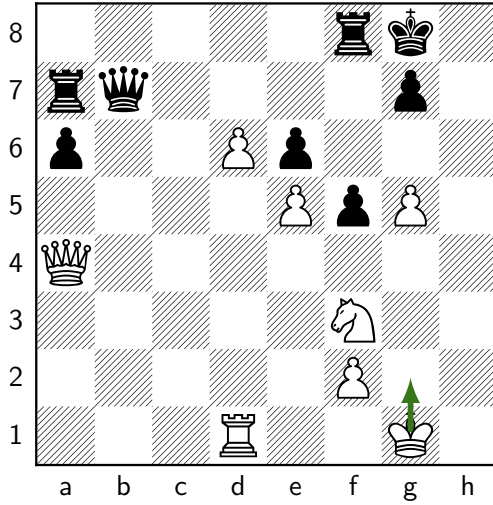
Figure 35: **Move 41, Black: Qh5.** (a) Our position; Melanie plays Qh5 (g4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1500` (ply 90) [structural distance $d=57$, QAP obj=49]: the master played Qe5; under μ our Qg5 plays the role of that game’s Qg5; cross-type roles: our Pa7 as their Ra2, our Na6 as their Pf3; transferred move g4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3003` (ply 83) [structural distance $d=58$, QAP obj=37]: the master played Qa5; under μ our Qg5 plays the role of that game’s Qb4; cross-type roles: our Na6 as their Ph6, our rc1 as their pg3; transferred move g4h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3481` (ply 94) [structural distance $d=56$, QAP obj=37]: the master played Qe5+; under μ our Qg5 plays the role of that game’s Qd5; cross-type roles: our Pb7 as their Rf1, our Na6 as their Pb3; transferred move g4d4. Synthesis: 2 retrieved analogs transferred to Qh5 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



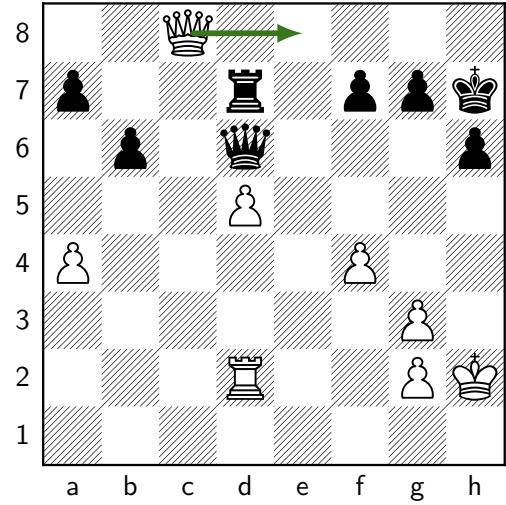
(a)



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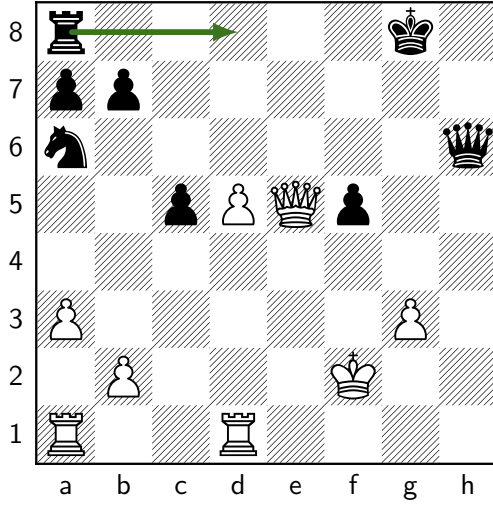


(c)

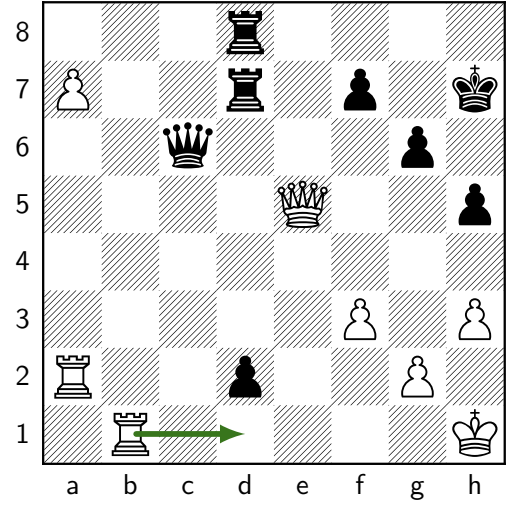


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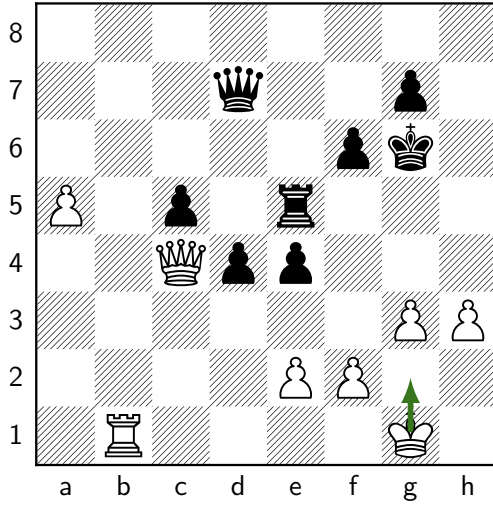
Figure 36: **Move 42, Black: Qh6.** (a) Our position; Melanie plays Qh6 (h5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#84` (ply 83) [structural distance $d=43$, QAP obj=35]: the master played Qc7+; under μ our Qh4 plays the role of that game’s Qc7; cross-type roles: our Na6 as their Ph6, our rd1 as their pb3; transferred move h5h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3088` (ply 78) [structural distance $d=50$, QAP obj=35]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our pg3 as their Rd1, our rc1 as their Pf2; transferred move g8f7. (d) Retrieved from `ccr1_top_p0002#642` (ply 84) [structural distance $d=48$, QAP obj=30]: the master played Qe8; under μ our Qh4 plays the role of that game’s Qc8; cross-type roles: our Pb7 as their Rd2, our Na6 as their Pg3; transferred move h5h2. Synthesis: 1 retrieved analog transferred to Qh6 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



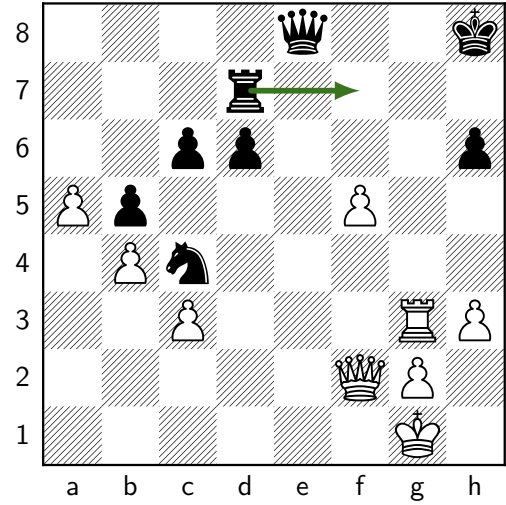
(a)



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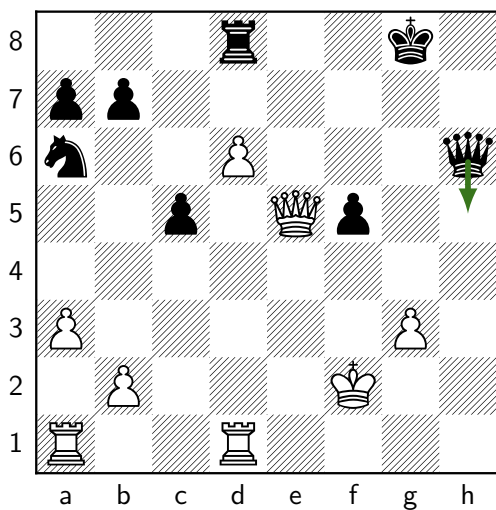


(c)

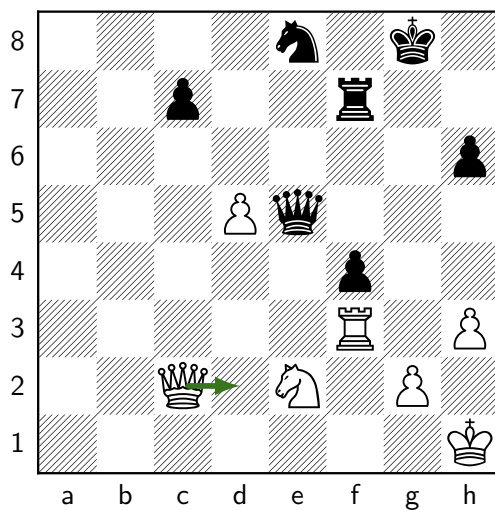


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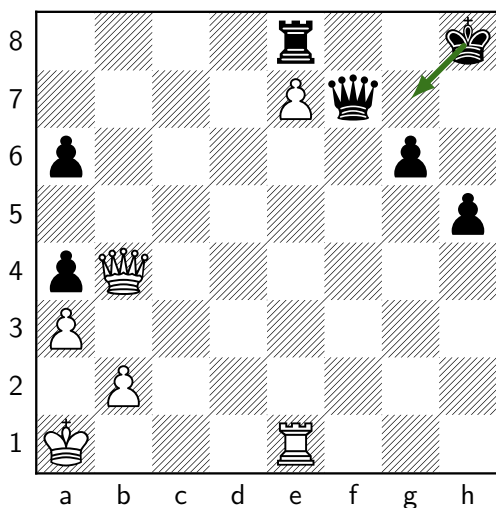
Figure 37: **Move 43, Black: Rd8.** (a) Our position; Melanie plays Rd8 (a8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1500` (ply 92) [structural distance $d=59$, QAP obj=11]: the master played Rd1; under μ our *Ra1* plays the role of that game’s *Rb1*; cross-type roles: our *Na6* as their *Ra2*, our *ra1* as their *pf7*; transferred move **a8d8 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#3200` (ply 74) [structural distance $d=66$, QAP obj=37]: the master played Kg2; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *rd1* as their *Pg3*, our *Na6* as their *Ph3*; transferred move **g8f7**. (d) Retrieved from `ccrl_top_p0001#3101` (ply 101) [structural distance $d=50$, QAP obj=12]: the master played Rf7; under μ our *Ra1* plays the role of that game’s *Rd2*; cross-type roles: our *Na6* as their *Pb5*, our *qe5* as their *Nc4*; transferred move **a8c8**. Synthesis: 1 retrieved analog transferred to Rd8 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



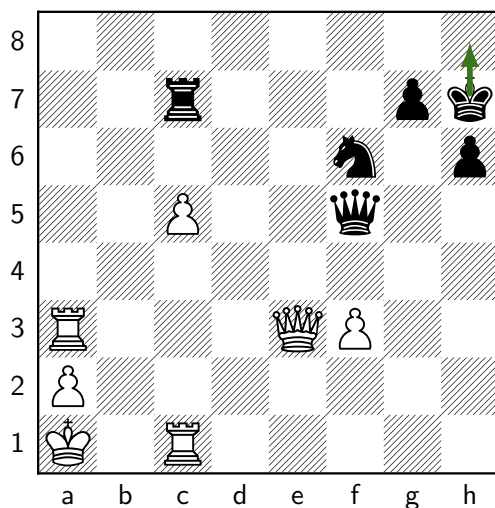
(a)



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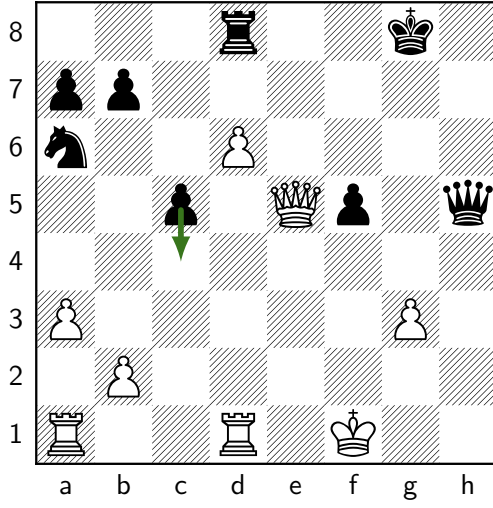


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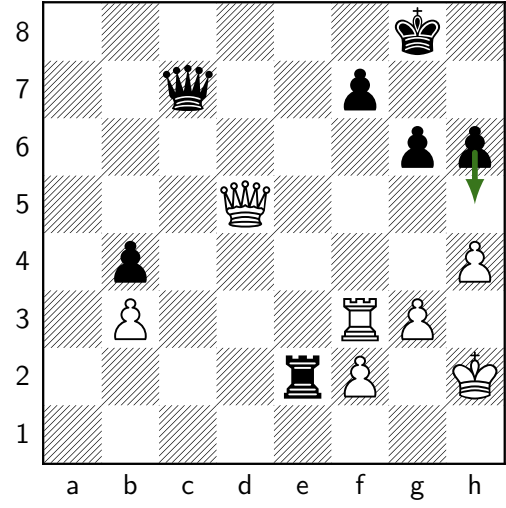


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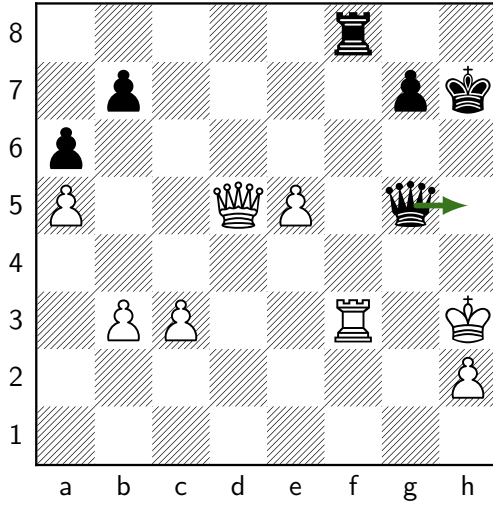
Figure 38: **Move 44, Black: Qh5**. (a) Our position; Melanie plays **Qh5** (**h6h5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1032` (ply 72) [structural distance $d=54$, QAP obj=59]: the master played **Qd2**; under μ our **Qh3** plays the role of that game’s **Qc2**; cross-type roles: our **Pf5** as their **Ne2**, our **Na6** as their **Pg2**; transferred move **h6h5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1682` (ply 95) [structural distance $d=63$, QAP obj=61]: the master played **Kg7**; under μ our **Kg1** plays the role of that game’s **Kh1**; cross-type roles: our **Na6** as their **Pa6**; transferred move **g8h7**. (d) Retrieved from `ccr1_top_p0003#1424` (ply 95) [structural distance $d=65$, QAP obj=56]: the master played **Kh8**; under μ our **Kg1** plays the role of that game’s **Kh2**; cross-type roles: our **Pc5** as their **ra3**; transferred move **g8f7**. Synthesis: 1 retrieved analog transferred to **Qh5** (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



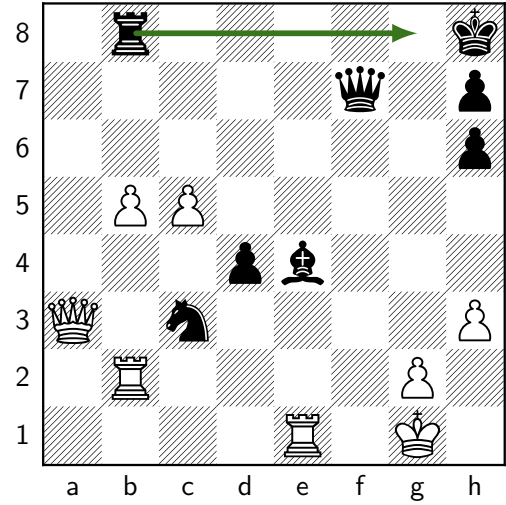
(a)



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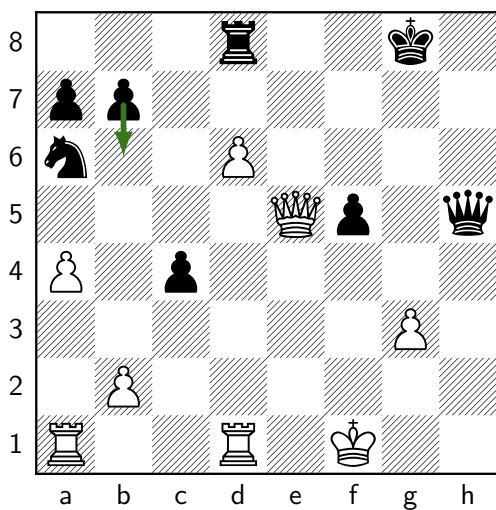


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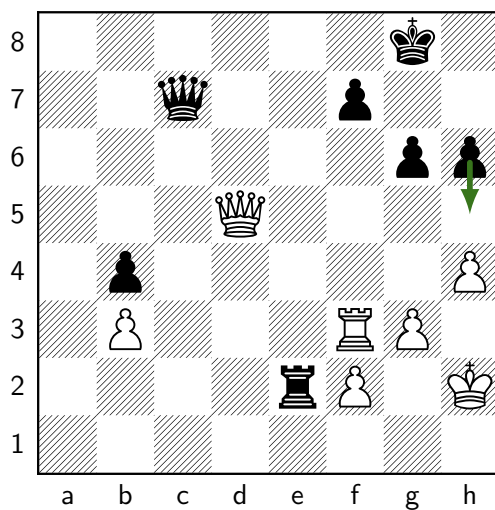


(d)

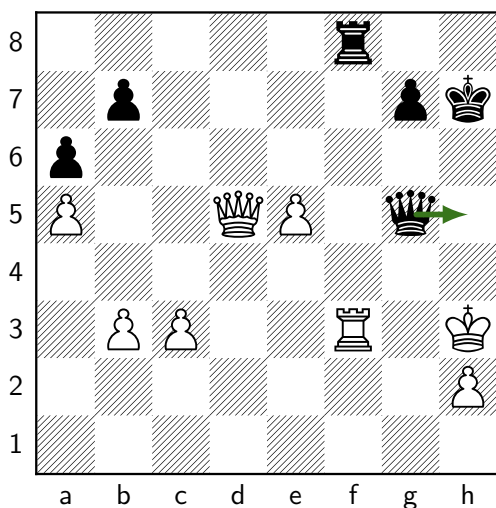
Figure 39: **Move 45, Black: c4**. (a) Our position; Melanie plays c4 (c5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#84` (ply 87) [structural distance $d=58$, QAP obj=54]: the master played h5; under μ our *Pc4* plays the role of that game’s *Ph3*; cross-type roles: our *rd1* as their *Qc7*, our *Na6* as their *Pg6*; transferred move c5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#88` (ply 75) [structural distance $d=63$, QAP obj=47]: the master played Qh5+; under μ our *Rd1* plays the role of that game’s *Qg4* (a **cross-type** correspondence); cross-type roles: our *Qh5* as their *Rf8*, our *Na6* as their *Pg7*; transferred move d8d7. (d) Retrieved from `ccr1_top_p0003#1023` (ply 95) [structural distance $d=63$, QAP obj=40]: the master played Rg8; under μ our *Rd1* plays the role of that game’s *Rb1*; cross-type roles: our *pa3* as their *Qf7*, our *Na6* as their *Pd4*; transferred move d8f8. Synthesis: 1 retrieved analog transferred to c4 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



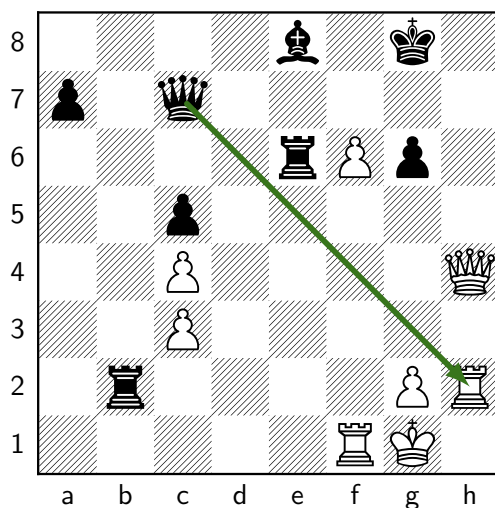
(a)



(b)

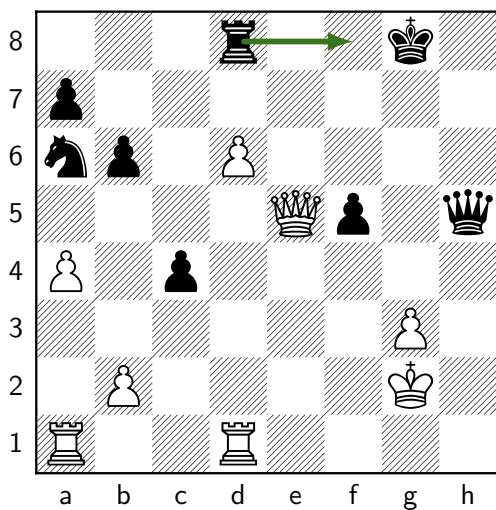


(c)

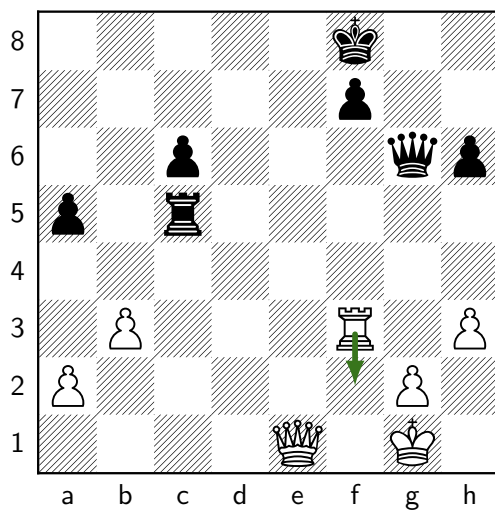


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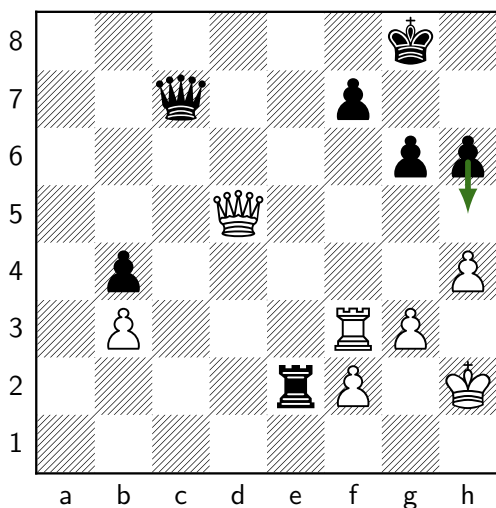
Figure 40: **Move 46, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#84` (ply 87) [structural distance $d=67$, QAP obj=54]: the master played h5; under μ our *Pb2* plays the role of that game’s *Ph3*; cross-type roles: our *rd1* as their *Qc7*, our *Na6* as their *Pg6*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#88` (ply 75) [structural distance $d=62$, QAP obj=47]: the master played Qh5+; under μ our *Rd1* plays the role of that game’s *Qg4* (a **cross-type** correspondence); cross-type roles: our *Qh5* as their *Rf8*, our *Na6* as their *Pg7*; transferred move d8d7. (d) Retrieved from `ccr1_top_p0003#2799` (ply 91) [structural distance $d=69$, QAP obj=31]: the master played Qxh2+; under μ our *Qh4* plays the role of that game’s *Qc2*; cross-type roles: our *Na6* as their *Be8*, our *qe5* as their *Re6*; transferred move h5e2. Synthesis: 1 retrieved analog transferred to b6 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



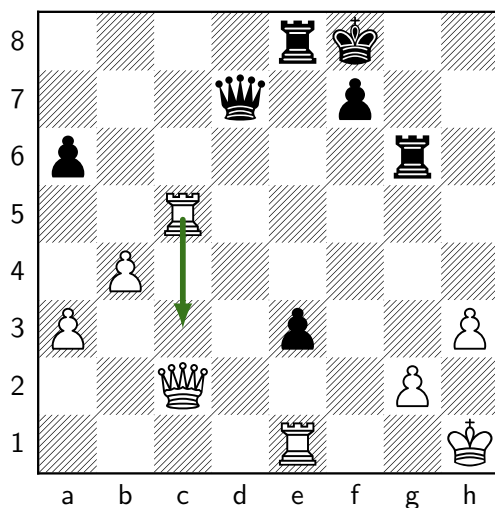
(a)



(b)

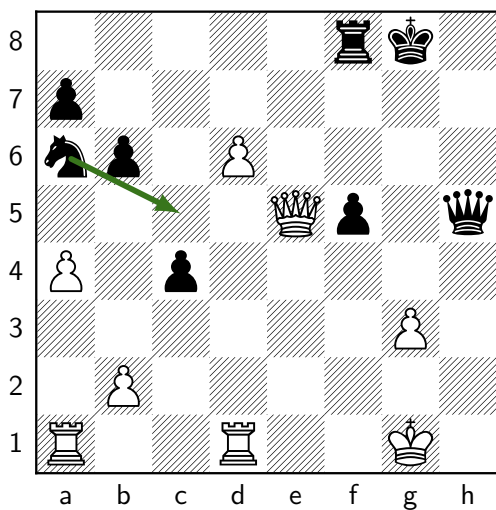


(c)

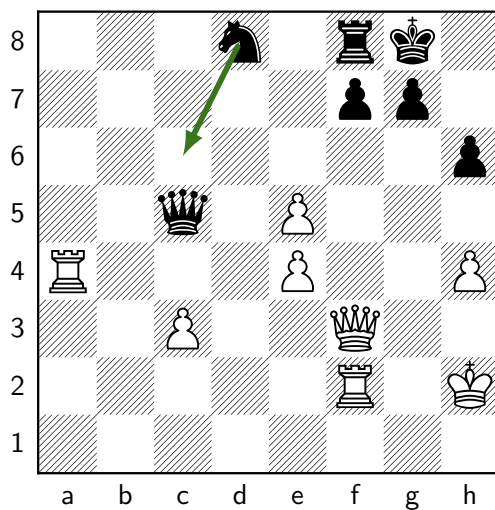


(d)

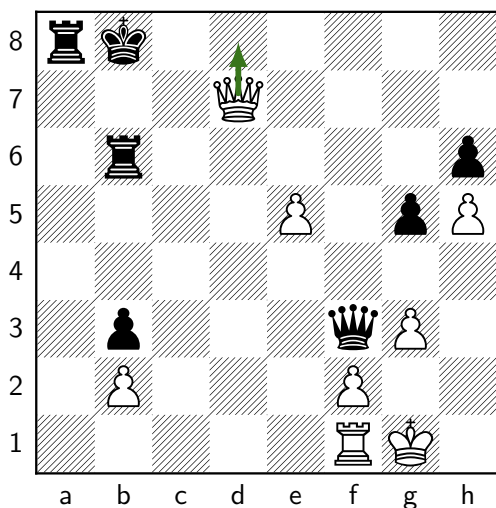
Figure 41: **Move 47, Black: Rf8.** (a) Our position; Melanie plays Rf8 (d8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1278` (ply 100) [structural distance $d=54$, QAP obj=16]: the master played Rf2; under μ our Rd1 plays the role of that game’s Rf3; transferred move d8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#84` (ply 87) [structural distance $d=43$, QAP obj=54]: the master played h5; under μ our Pc5 plays the role of that game’s Ph3; cross-type roles: our rd1 as their Qc7, our ra1 as their Pg6; transferred move c4c3. (d) Retrieved from `ccr1_top_p0003#1707` (ply 80) [structural distance $d=60$, QAP obj=42]: the master played Rc3; under μ our Pf4 plays the role of that game’s Rc5 (a **cross-type** correspondence); cross-type roles: our ra1 as their pa6, our Pa7 as their re8; transferred move f5f4. Synthesis: 1 retrieved analog transferred to Rf8 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



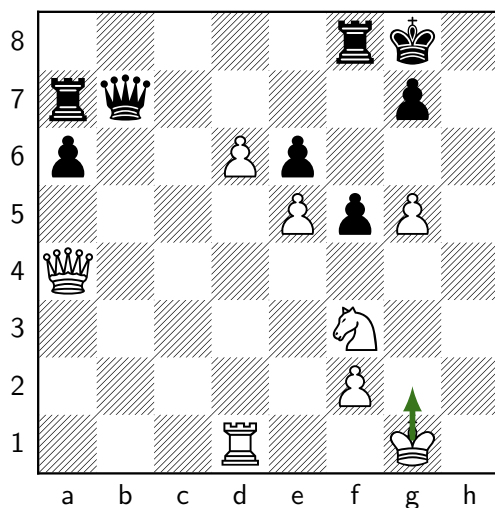
(a)



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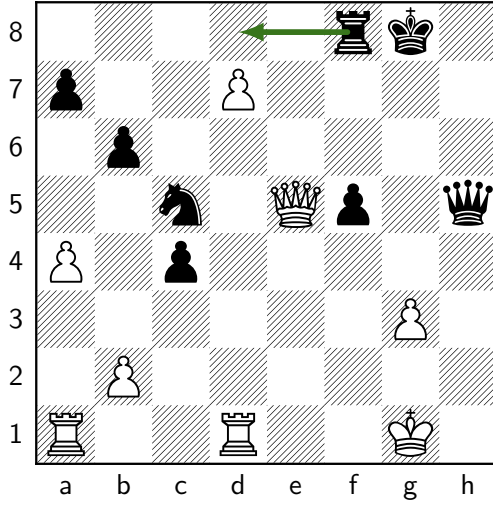


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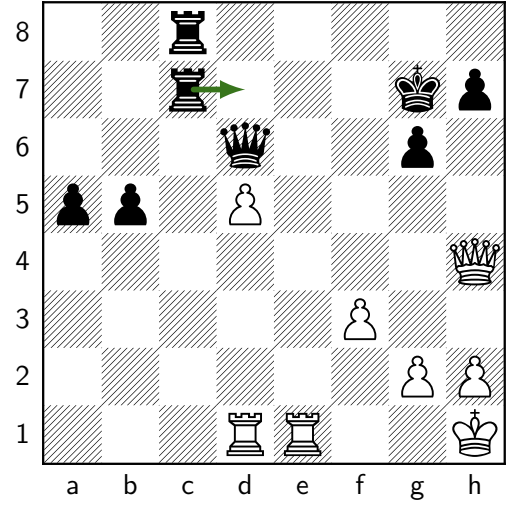


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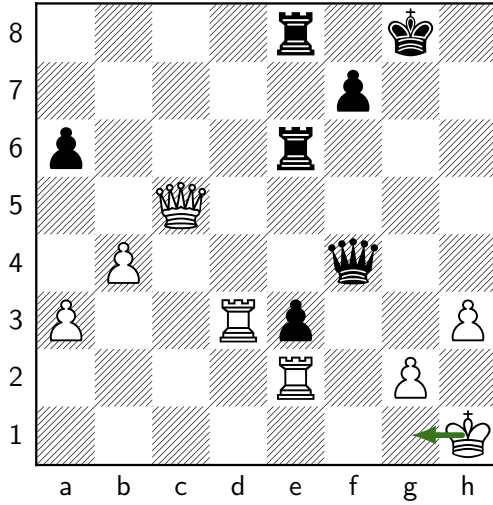
Figure 42: **Move 48, Black: Nc5.** (a) Our position; Melanie plays Nc5 (a6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1827` (ply 85) [structural distance $d=59$, QAP obj=29]: the master played Nc6; under μ our Na3 plays the role of that game’s Nd1; cross-type roles: our rd1 as their pe5, our Pa7 as their ra4; transferred move a6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1079` (ply 104) [structural distance $d=54$, QAP obj=57]: the master played Qd8+; under μ our Qh4 plays the role of that game’s Qd7; cross-type roles: our pd6 as their ra8; transferred move h5e2. (d) Retrieved from `ccr1_top_p0002#3088` (ply 78) [structural distance $d=61$, QAP obj=38]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our pd6 as their Rd1, our rd1 as their Nf3; transferred move g8h7. Synthesis: 1 retrieved analog transferred to Nc5 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



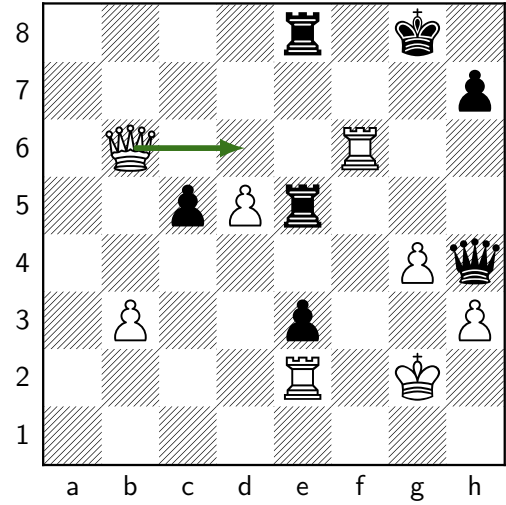
(a)



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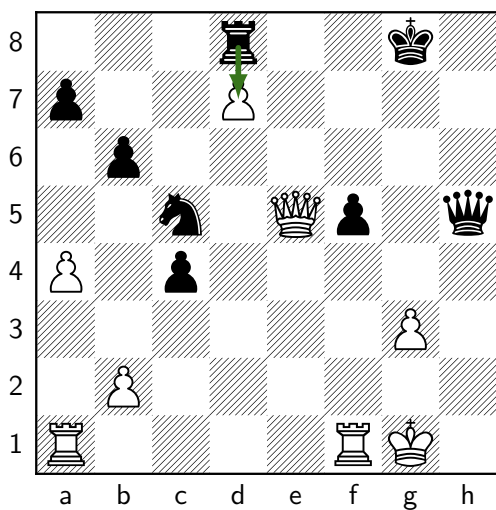


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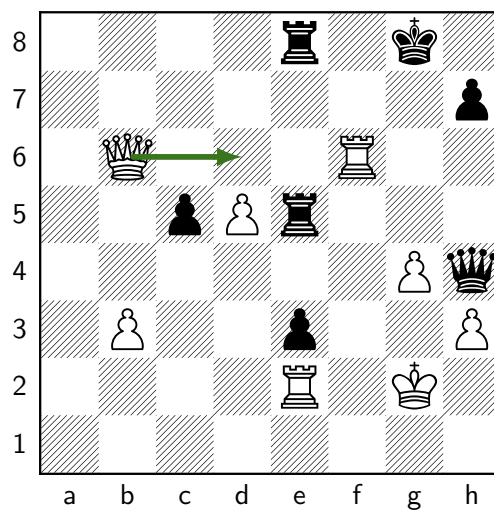


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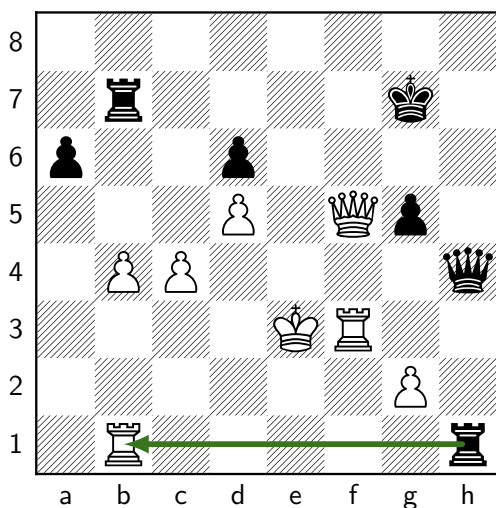
Figure 43: **Move 49, Black: Rd8.** (a) Our position; Melanie plays Rd8 (f8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1770` (ply 89) [structural distance $d=57$, QAP obj=44]: the master played Rd7; under μ our Rf1 plays the role of that game’s Rc2; cross-type roles: our rd1 as their Ph7, our Nc5 as their ph2; transferred move f8d8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1707` (ply 88) [structural distance $d=54$, QAP obj=54]: the master played Kg1; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our qe5 as their Re2, our qe5 as their Rd3; transferred move g8h7. (d) Retrieved from `ccrl_top_p0000#854` (ply 90) [structural distance $d=53$, QAP obj=50]: the master played Qd6; under μ our Qh4 plays the role of that game’s Qb6; cross-type roles: our qe5 as their Re2, our rd1 as their pc5; transferred move h5f3. Synthesis: 1 retrieved analog transferred to Rd8 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



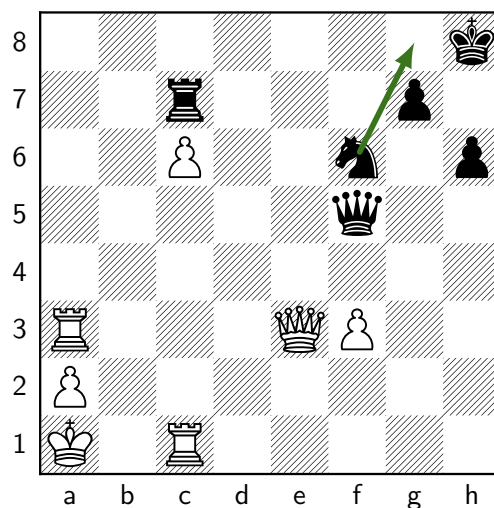
(a)



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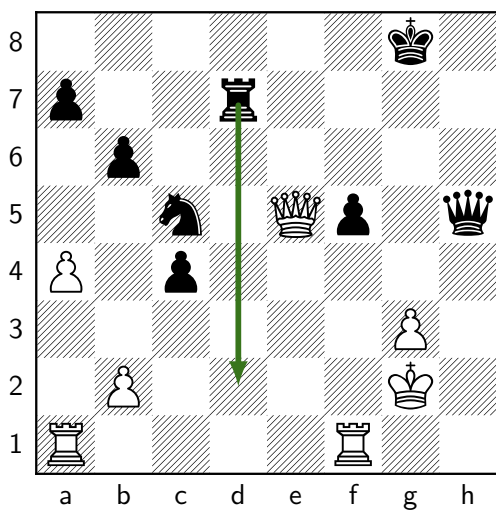


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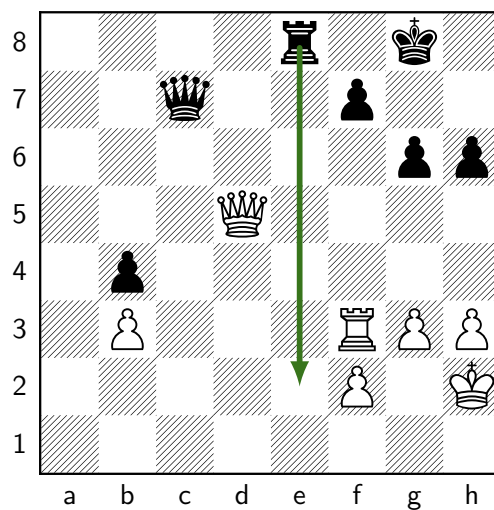


(d)

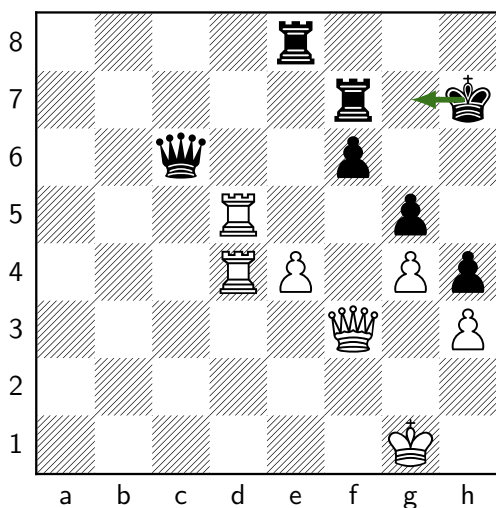
Figure 44: **Move 50, Black: Rxd7**. (a) Our position; Melanie plays Rxd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#854` (ply 90) [structural distance $d=49$, QAP obj=51]: the master played Qd6; under μ our *Rd1* plays the role of that game’s *Qb6* (a **cross-type** correspondence); cross-type roles: our *qe5* as their *Re2*, our *Qh5* as their *re5*; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#1280` (ply 101) [structural distance $d=54$, QAP obj=34]: the master played Rxb1; under μ our *Rd1* plays the role of that game’s *Rh8*; cross-type roles: our *qe5* as their *Rb7*, our *Nc5* as their *Pg5*; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#1424` (ply 97) [structural distance $d=60$, QAP obj=45]: the master played Ng8; under μ our *Qh4* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *qe5* as their *Rc7*, our *Rd8* as their *Qf5*; transferred move h5h7. Synthesis: 2 retrieved analogs transferred to Rxd7 (summed QAP objective 84, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



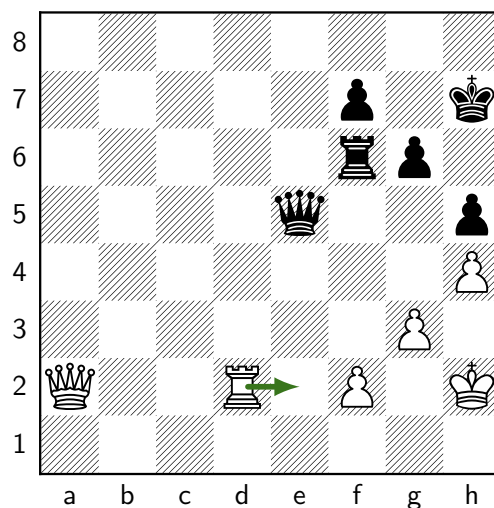
(a)



(b)



(c)



(d)

Figure 45: **Move 51, Black: Rd2+.** (a) Our position; Melanie plays Rd2+ (d7d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#84` (ply 85) [structural distance $d=44$, QAP obj=26]: the master played Re2; under μ our Rd2 plays the role of that game’s Re1; cross-type roles: our Nc5 as their Ph6, our ra1 as their pf2; transferred move d7d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3564` (ply 115) [structural distance $d=50$, QAP obj=33]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kh2; cross-type roles: our Qh5 as their Rf7, our Nc5 as their rd5; transferred move g8h7. (d) Retrieved from `ccr1_top_p0000#2481` (ply 114) [structural distance $d=52$, QAP obj=23]: the master played Re2; under μ our Rd2 plays the role of that game’s Rd2; cross-type roles: our Nc5 as their Qa2, our Qh5 as their Pf2; transferred move d7c7. Synthesis: 1 retrieved analog transferred to Rd2+ (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

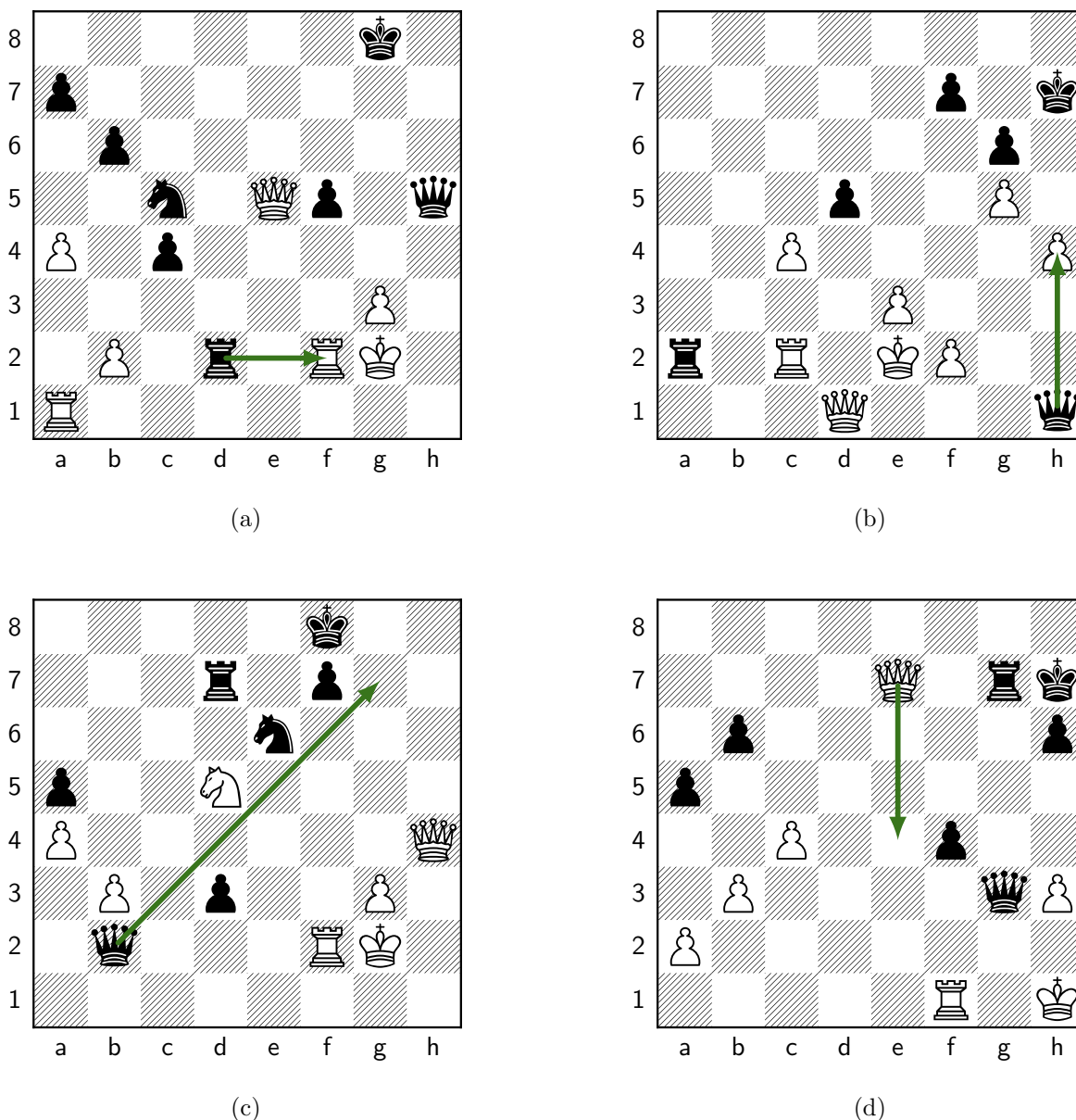
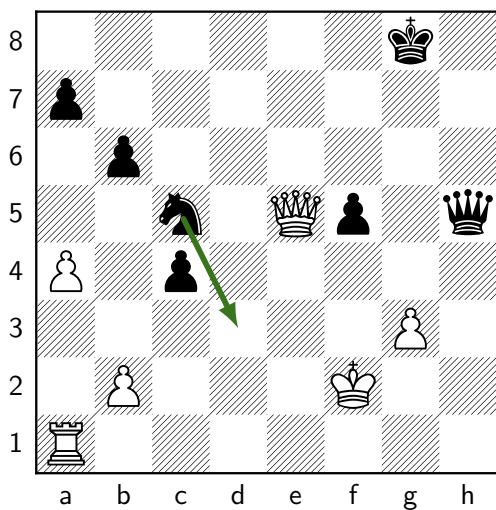
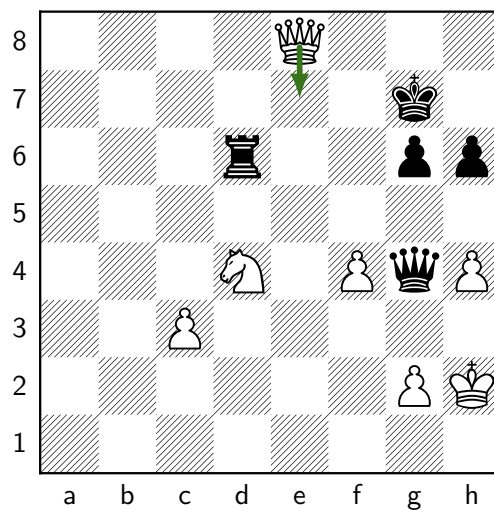


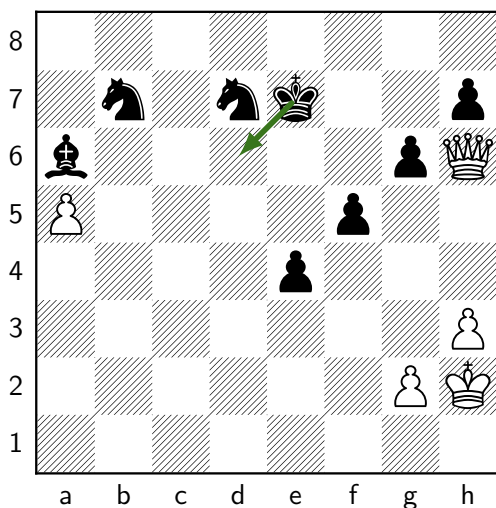
Figure 46: **Move 52, Black: $Rxf2+$.** (a) Our position; Melanie plays $Rxf2+$ ($d2f2$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3183` (ply 91) [structural distance $d=55$, QAP obj=102]: the master played $Qxh4$; under μ our $Rd7$ plays the role of that game's $Qh8$ (a **cross-type** correspondence); cross-type roles: our $Nc5$ as their $Pg6$, our $qe5$ as their $pc4$; transferred move $d2c2$. (c) Retrieved from `ccr1_top_p0001#3114` (ply 107) [structural distance $d=55$, QAP obj=59]: the master played $Qg7$; under μ our $Rd7$ plays the role of that game's $Qb7$ (a **cross-type** correspondence); cross-type roles: our $qe5$ as their $Rd7$, our $pg3$ as their $Ne6$; transferred move $d2d7$. (d) Retrieved from `ccr1_top_p0002#3481` (ply 118) [structural distance $d=50$, QAP obj=56]: the master played $Qe4+$; under μ our $Rd7$ plays the role of that game's $Qe7$ (a **cross-type** correspondence); cross-type roles: our $Qh5$ as their $Rf1$, our $Nc5$ as their $pa5$; transferred move $d2d4$. Synthesis: no retrieved analog transferred to $Rxf2+$ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



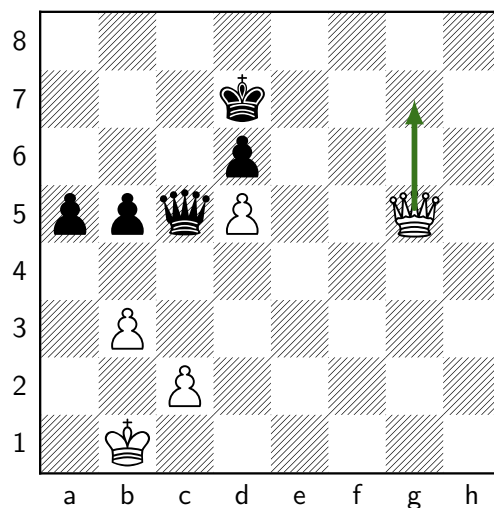
(a)



(b)

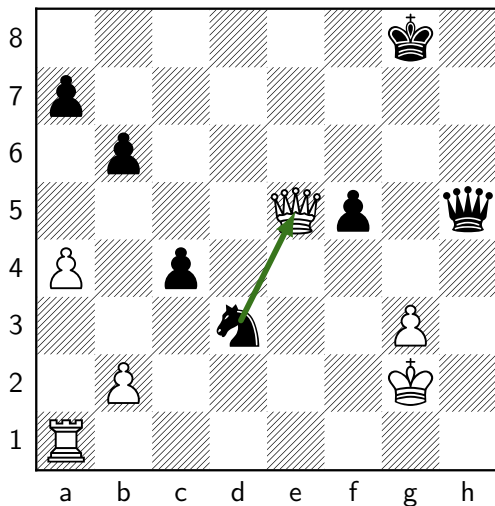


(c)

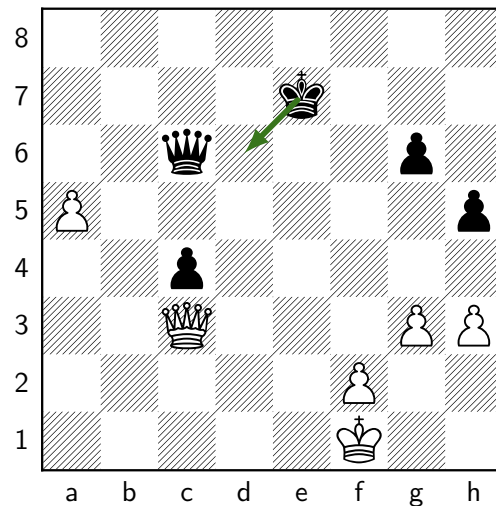


(d)

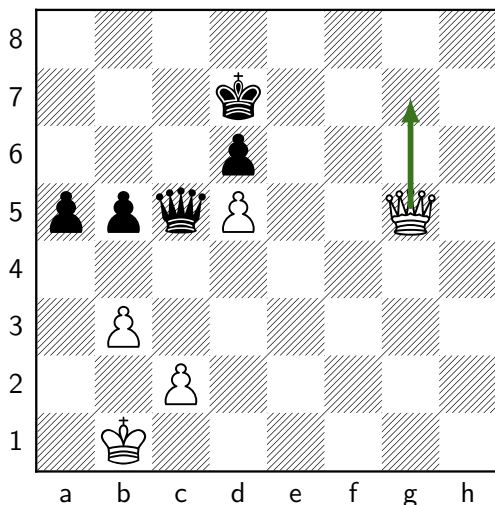
Figure 47: **Move 53, Black: Nd3+.** (a) Our position; Melanie plays Nd3+ (c5d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1058` (ply 118) [structural distance $d=33$, QAP obj=80]: the master played Qe7+; under μ our Qh4 plays the role of that game’s Qe8; cross-type roles: our Nc5 as their Pg2, our Nc5 as their Ph4; transferred move h5h2. (c) Retrieved from `ccr1_top_p0001#3622` (ply 105) [structural distance $d=32$, QAP obj=66]: the master played Kd6; under μ our Kg1 plays the role of that game’s Ke2; cross-type roles: our Pa7 as their Nd7, our Nc5 as their Ph7; transferred move g8h7. (d) Retrieved from `ccr1_top_p0002#304` (ply 90) [structural distance $d=31$, QAP obj=53]: the master played Qg7+; under μ our Qh4 plays the role of that game’s Qg5; cross-type roles: our Nc5 as their Pc2; transferred move h5h2. Synthesis: no retrieved analog transferred to Nd3+ – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



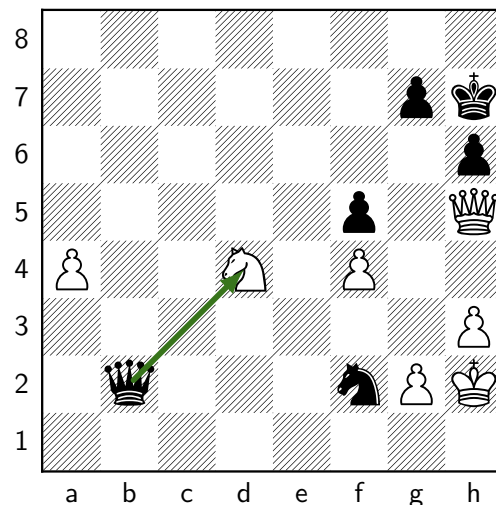
(a)



(b)

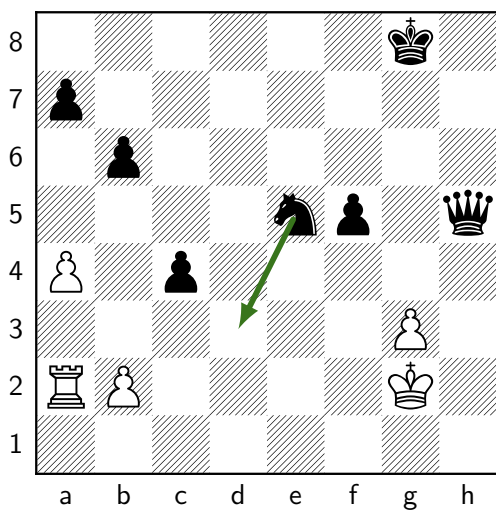


(c)

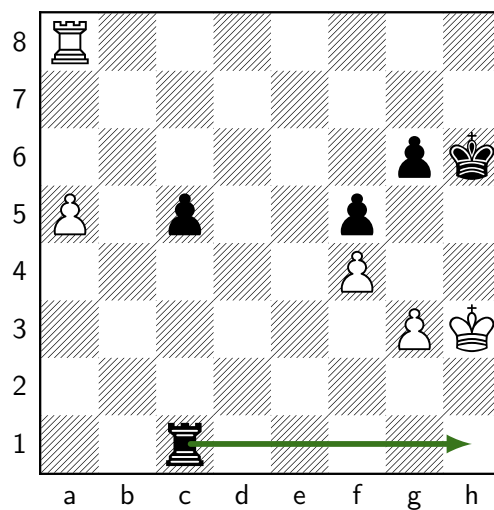


(d)

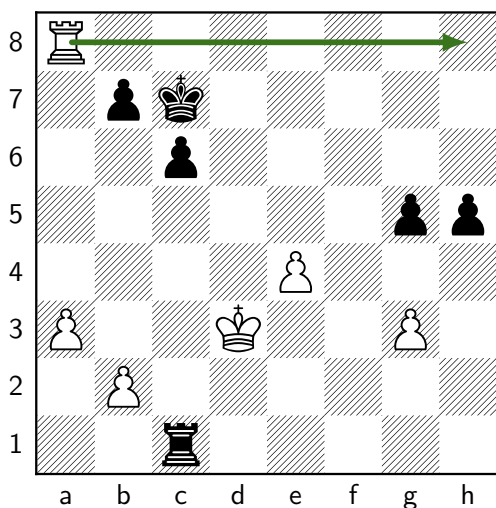
Figure 48: **Move 54, Black: Nxe5**. (a) Our position; Melanie plays Nxe5 (d3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#34` (ply 105) [structural distance $d=39$, QAP obj=35]: the master played Kd6; under μ our *Kg1* plays the role of that game’s *Ke2*; transferred move g8h7. (c) Retrieved from `ccr1_top_p0002#304` (ply 90) [structural distance $d=37$, QAP obj=33]: the master played Qg7+; under μ our *Qh4* plays the role of that game’s *Qg5*; transferred move h5g5. (d) Retrieved from `ccr1_top_p0001#1686` (ply 97) [structural distance $d=39$, QAP obj=33]: the master played Qxd4; under μ our *Nd6* plays the role of that game’s *Qb7* (a **cross-type** correspondence); cross-type roles: our *qe5* as their *nd4*, our *Qh5* as their *Nf2*; transferred move d3b2. Synthesis: no retrieved analog transferred to Nxe5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



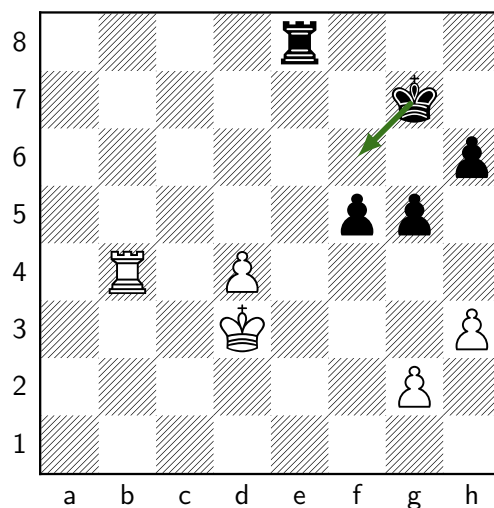
(a)



(b)

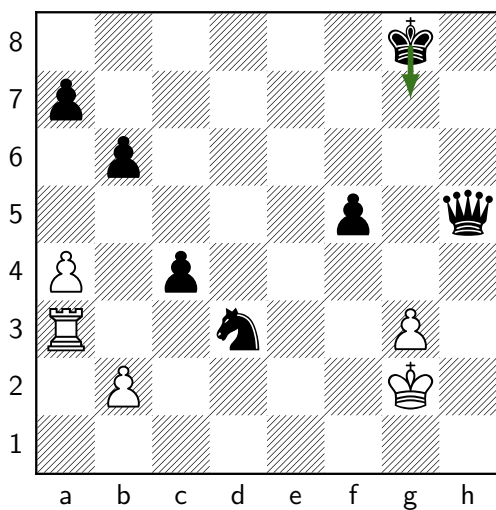


(c)

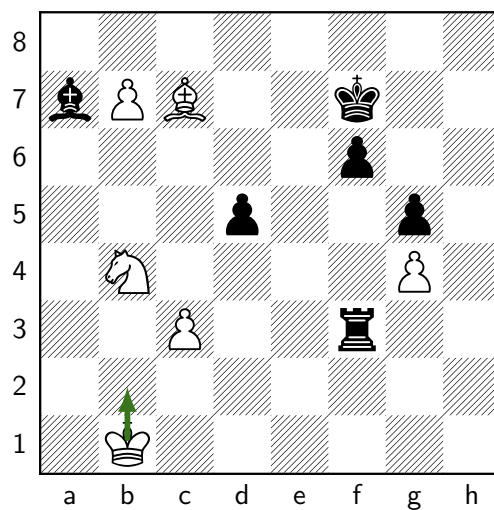


(d)

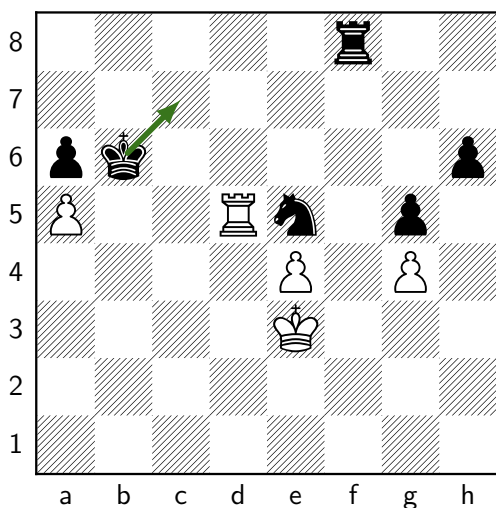
Figure 49: **Move 55, Black: Nd3**. (a) Our position; Melanie plays Nd3 (e5d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#9` (ply 95) [structural distance $d=23$, QAP obj=4]: the master played Rh1+; under μ our Ne4 plays the role of that game’s Rc8 (a **cross-type** correspondence); cross-type roles: our Qh5 as their ra8; transferred move e5d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3960` (ply 116) [structural distance $d=23$, QAP obj=3]: the master played Rh8; under μ our Ne4 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our Pa7 as their rc1, our ra2 as their Pb2; transferred move e5d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3165` (ply 103) [structural distance $d=23$, QAP obj=5]: the master played Kf6; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Qh5 as their Re8, our pa4 as their rb4; transferred move g8f8. Synthesis: 2 retrieved analogs transferred to Nd3 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



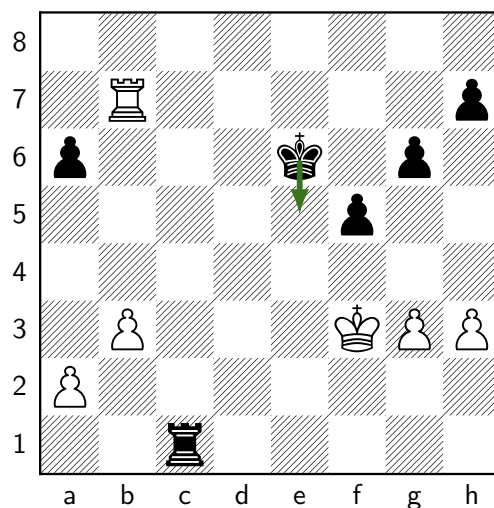
(a)



(b)

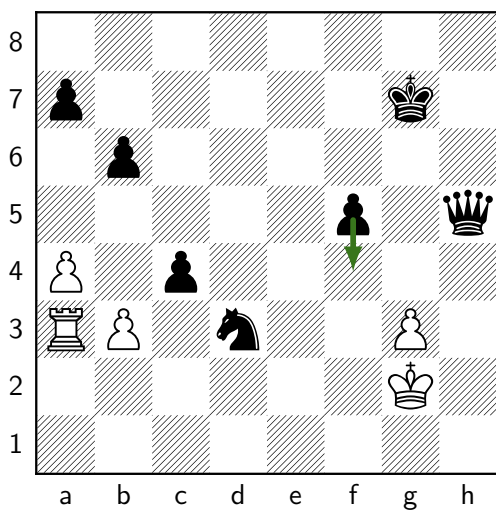


(c)

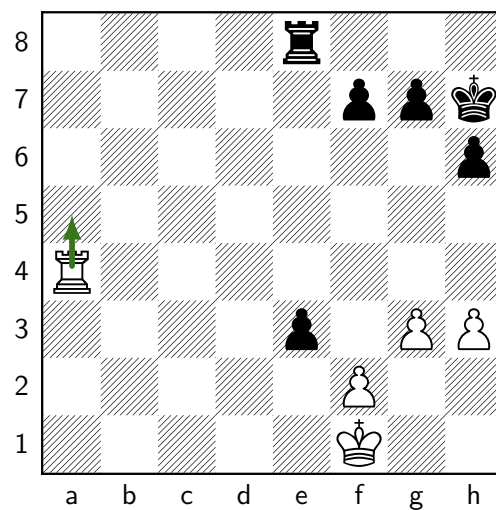


(d)

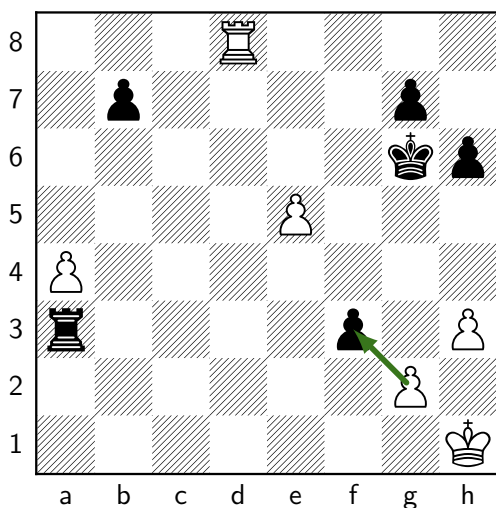
Figure 50: **Move 56, Black: Kg7 (g8g7)**. (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2709` (ply 116) [structural distance $d=23$, QAP obj=14]: the master played Kb2; under μ our Kg1 plays the role of that game’s Kb1; cross-type roles: our Nd3 as their rf3, our ra3 as their Nb4; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#199` (ply 95) [structural distance $d=25$, QAP obj=4]: the master played Kc7; under μ our Kg1 plays the role of that game’s Kb3; cross-type roles: our Qh5 as their Rf8, our Pf5 as their rd5; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3433` (ply 97) [structural distance $d=22$, QAP obj=2]: the master played Ke5; under μ our Kg1 plays the role of that game’s Ke3; cross-type roles: our Pb6 as their rb7, our Nd3 as their Pg6; transferred move g8g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg7 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



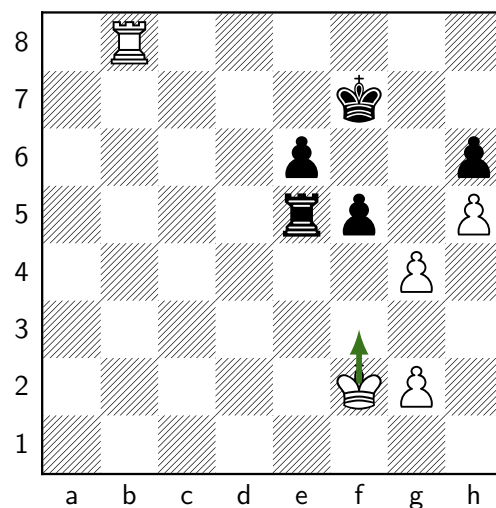
(a)



(b)



(c)



(d)

Figure 51: **Move 57, Black: f4** (f5f4). (a) Our position; Melanie plays f4 (f5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3022` (ply 100) [structural distance $d=26$, QAP obj=41]: the master played Ra5; under μ our Pf4 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Nd3 as their Pg3, our Qh5 as their ph6; transferred move f5f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1768` (ply 96) [structural distance $d=25$, QAP obj=48]: the master played gxf3; under μ our Pc5 plays the role of that game’s Pg2; cross-type roles: our pb3 as their ra3, our Nd3 as their Ph3; transferred move c4b3. (d) Retrieved from `ccr1_top_p0000#2046` (ply 108) [structural distance $d=24$, QAP obj=44]: the master played Kf3; under μ our Kg2 plays the role of that game’s Kf2; cross-type roles: our Nd3 as their Ph5, our Qh5 as their Rb8; transferred move g7f6. Synthesis: 1 retrieved analog transferred to f4 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

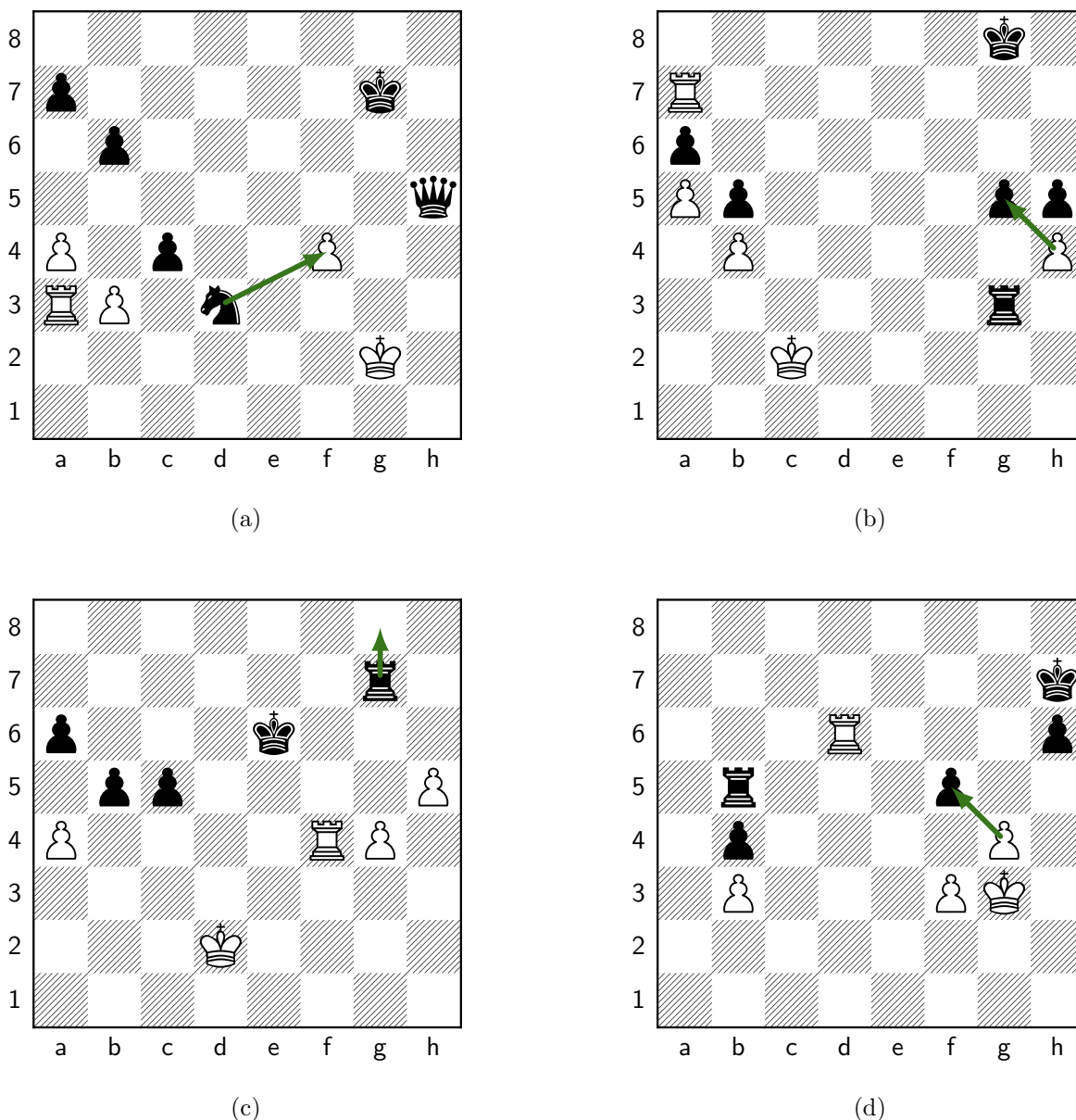
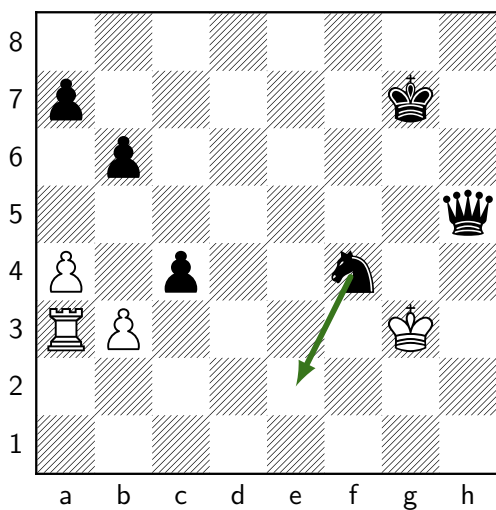
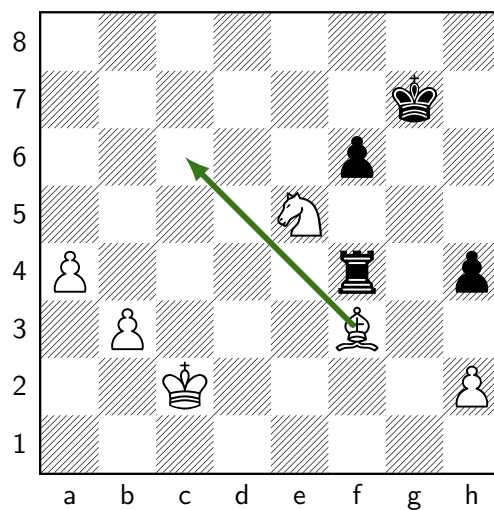


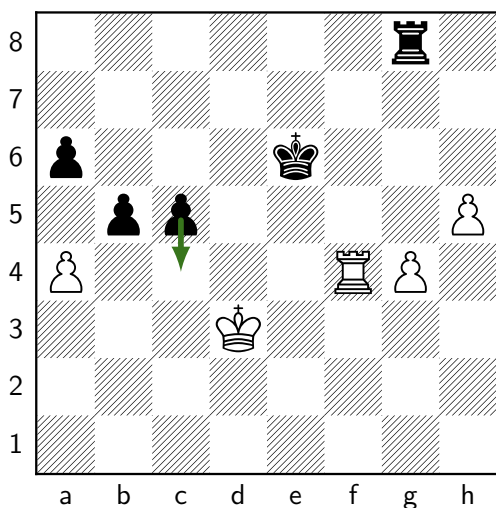
Figure 52: **Move 58, Black: Nxf4+.** (a) Our position; Melanie plays Nxf4+ (d3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3535` (ply 102) [structural distance $d=28$, QAP obj=49]: the master played **hxxg5**; under μ our *Pc5* plays the role of that game’s *Ph4*; cross-type roles: our *Nd3* as their *Pa5*, our *Pc4* as their *Ra7*; transferred move **c4b3**. (c) Retrieved from `ccr1_top_p0001#3807` (ply 105) [structural distance $d=28$, QAP obj=44]: the master played **Rg8**; under μ our *Nd6* plays the role of that game’s *Rg2* (a **cross-type** correspondence); cross-type roles: our *ra3* as their *Pa6*, our *Qh5* as their *ph5*; transferred move **d3b4**. (d) Retrieved from `ccr1_top_p0000#1112` (ply 112) [structural distance $d=34$, QAP obj=44]: the master played **gxf5**; under μ our *Pc5* plays the role of that game’s *Pg4*; cross-type roles: our *Nd3* as their *Rd6*; transferred move **c4b3**. Synthesis: no retrieved analog transferred to Nxf4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



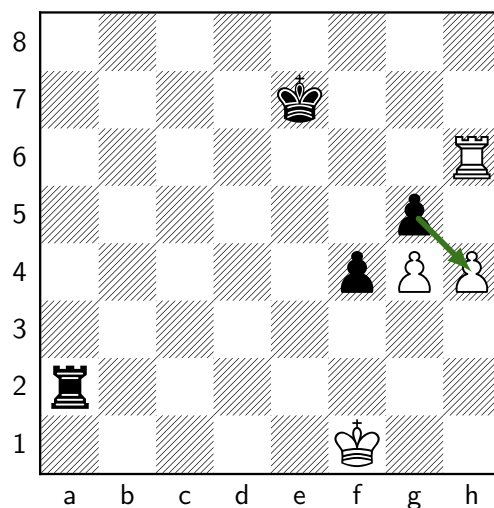
(a)



(b)



(c)



(d)

Figure 53: **Move 59, Black: Ne2+.** (a) Our position; Melanie plays Ne2+ (f4e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2628` (ply 104) [structural distance $d=31$, QAP obj=24]: the master played Bc6; under μ our Nf5 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Qh5 as their Pb3, our kg3 as their rf4; transferred move f4d3. (c) Retrieved from `ccrl_top_p0001#3807` (ply 107) [structural distance $d=32$, QAP obj=15]: the master played c4+; under μ our Qh4 plays the role of that game’s Pc4 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Rg8, our Nf4 as their Pa6; transferred move h5f3. (d) Retrieved from `ccrl_top_p0001#3425` (ply 113) [structural distance $d=31$, QAP obj=15]: the master played gxh4; under μ our Pc5 plays the role of that game’s Pg4; cross-type roles: our Qh5 as their Ra2; transferred move c4b3. Synthesis: no retrieved analog transferred to Ne2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

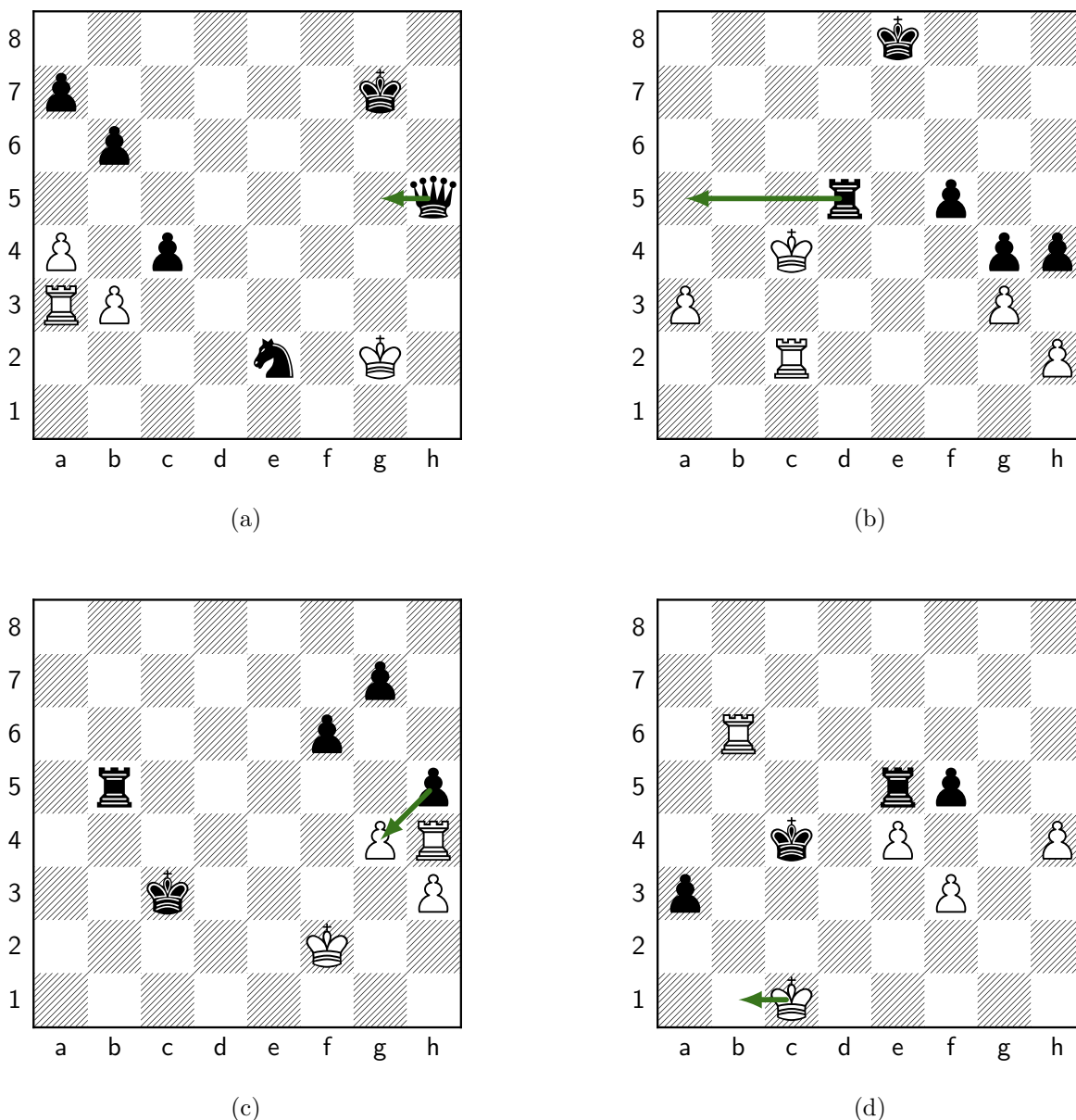
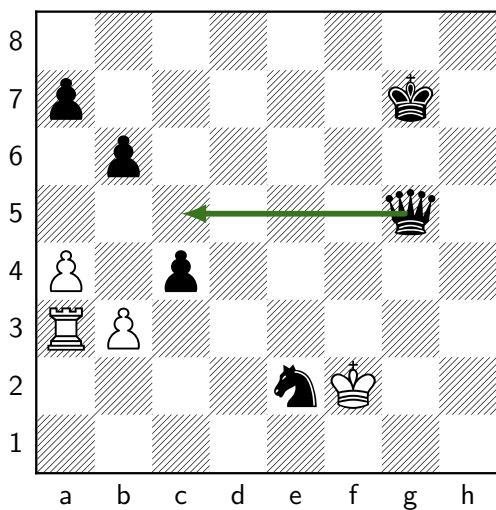
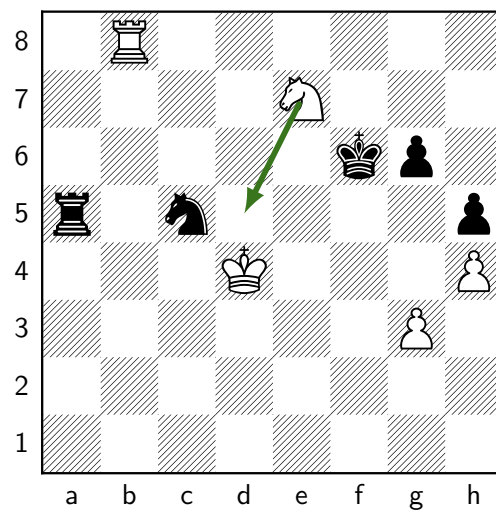


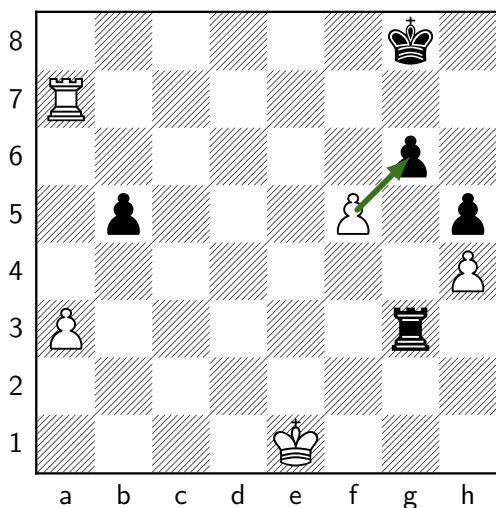
Figure 54: **Move 60, Black: Qg5+.** (a) Our position; Melanie plays Qg5+ (h5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#421` (ply 105) [structural distance $d=27$, QAP obj=1]: the master played Ra5; under μ our Qh4 plays the role of that game’s Rd4 (a **cross-type** correspondence); cross-type roles: our Ne2 as their rc2, our ra3 as their ph2; transferred move h5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3100` (ply 105) [structural distance $d=27$, QAP obj=53]: the master played hxg4; under μ our Pc5 plays the role of that game’s Ph4; cross-type roles: our pa4 as their Rb5, our pb3 as their rh4; transferred move c4b3. (d) Retrieved from `ccr1_top_p0001#2498` (ply 118) [structural distance $d=28$, QAP obj=52]: the master played Kb1; under μ our Kg2 plays the role of that game’s Kc1; cross-type roles: our ra3 as their Pf3, our Pc4 as their re5; transferred move g7h8. Synthesis: 1 retrieved analog transferred to Qg5+ (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



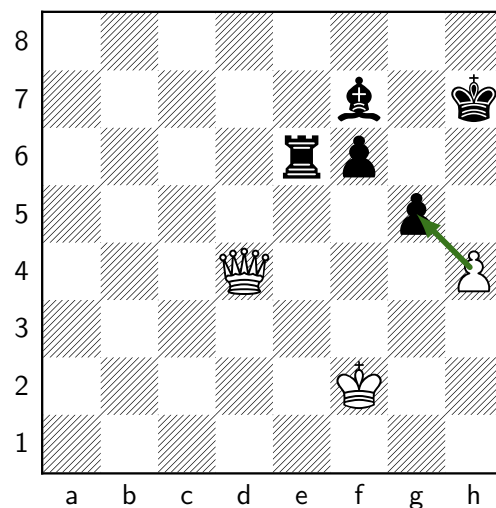
(a)



(b)

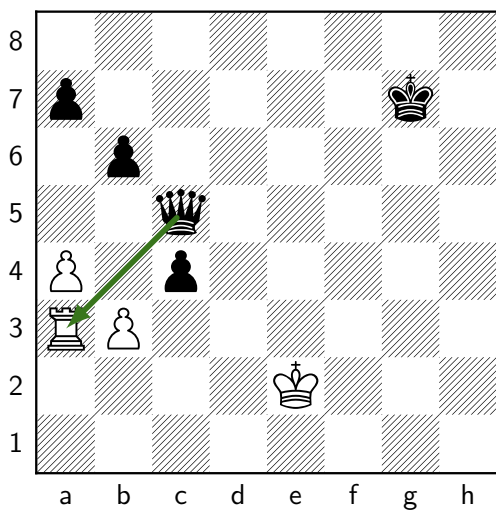


(c)

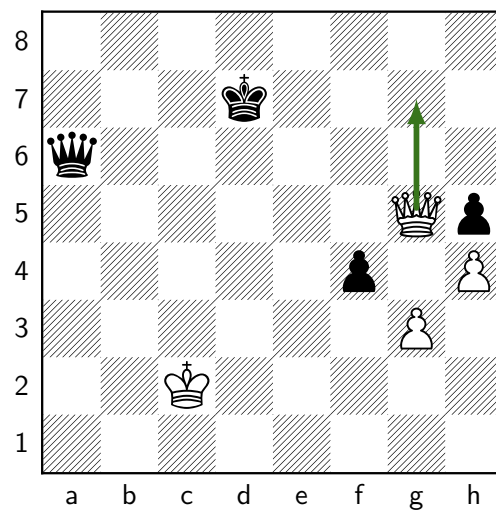


(d)

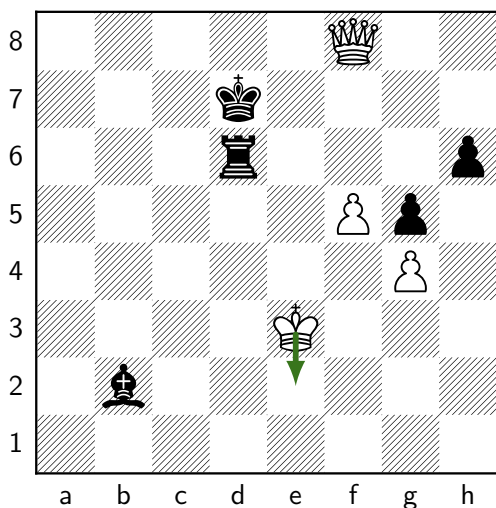
Figure 55: **Move 61, Black: Qc5+.** (a) Our position; Melanie plays Qc5+ (g5c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#2955` (ply 114) [structural distance $d=31$, QAP obj=19]: the master played Nd5+; under μ our Ne7 plays the role of that game's Ne7; cross-type roles: our Pa7 as their ra5, our ra3 as their nc5; transferred move e2f4. (c) Retrieved from `ccr1_top.p0002#2982` (ply 134) [structural distance $d=28$, QAP obj=18]: the master played fxg6; under μ our Pc5 plays the role of that game's Pf5; cross-type roles: our Qg5 as their Ph4, our Ne2 as their Ra7; transferred move c4b3. (d) Retrieved from `ccr1_top.p0001#1694` (ply 114) [structural distance $d=29$, QAP obj=15]: the master played hxg5; under μ our Pc5 plays the role of that game's Ph4; cross-type roles: our pa4 as their re6, our ra3 as their pf6; transferred move c4b3. Synthesis: no retrieved analog transferred to Qc5+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



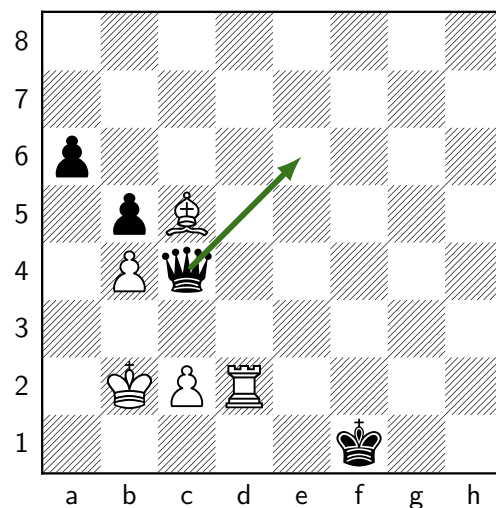
(a)



(b)

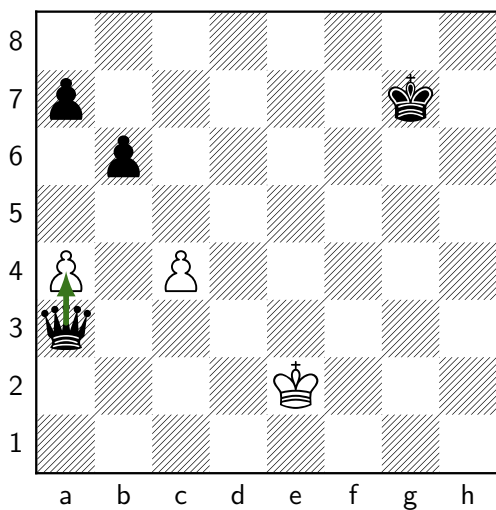


(c)

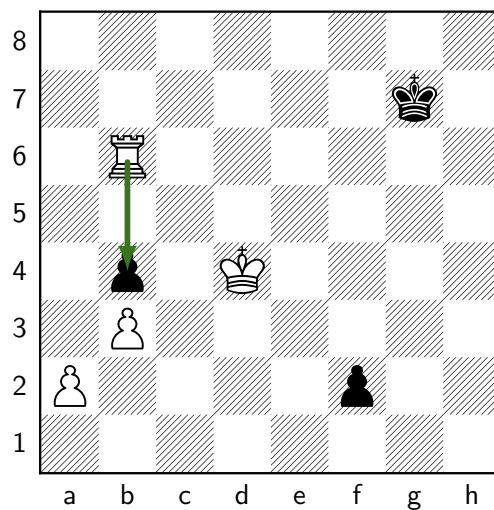


(d)

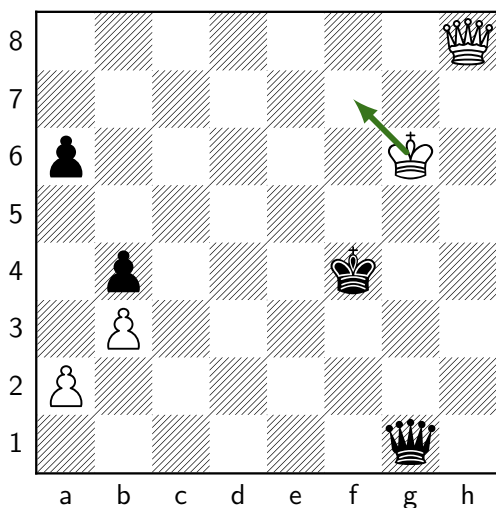
Figure 56: **Move 62, Black: Qxa3**. (a) Our position; Melanie plays **Qxa3** (c5a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3861` (ply 136) [structural distance $d=34$, QAP obj=28]: the master played **Qg7+**; under μ our *Qc4* plays the role of that game’s *Qg5*; cross-type roles: our *ra3* as their *ph5*, our *pa4* as their *qa6*; transferred move **c5a3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3934` (ply 126) [structural distance $d=26$, QAP obj=21]: the master played **Ke2**; under μ our *Kg2* plays the role of that game’s *Ke3*; cross-type roles: our *pb3* as their *bb2*; transferred move **g7h7**. (d) Retrieved from `ccr1_top_p0003#1612` (ply 115) [structural distance $d=26$, QAP obj=19]: the master played **Qe6**; under μ our *Qc4* plays the role of that game’s *Qc5*; cross-type roles: our *ra3* as their *bc5*, our *ra3* as their *pb4*; transferred move **c5d6**. Synthesis: 1 retrieved analog transferred to **Qxa3** (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



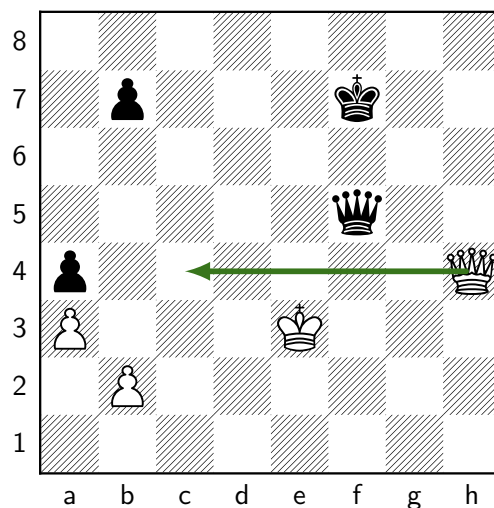
(a)



(b)

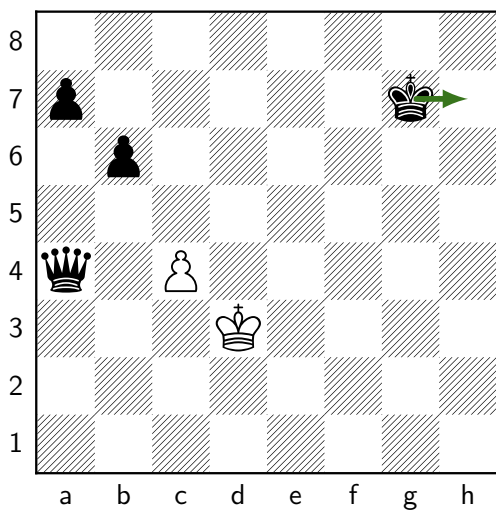


(c)

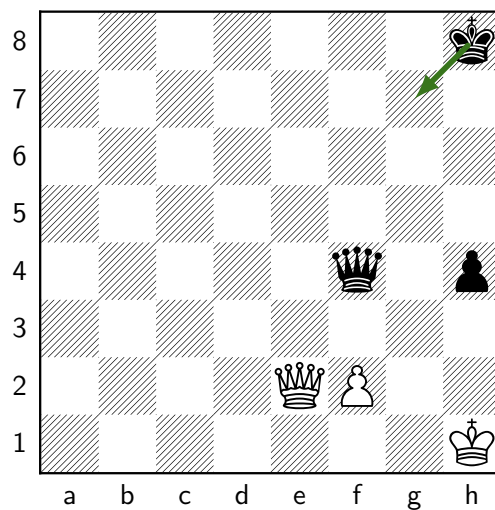


(d)

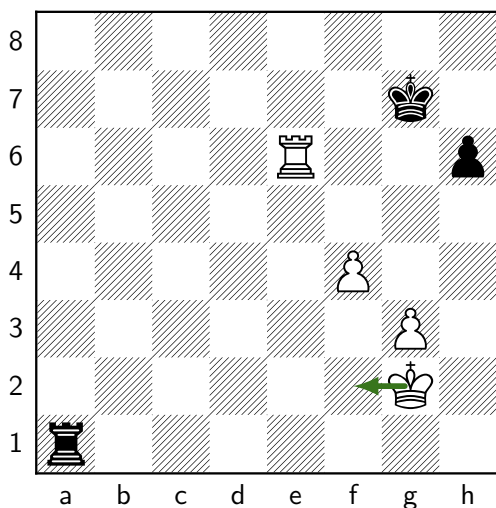
Figure 57: **Move 63, Black: Qxa4.** (a) Our position; Melanie plays **Qxa4** (a3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2995` (ply 114) [structural distance $d=5$, QAP obj=12]: the master played **Rxb4**; under μ our *Qa6* plays the role of that game’s *Rb6* (a **cross-type** correspondence); transferred move **a3a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#580` (ply 110) [structural distance $d=7$, QAP obj=2]: the master played **Kf7**; under μ our *Kg2* plays the role of that game’s *Kg6*; cross-type roles: our *pc4* as their *qg1*; transferred move **g7h6**. (d) Retrieved from `ccr1_top_p0001#3530` (ply 130) [structural distance $d=8$, QAP obj=2]: the master played **Qc4+**; under μ our *Qa6* plays the role of that game’s *Qh4*; cross-type roles: our *pc4* as their *qf5*; transferred move **a3d6**. Synthesis: 1 retrieved analog transferred to **Qxa4** (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



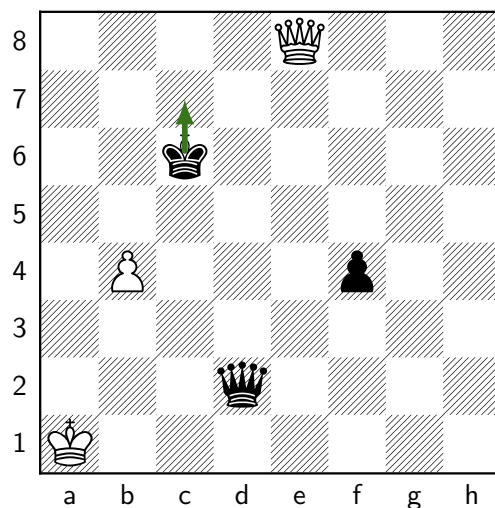
(a)



(b)



(c)



(d)

Figure 58: **Move 64, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#1345` (ply 137) [structural distance $d=11$, QAP obj=20]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our Pb6 as their qe2; transferred move g7h7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#841` (ply 130) [structural distance $d=12$, QAP obj=17]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Qa4 as their Re6; transferred move g7h7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#1751` (ply 143) [structural distance $d=12$, QAP obj=3]: the master played Kc7; under μ our Kg2 plays the role of that game’s Kc3; cross-type roles: our Pb6 as their qe8; transferred move g7h7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh7 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

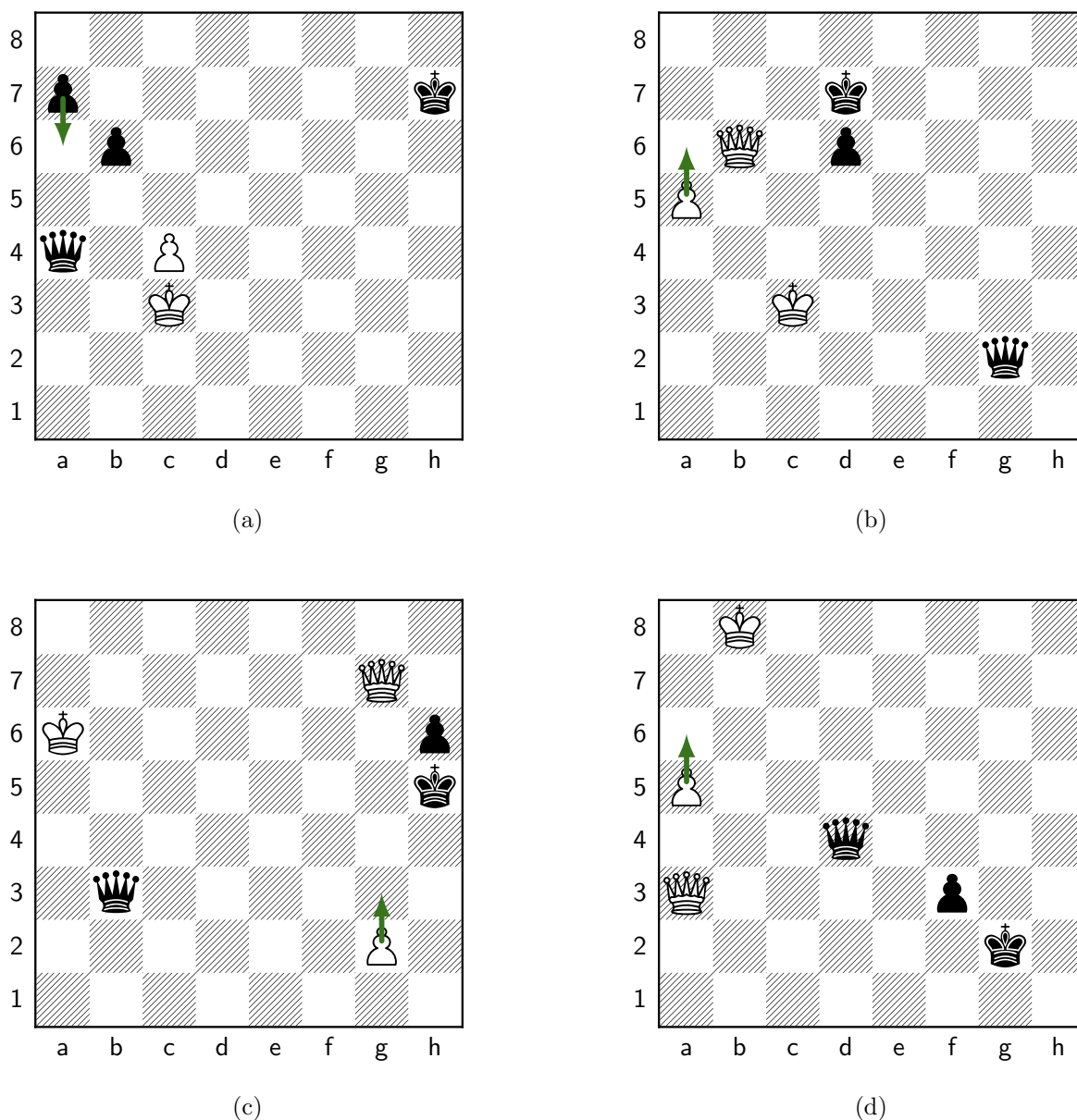
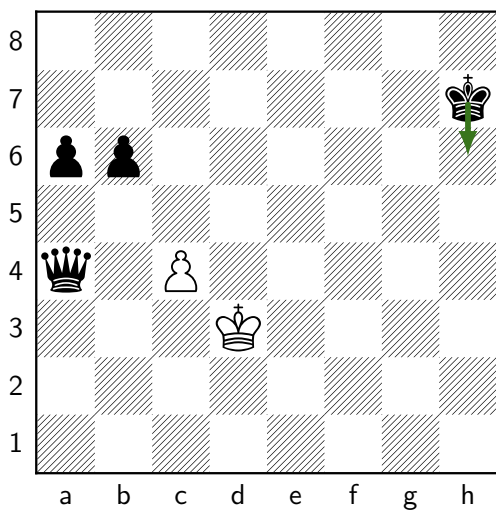
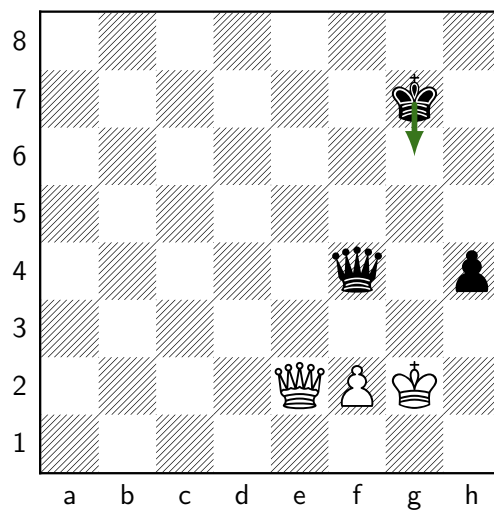


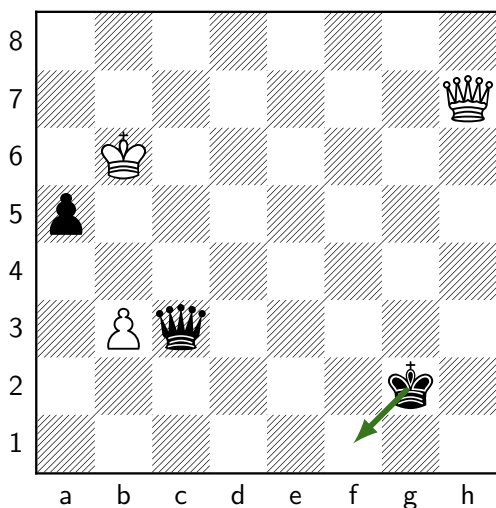
Figure 59: **Move 65, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#146 (ply 114) [structural distance $d=6$, QAP obj=23]: the master played a6; under μ our $Pa2$ plays the role of that game's $Pa5$; cross-type roles: our $Pb6$ as their $qg2$; transferred move a7a6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1098 (ply 140) [structural distance $d=12$, QAP obj=13]: the master played g3; under μ our $Pa2$ plays the role of that game's $Pg2$; cross-type roles: our $Pb6$ as their $qb3$; transferred move a7a6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3119 (ply 140) [structural distance $d=9$, QAP obj=10]: the master played a6; under μ our $Pa2$ plays the role of that game's $Pa5$; cross-type roles: our $Pb6$ as their $qd4$; transferred move a7a6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to a6 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



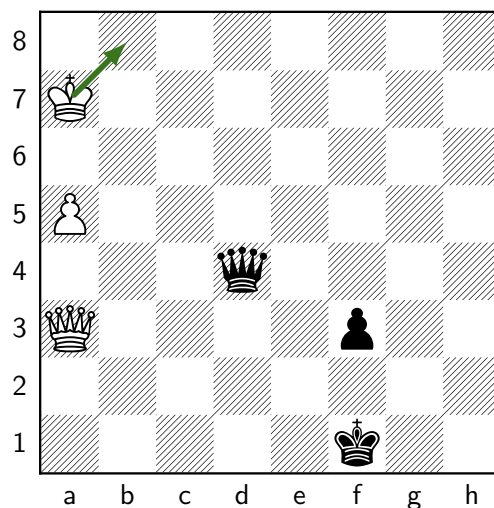
(a)



(b)

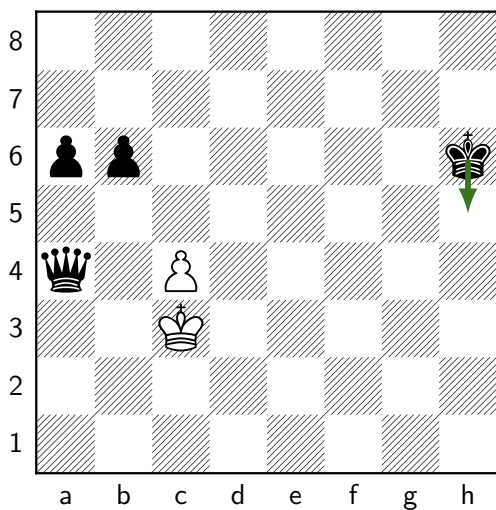


(c)

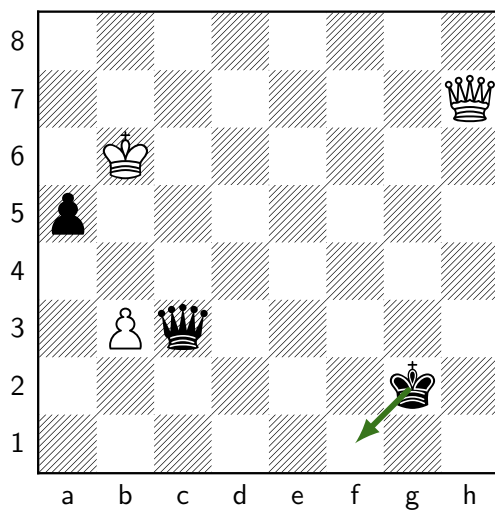


(d)

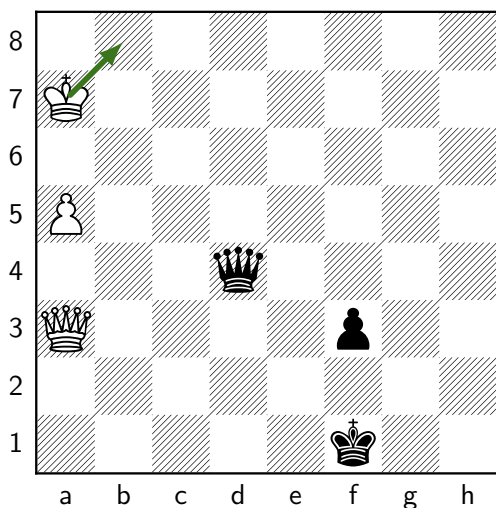
Figure 60: **Move 66, Black: Kh6.** (a) Our position; Melanie plays Kh6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#1345` (ply 139) [structural distance $d=10$, QAP obj=23]: the master played Kg6; under μ our Kh2 plays the role of that game's Kg2; cross-type roles: our Pb6 as their qe2; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#1170` (ply 143) [structural distance $d=9$, QAP obj=13]: the master played Kf1; under μ our Kh2 plays the role of that game's Kg7; cross-type roles: our Pb6 as their qh7; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccl1_top_p0001#3119` (ply 138) [structural distance $d=11$, QAP obj=2]: the master played Kb8; under μ our Kh2 plays the role of that game's Ka7; cross-type roles: our pc4 as their qd4; transferred move h7h6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh6 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



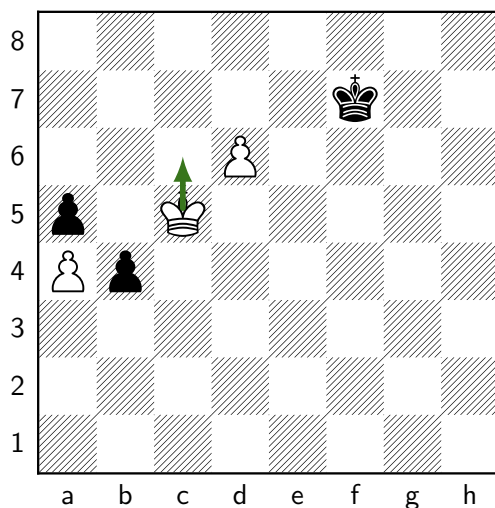
(a)



(b)

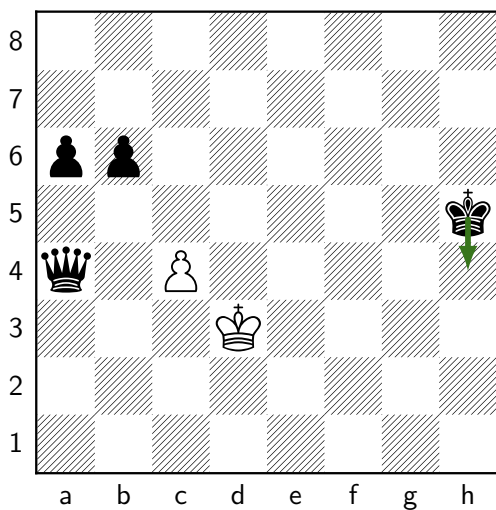


(c)

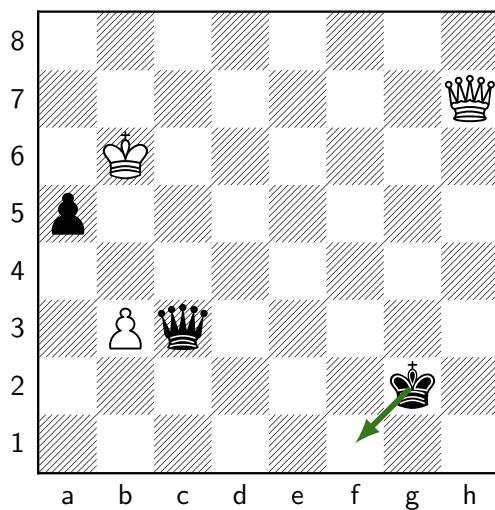


(d)

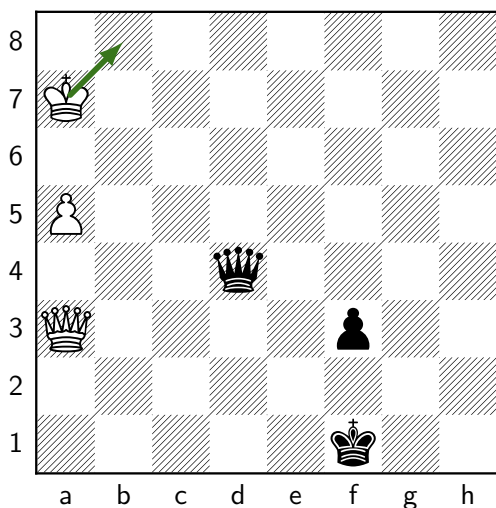
Figure 61: **Move 67, Black: Kh5.** (a) Our position; Melanie plays Kh5 (h6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1170` (ply 143) [structural distance $d=9$, QAP obj=14]: the master played Kf1; under μ our *Kh3* plays the role of that game’s *Kg7*; cross-type roles: our *Pb6* as their *qh7*; transferred move h6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3119` (ply 138) [structural distance $d=12$, QAP obj=2]: the master played Kb8; under μ our *Kh3* plays the role of that game’s *Ka7*; cross-type roles: our *pc4* as their *qd4*; transferred move h6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2602` (ply 118) [structural distance $d=12$, QAP obj=2]: the master played Kc6; under μ our *Kh3* plays the role of that game’s *Kc5*; cross-type roles: our *Qa4* as their *Pa4*; transferred move h6h5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kh5 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



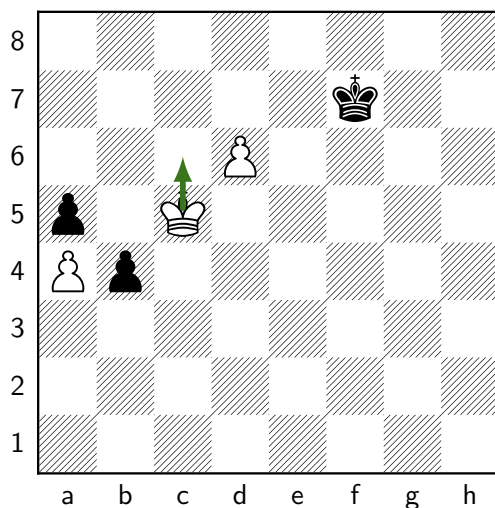
(a)



(b)

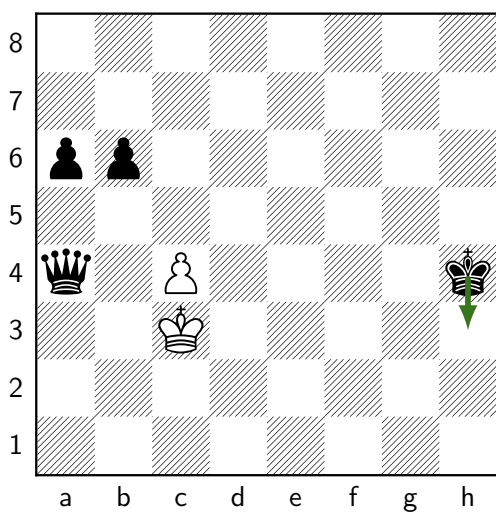


(c)

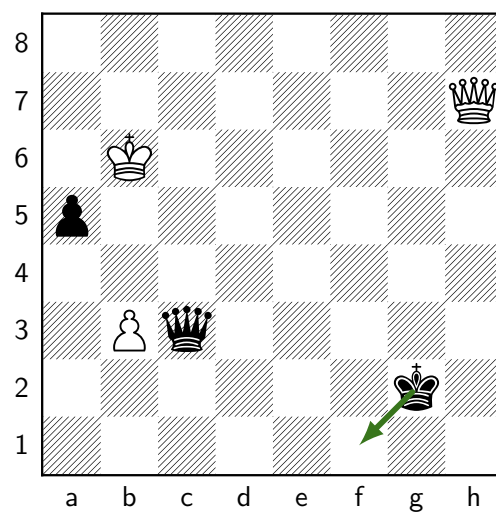


(d)

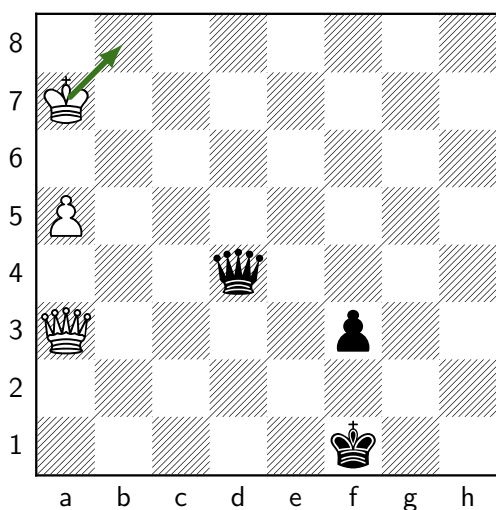
Figure 62: **Move 68, Black: Kh4**. (a) Our position; Melanie plays Kh4 (h5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1170 (ply 143) [structural distance $d=9$, QAP obj=13]: the master played Kf1; under μ our Kh4 plays the role of that game’s Kg7; cross-type roles: our Pb6 as their qh7; transferred move h5h4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3119 (ply 138) [structural distance $d=11$, QAP obj=2]: the master played Kb8; under μ our Kh4 plays the role of that game’s Ka7; cross-type roles: our pc4 as their qd4; transferred move h5h4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2602 (ply 118) [structural distance $d=11$, QAP obj=2]: the master played Kc6; under μ our Kh4 plays the role of that game’s Kc5; cross-type roles: our Qa4 as their Pa4; transferred move h5h4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kh4 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



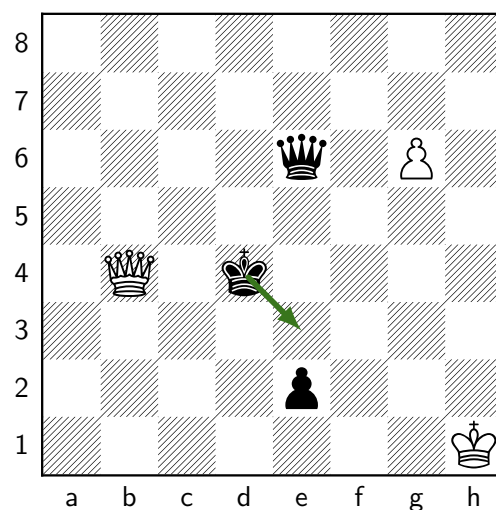
(a)



(b)

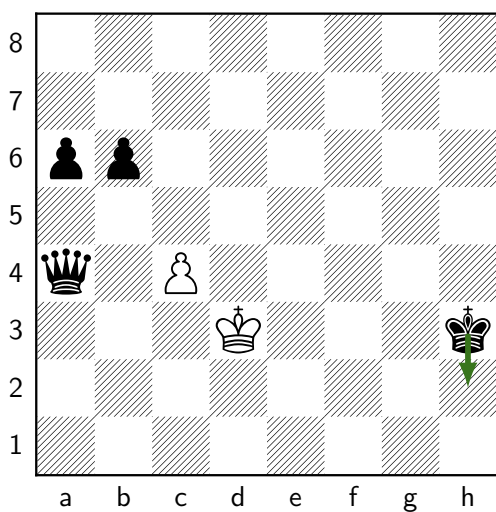


(c)

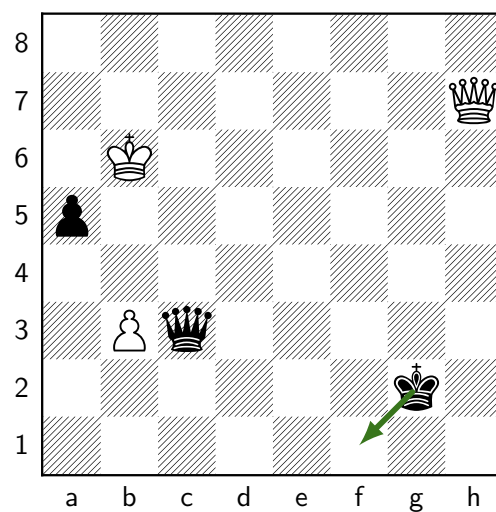


(d)

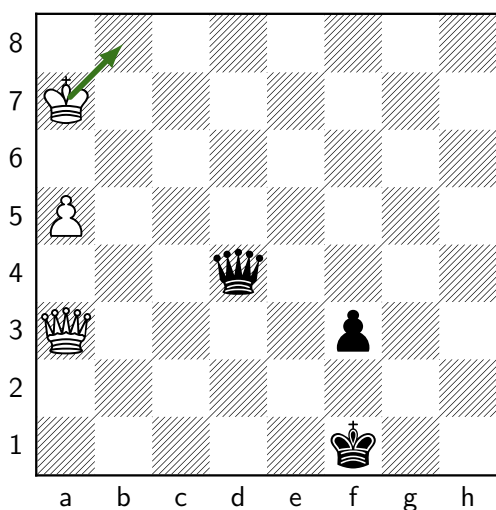
Figure 63: **Move 69, Black: Kh3.** (a) Our position; Melanie plays Kh3 (h4h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1170 (ply 143) [structural distance $d=8$, QAP obj=14]: the master played Kf1; under μ our Kh5 plays the role of that game's Kg7; cross-type roles: our Pb6 as their qh7; transferred move h4h3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3119 (ply 138) [structural distance $d=12$, QAP obj=2]: the master played Kb8; under μ our Kh5 plays the role of that game's Ka7; cross-type roles: our pc4 as their qd4; transferred move h4h3 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#392 (ply 135) [structural distance $d=11$, QAP obj=2]: the master played Ke3; under μ our Kh5 plays the role of that game's Kd5; cross-type roles: our Pb6 as their qb4; transferred move h4h3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh3 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



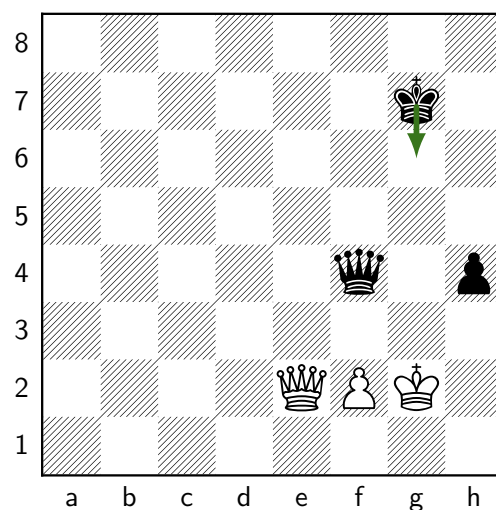
(a)



(b)

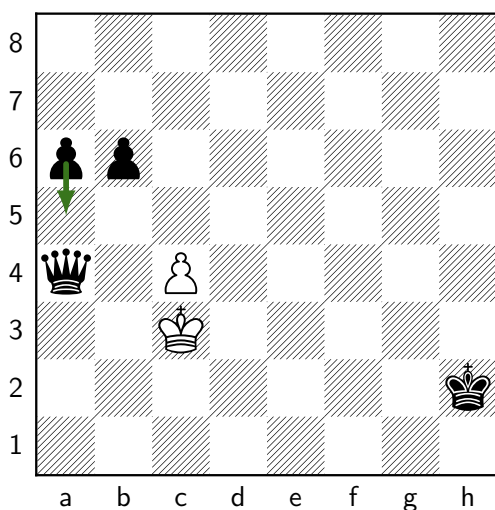


(c)

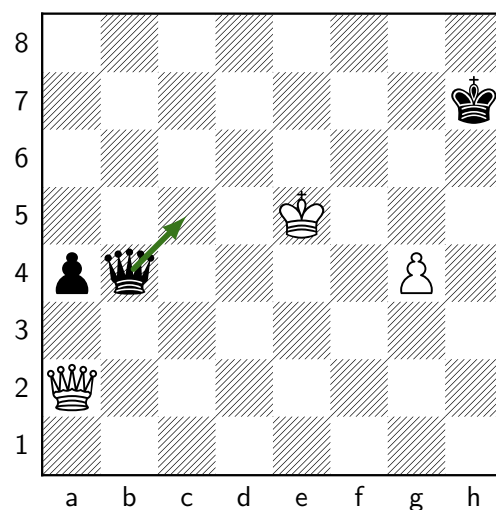


(d)

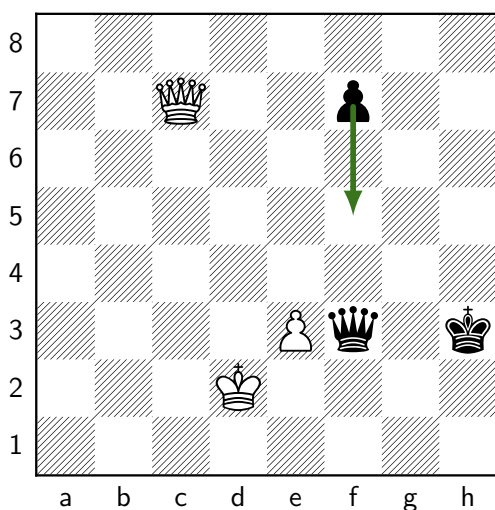
Figure 64: **Move 70, Black: Kh2.** (a) Our position; Melanie plays Kh2 (h3h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1170` (ply 143) [structural distance $d=9$, QAP obj=13]: the master played Kf1; under μ our *Kh6* plays the role of that game’s *Kg7*; cross-type roles: our *Pb6* as their *qh7*; transferred move h3h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3119` (ply 138) [structural distance $d=11$, QAP obj=2]: the master played Kb8; under μ our *Kh6* plays the role of that game’s *Ka7*; cross-type roles: our *pc4* as their *qd4*; transferred move h3h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1345` (ply 139) [structural distance $d=10$, QAP obj=23]: the master played Kg6; under μ our *Kh6* plays the role of that game’s *Kg2*; cross-type roles: our *Pb6* as their *qe2*; transferred move h3h4. Synthesis: 3 retrieved analogs transferred to Kh2 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



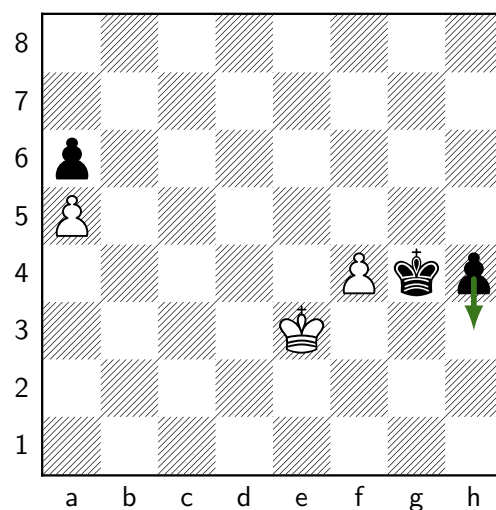
(a)



(b)

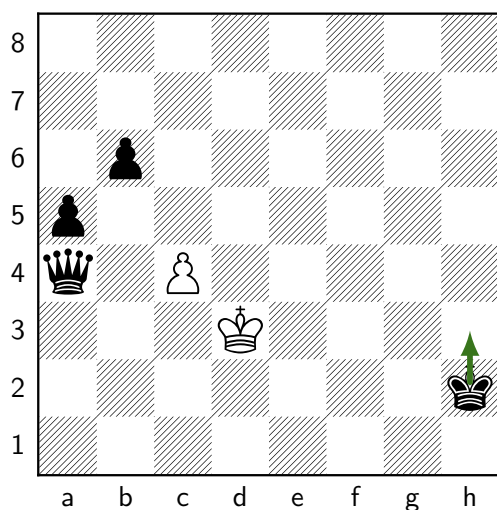


(c)

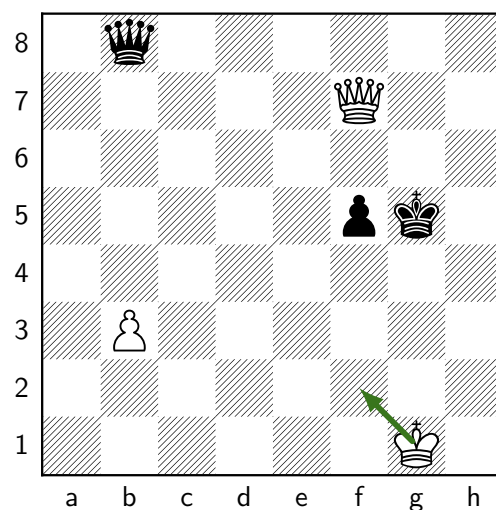


(d)

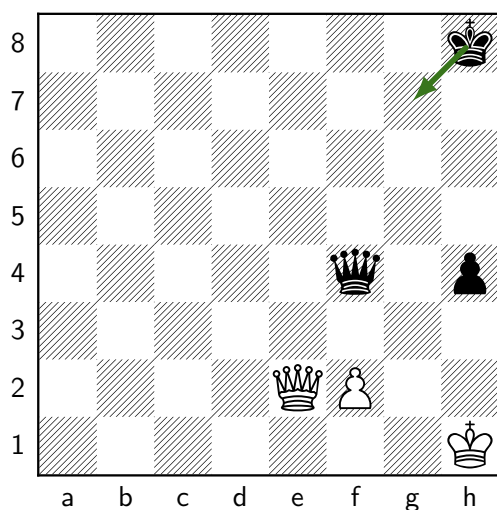
Figure 65: **Move 71, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3123 (ply 153) [structural distance $d=12$, QAP obj=18]: the master played Qc5+; under μ our Pa3 plays the role of that game’s Qb5 (a **cross-type** correspondence); transferred move a6a5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1278 (ply 131) [structural distance $d=12$, QAP obj=12]: the master played f5; under μ our Pa3 plays the role of that game’s Pf2; cross-type roles: our Pb6 as their qc7; transferred move a6a5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3736 (ply 135) [structural distance $d=10$, QAP obj=3]: the master played h3; under μ our Pa3 plays the role of that game’s Ph5; cross-type roles: our Qa4 as their pa5; transferred move a6a5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a5 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



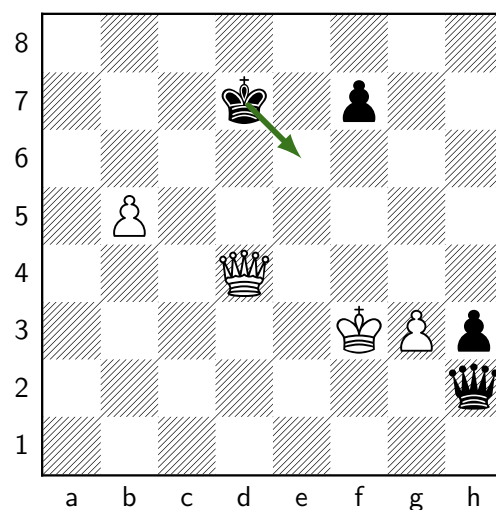
(a)



(b)



(c)



(d)

Figure 66: **Move 72, Black: Kh3**. (a) Our position; Melanie plays Kh3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#1723` (ply 158) [structural distance $d=12$, QAP obj=22]: the master played Kf2; under μ our Kh7 plays the role of that game's Kg1; cross-type roles: our Pb6 as their qb8; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#1345` (ply 137) [structural distance $d=12$, QAP obj=20]: the master played Kg7; under μ our Kh7 plays the role of that game's Kh1; cross-type roles: our Pb6 as their qe2; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#1182` (ply 129) [structural distance $d=13$, QAP obj=6]: the master played Ke6; under μ our Kh7 plays the role of that game's Kd2; transferred move h2h3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kh3 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

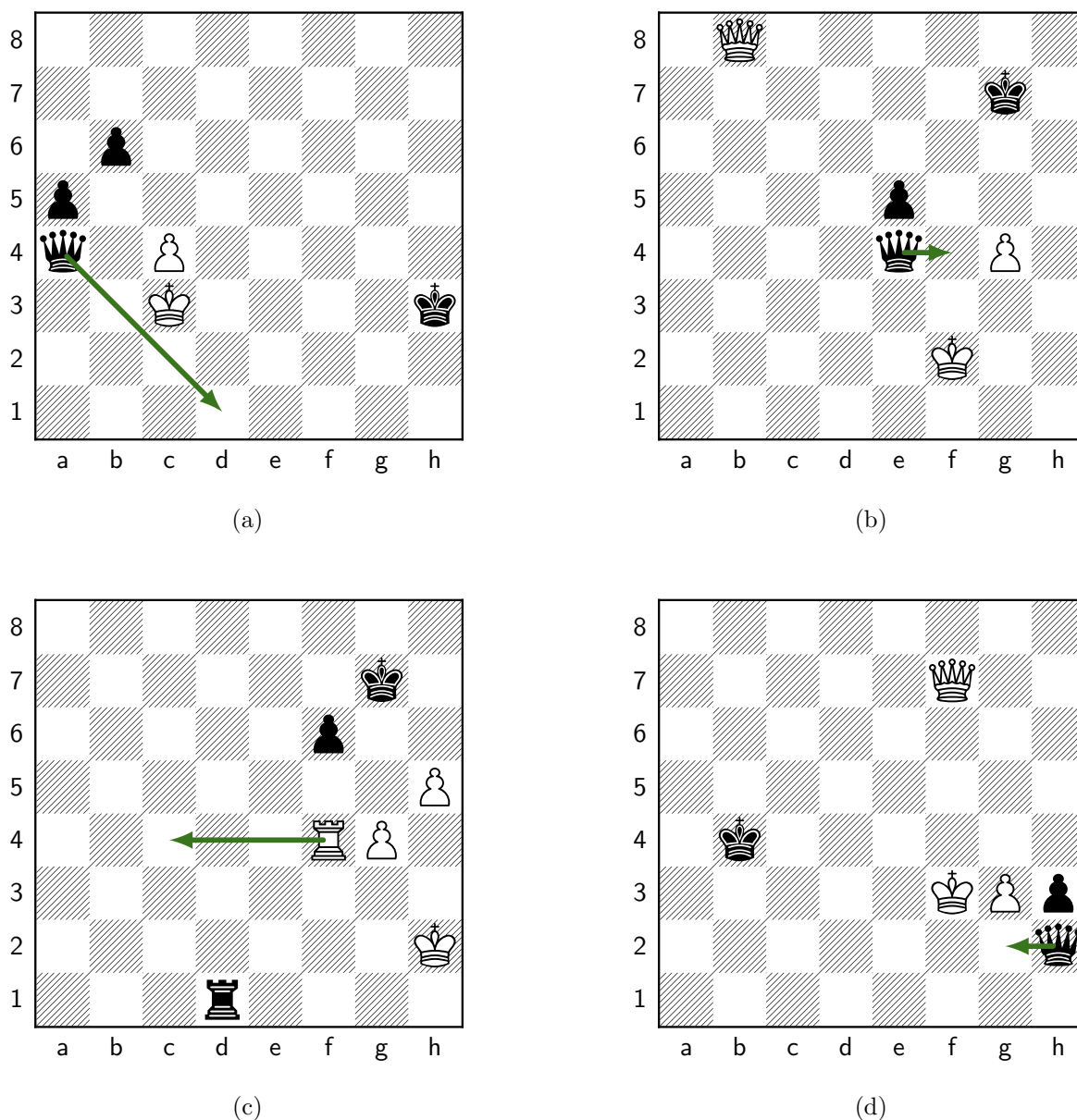
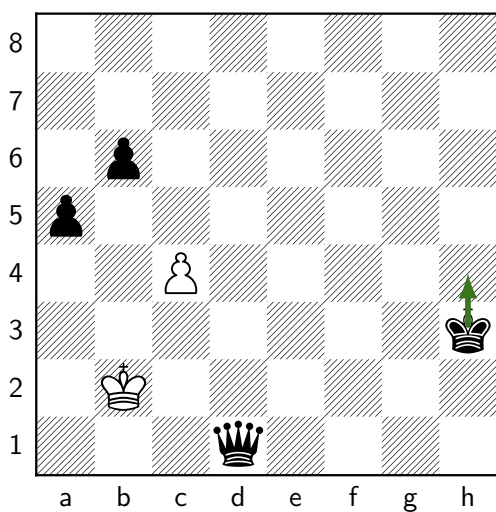
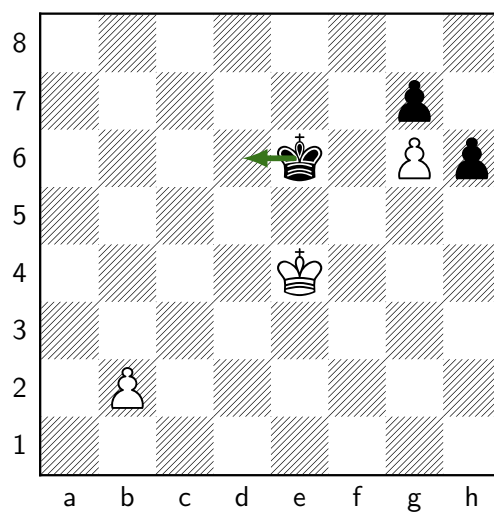


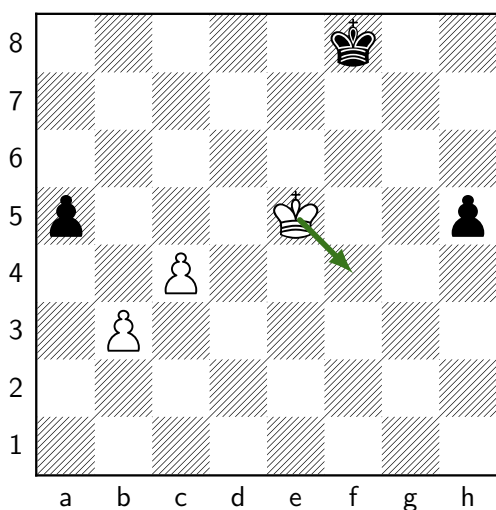
Figure 67: **Move 73, Black: Qd1.** (a) Our position; Melanie plays Qd1 (a4d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2241` (ply 145) [structural distance $d=13$, QAP obj=24]: the master played Qf4+; under μ our Qa5 plays the role of that game’s Qe5; cross-type roles: our Pb6 as their qb8; transferred move a4c6. (c) Retrieved from `ccr1_top_p0001#3927` (ply 150) [structural distance $d=15$, QAP obj=23]: the master played Rc4; under μ our Qa5 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Pb6 as their rd1; transferred move a4c6. (d) Retrieved from `ccr1_top_p0003#1182` (ply 159) [structural distance $d=10$, QAP obj=15]: the master played Qg2+; under μ our Qa5 plays the role of that game’s Qh7; cross-type roles: our Pb6 as their qf7; transferred move a4b3. Synthesis: no retrieved analog transferred to Qd1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



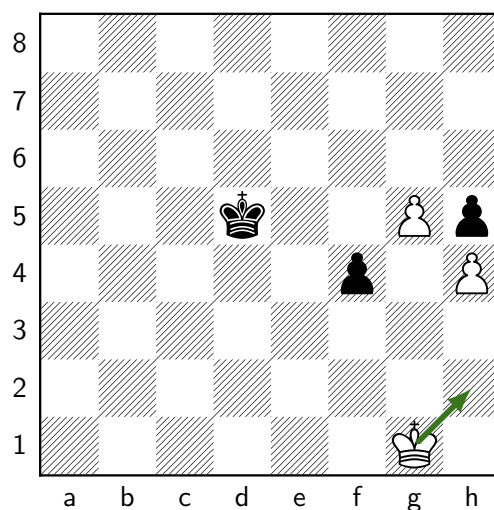
(a)



(b)

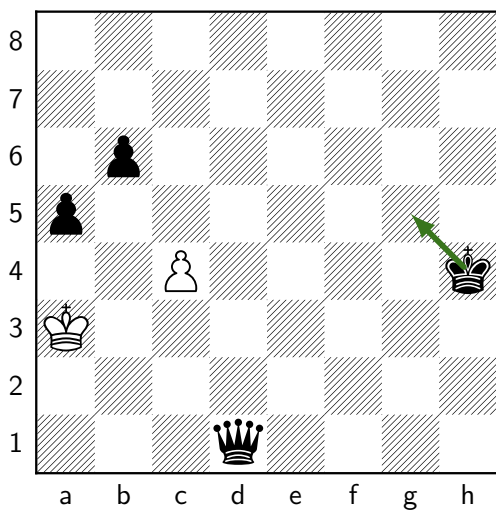


(c)

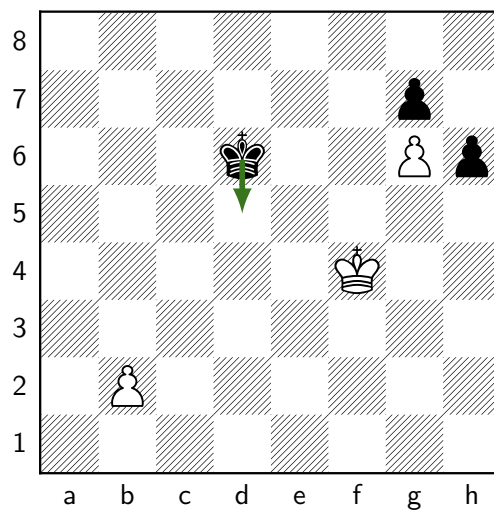


(d)

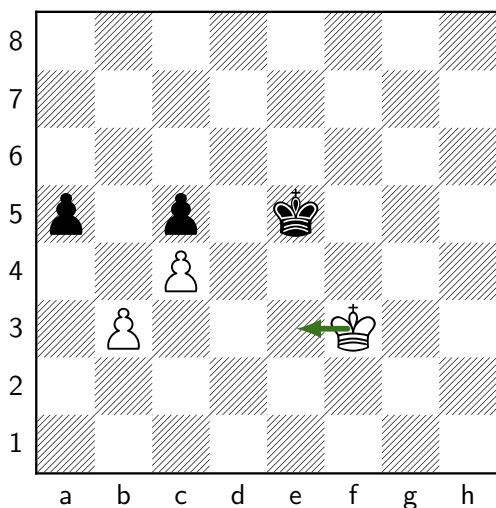
Figure 68: **Move 74, Black: Kh4**. (a) Our position; Melanie plays Kh4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2810 (ply 147) [structural distance $d=4$, QAP obj=2]: the master played Kd6; under μ our Kh6 plays the role of that game’s Ke3; cross-type roles: our Qd1 as their pb2; transferred move h3h4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1199 (ply 154) [structural distance $d=4$, QAP obj=2]: the master played Kf4; under μ our Kh6 plays the role of that game’s Ke5; cross-type roles: our Qd1 as their ph5; transferred move h3h4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2494 (ply 146) [structural distance $d=4$, QAP obj=2]: the master played Kh2; under μ our Kh6 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their pf4; transferred move h3h4 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kh4 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



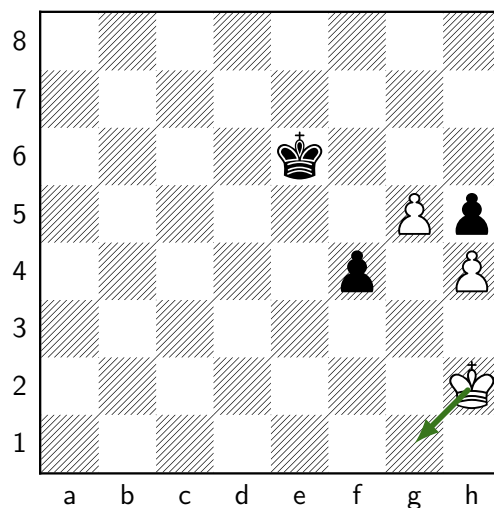
(a)



(b)

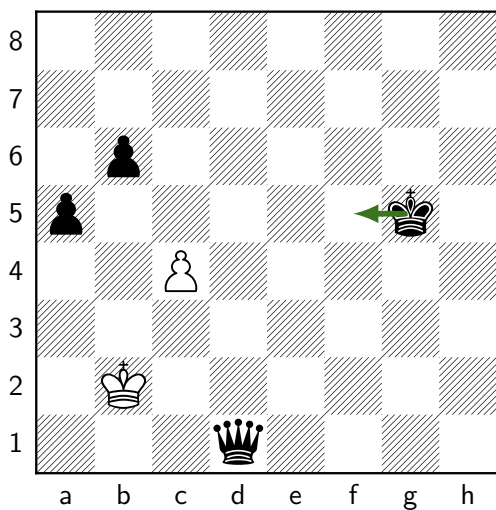


(c)

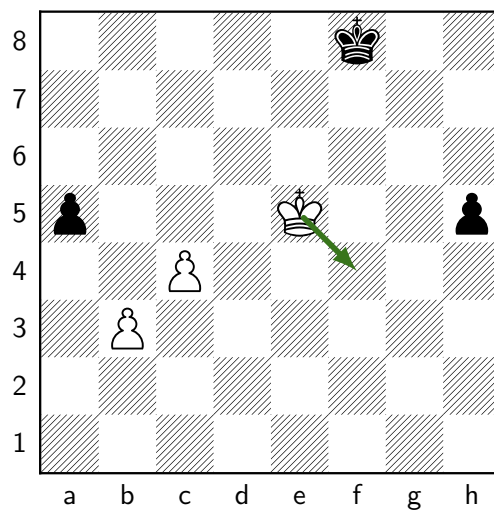


(d)

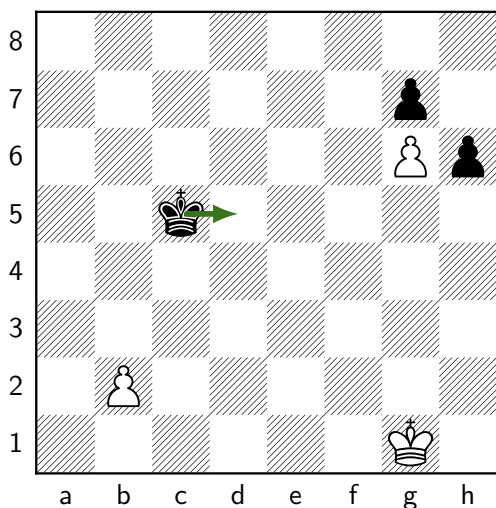
Figure 69: **Move 75, Black: Kg5.** (a) Our position; Melanie plays Kg5 (**h4g5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2810` (ply 149) [structural distance $d=3$, QAP obj=2]: the master played Kd5; under μ our *Kh5* plays the role of that game’s *Kd3*; cross-type roles: our *Qd1* as their *pb2*; transferred move **h4g5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#1106` (ply 148) [structural distance $d=4$, QAP obj=1]: the master played Ke3; under μ our *Kh5* plays the role of that game’s *Kf3*; cross-type roles: our *Qd1* as their *pc5*; transferred move **h4g5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#2494` (ply 144) [structural distance $d=4$, QAP obj=1]: the master played Kg1; under μ our *Kh5* plays the role of that game’s *Kh2*; cross-type roles: our *Qd1* as their *pf4*; transferred move **h4g5 (endorses this move)**. Synthesis: 14 retrieved analogs transferred to Kg5 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



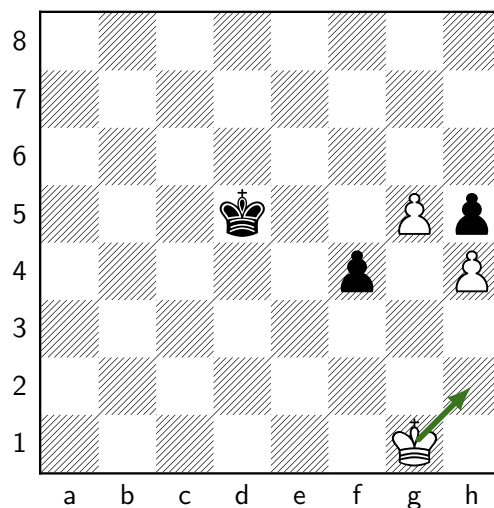
(a)



(b)

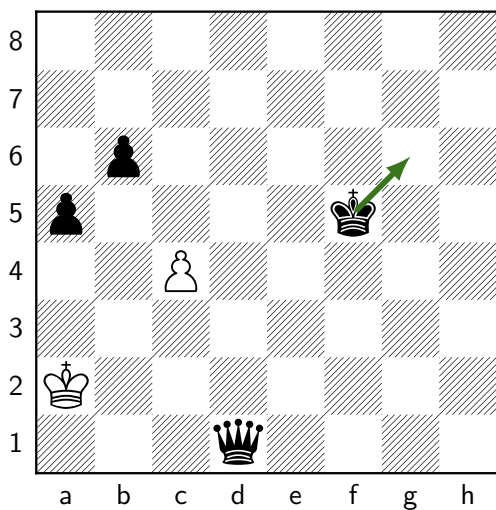


(c)

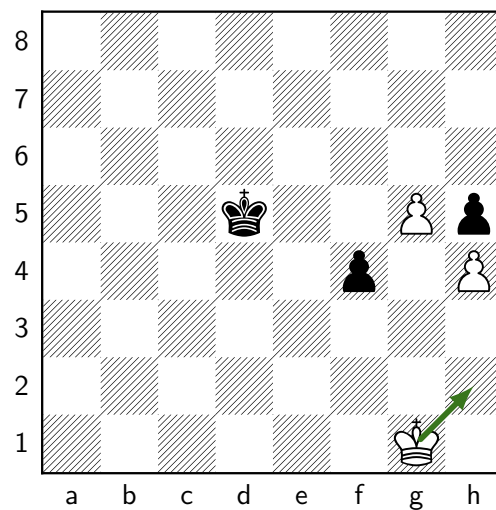


(d)

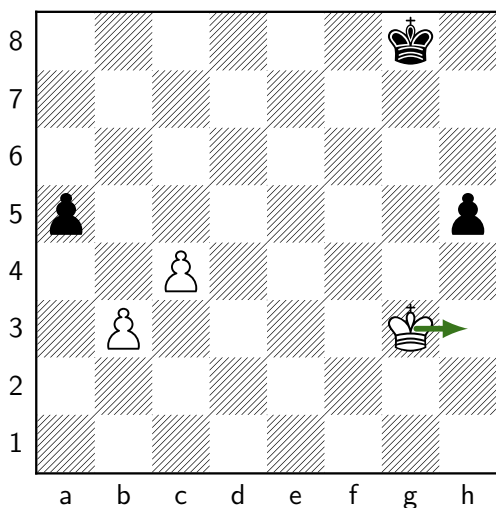
Figure 70: **Move 76, Black: Kf5**. (a) Our position; Melanie plays Kf5 (g5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 154) [structural distance $d=4$, QAP obj=2]: the master played Kf4; under μ our $Kg4$ plays the role of that game’s $Ke5$; cross-type roles: our $Qd1$ as their $ph5$; transferred move g5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 161) [structural distance $d=4$, QAP obj=2]: the master played Kd5; under μ our $Kg4$ plays the role of that game’s $Kc4$; cross-type roles: our $Qd1$ as their $pb2$; transferred move g5f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=2]: the master played Kh2; under μ our $Kg4$ plays the role of that game’s $Kg1$; cross-type roles: our $Qd1$ as their $pf4$; transferred move g5h6. Synthesis: 5 retrieved analogs transferred to Kf5 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



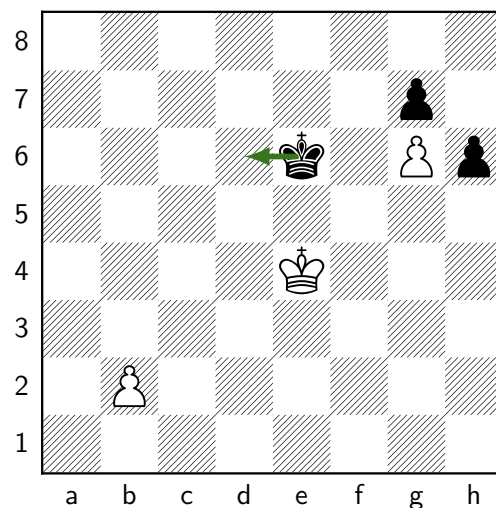
(a)



(b)

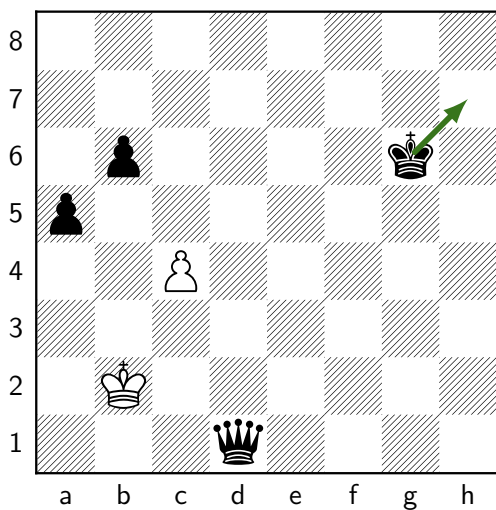


(c)

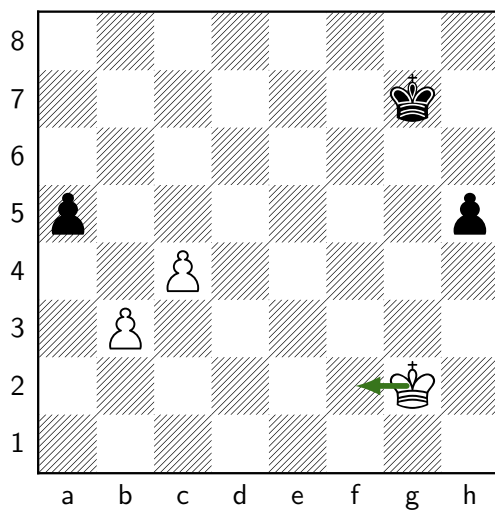


(d)

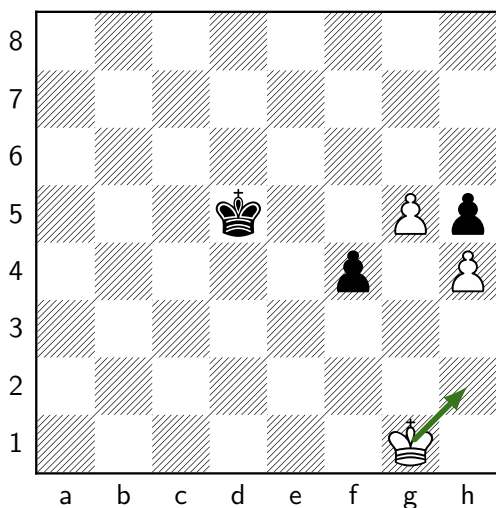
Figure 71: **Move 77, Black: Kg6.** (a) Our position; Melanie plays Kg6 (f5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=2]: the master played Kh2; under μ our $Kf4$ plays the role of that game's $Kg1$; cross-type roles: our $Qd1$ as their $pf4$; transferred move f5g6 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#1199` (ply 166) [structural distance $d=5$, QAP obj=1]: the master played Kh3; under μ our $Kf4$ plays the role of that game's $Kg3$; cross-type roles: our $Qd1$ as their $ph5$; transferred move f5g6 (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#2810` (ply 147) [structural distance $d=4$, QAP obj=2]: the master played Kd6; under μ our $Kf4$ plays the role of that game's $Ke3$; cross-type roles: our $Qd1$ as their $pb2$; transferred move f5f6. Synthesis: 5 retrieved analogs transferred to Kg6 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



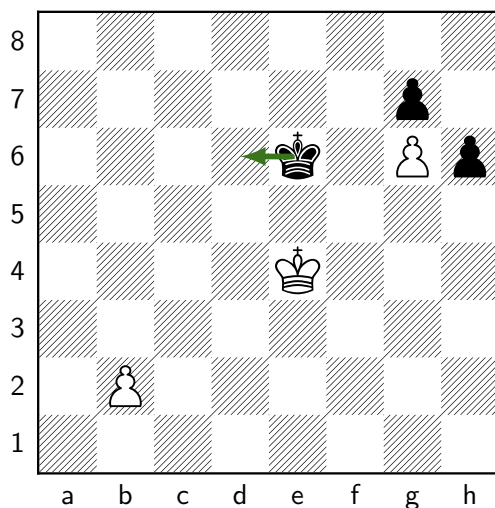
(a)



(b)

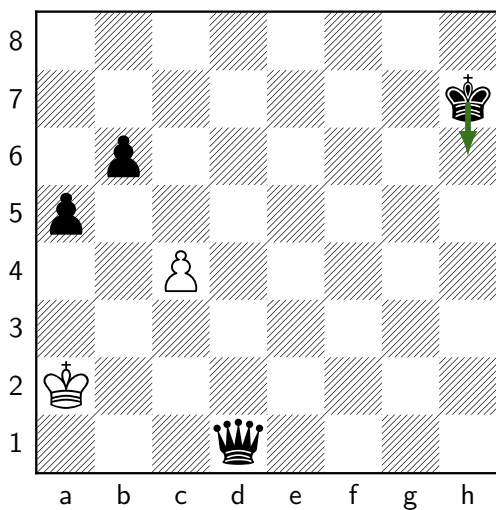


(c)

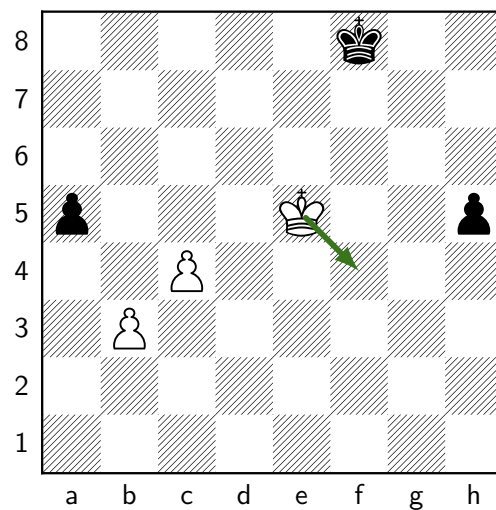


(d)

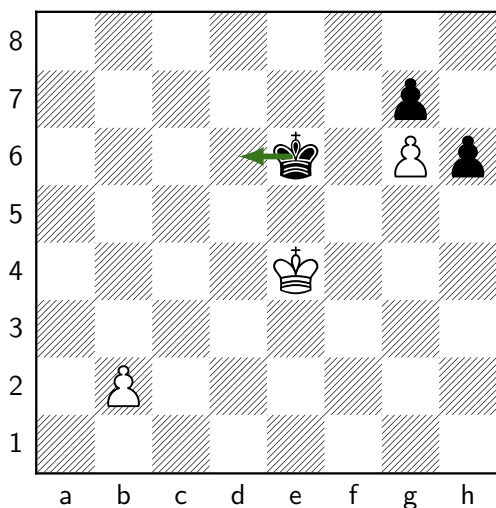
Figure 72: **Move 78, Black: Kh7**. (a) Our position; Melanie plays Kh7 (g6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 170) [structural distance $d=4$, QAP obj=2]: the master played Kf2; under μ our $Kg3$ plays the role of that game’s $Kg2$; cross-type roles: our $Qd1$ as their $ph5$; transferred move g6h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=1]: the master played Kh2; under μ our $Kg3$ plays the role of that game’s $Kg1$; cross-type roles: our $Qd1$ as their $ph5$; transferred move g6h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2810` (ply 147) [structural distance $d=4$, QAP obj=2]: the master played Kd6; under μ our $Kg3$ plays the role of that game’s $Ke3$; cross-type roles: our $Qd1$ as their $pb2$; transferred move g6f6. Synthesis: 4 retrieved analogs transferred to Kh7 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



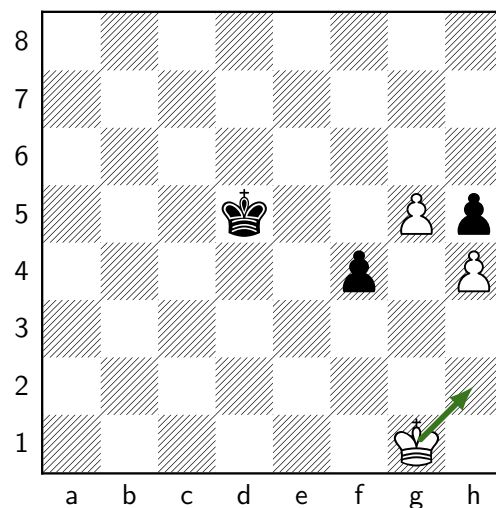
(a)



(b)

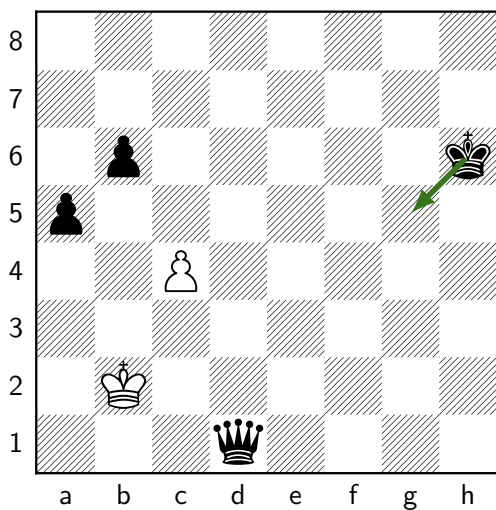


(c)

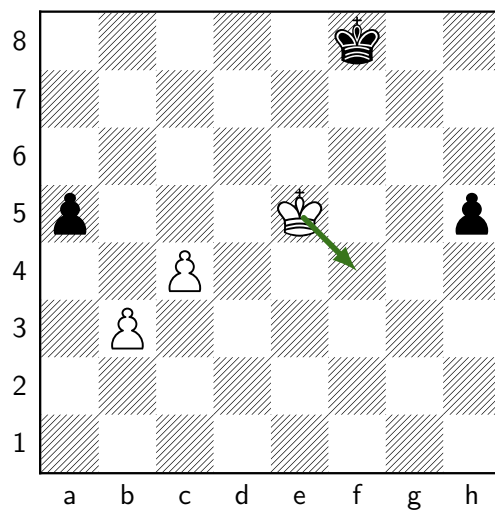


(d)

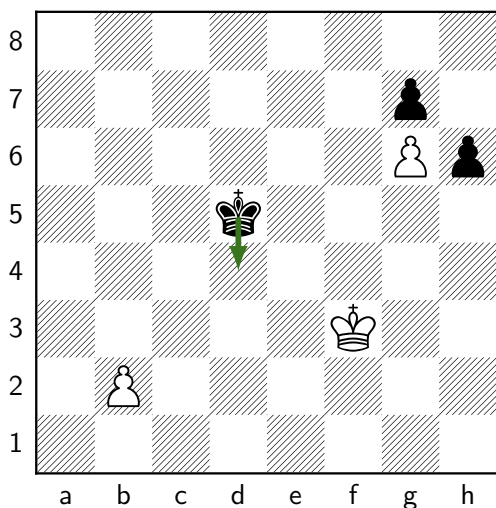
Figure 73: **Move 79, Black: Kh6**. (a) Our position; Melanie plays Kh6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 154) [structural distance $d=3$, QAP obj=2]: the master played Kf4; under μ our Kh2 plays the role of that game’s Ke5; cross-type roles: our Qd1 as their ph5; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 147) [structural distance $d=4$, QAP obj=2]: the master played Kd6; under μ our Kh2 plays the role of that game’s Ke3; cross-type roles: our Qd1 as their pb2; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=1]: the master played Kh2; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their ph5; transferred move h7g7. Synthesis: 13 retrieved analogs transferred to Kh6 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



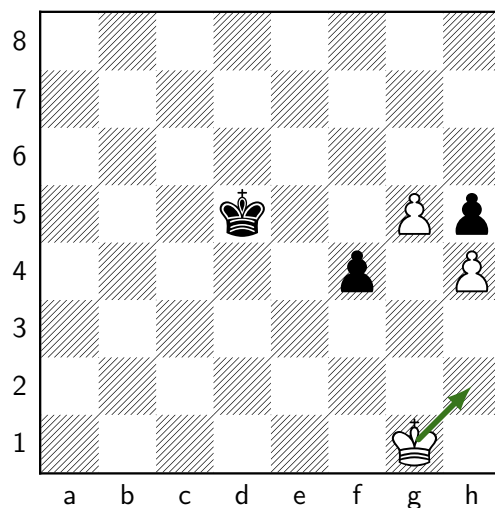
(a)



(b)



(c)



(d)

Figure 74: **Move 80, Black: Kg5.** (a) Our position; Melanie plays Kg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 154) [structural distance $d=4$, QAP obj=2]: the master played Kf4; under μ our *Kh3* plays the role of that game’s *Ke5*; cross-type roles: our *Qd1* as their *ph5*; transferred move h6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 151) [structural distance $d=3$, QAP obj=2]: the master played Kd4; under μ our *Kh3* plays the role of that game’s *Kd4*; cross-type roles: our *Qd1* as their *pb2*; transferred move h6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=1]: the master played Kh2; under μ our *Kh3* plays the role of that game’s *Kg1*; cross-type roles: our *Qd1* as their *ph5*; transferred move h6h7. Synthesis: 10 retrieved analogs transferred to Kg5 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

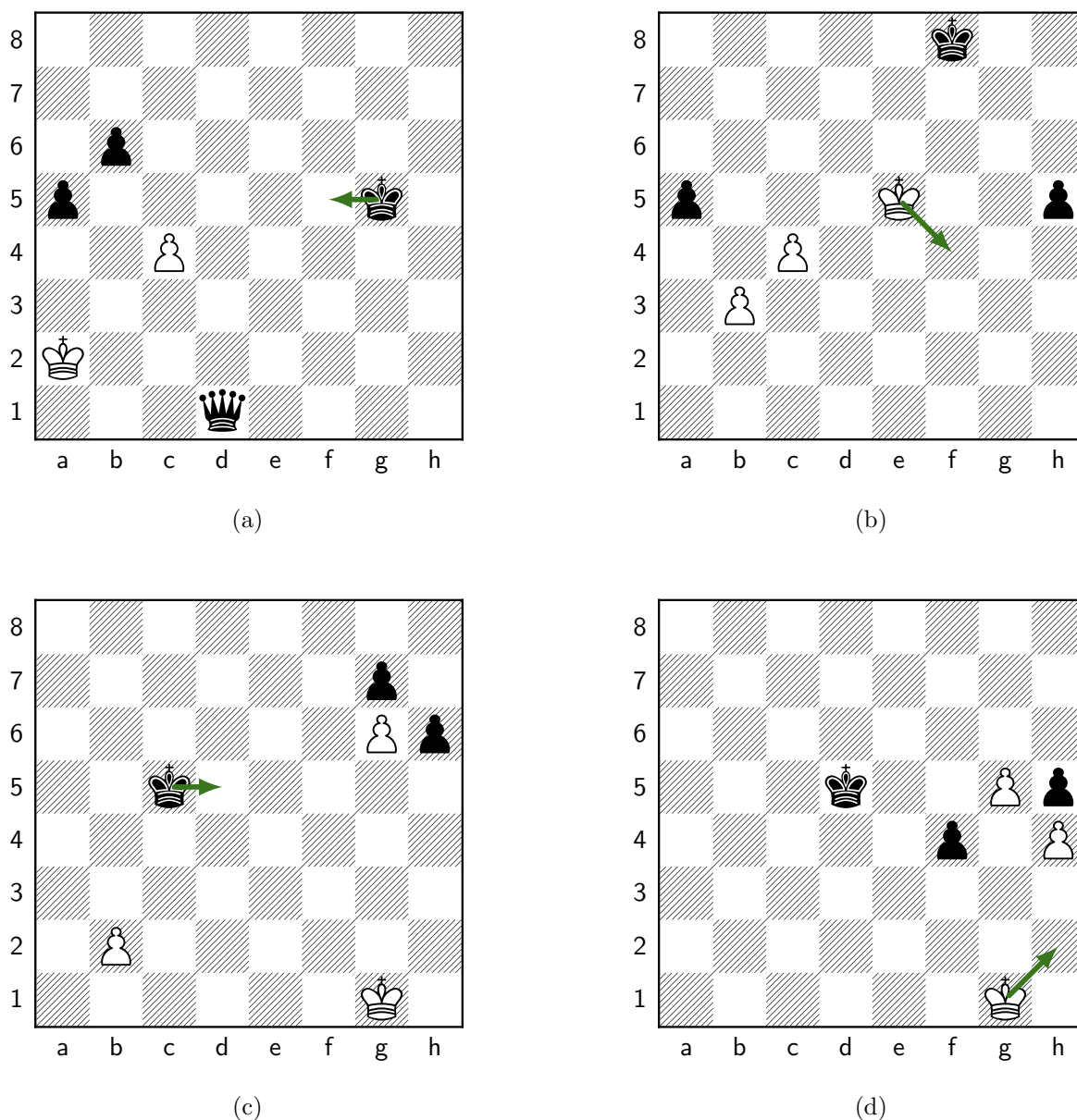
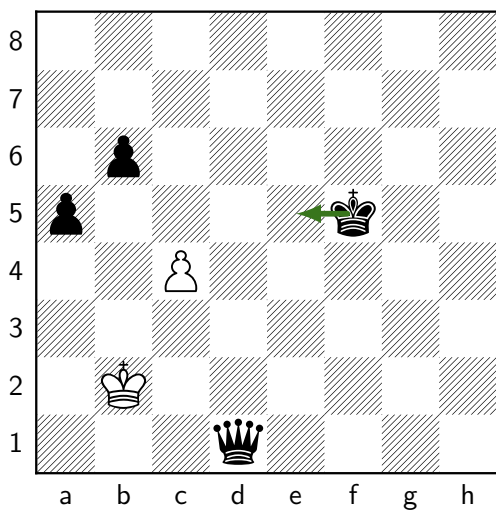
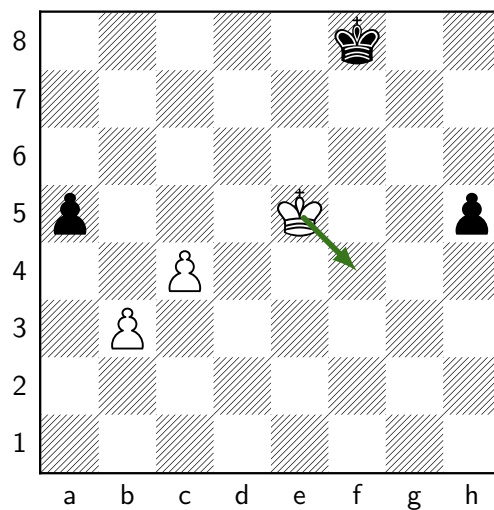


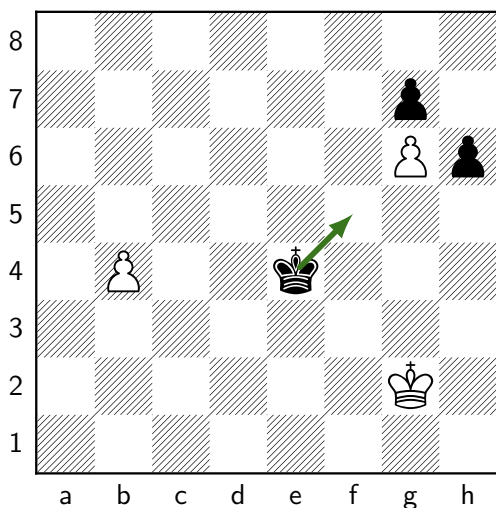
Figure 75: **Move 81, Black: Kf5.** (a) Our position; Melanie plays Kf5 (g5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 154) [structural distance $d=3$, QAP obj=2]: the master played Kf4; under μ our $Kg4$ plays the role of that game's $Ke5$; cross-type roles: our $Qd1$ as their $ph5$; transferred move g5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 161) [structural distance $d=3$, QAP obj=2]: the master played Kd5; under μ our $Kg4$ plays the role of that game's $Kc4$; cross-type roles: our $Qd1$ as their $pb2$; transferred move g5f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=2]: the master played Kh2; under μ our $Kg4$ plays the role of that game's $Kg1$; cross-type roles: our $Qd1$ as their $pf4$; transferred move g5h6. Synthesis: 5 retrieved analogs transferred to Kf5 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



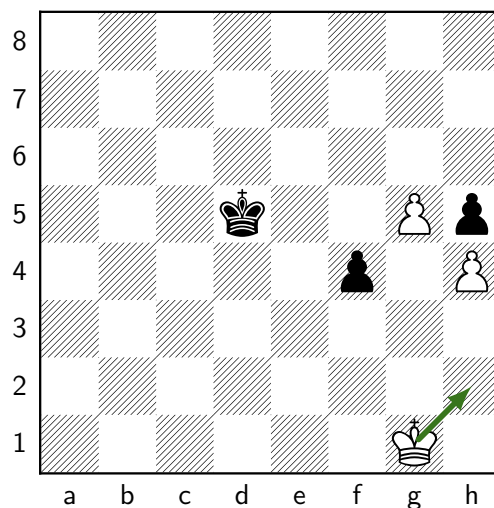
(a)



(b)

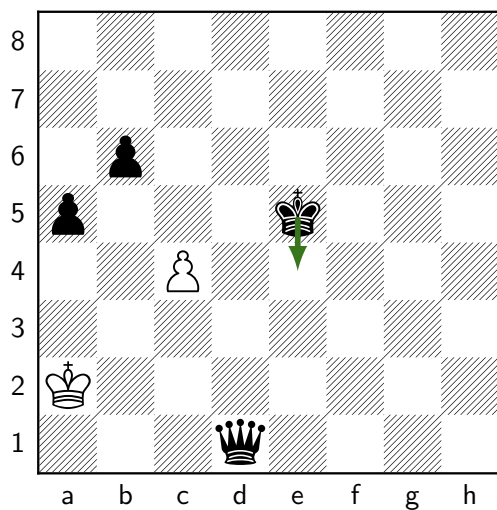


(c)

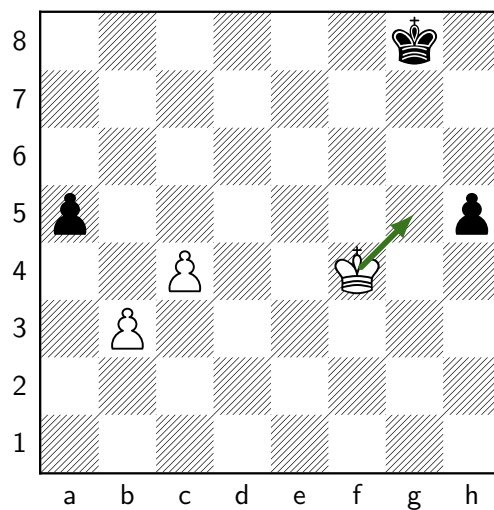


(d)

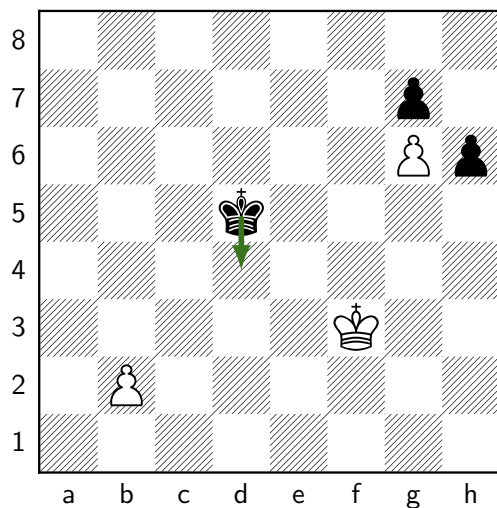
Figure 76: **Move 82, Black: Ke5.** (a) Our position; Melanie plays Ke5 (f5e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 154) [structural distance $d=4$, QAP obj=2]: the master played Kf4; under μ our Kf4 plays the role of that game's Ke5; cross-type roles: our Qd1 as their ph5; transferred move f5e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 179) [structural distance $d=3$, QAP obj=2]: the master played Kf5; under μ our Kf4 plays the role of that game's Ke5; cross-type roles: our Qd1 as their pb4; transferred move f5e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2494` (ply 146) [structural distance $d=4$, QAP obj=2]: the master played Kh2; under μ our Kf4 plays the role of that game's Kg1; cross-type roles: our Qd1 as their pf4; transferred move f5g6. Synthesis: 5 retrieved analogs transferred to Ke5 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

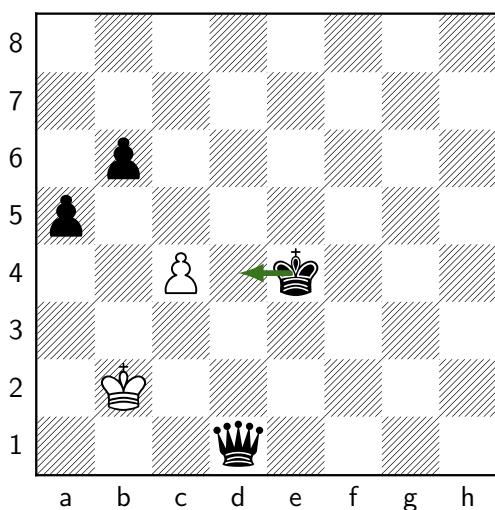


(b)

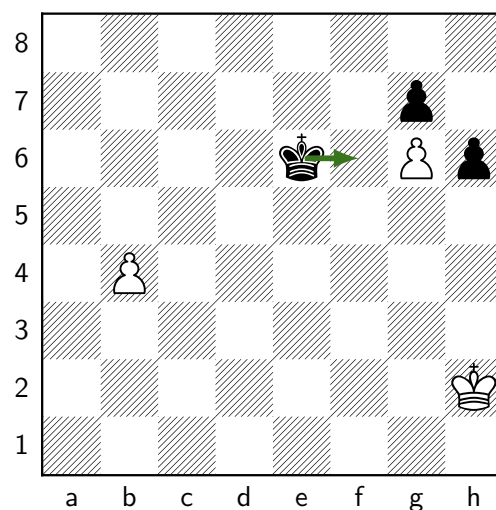


(c)

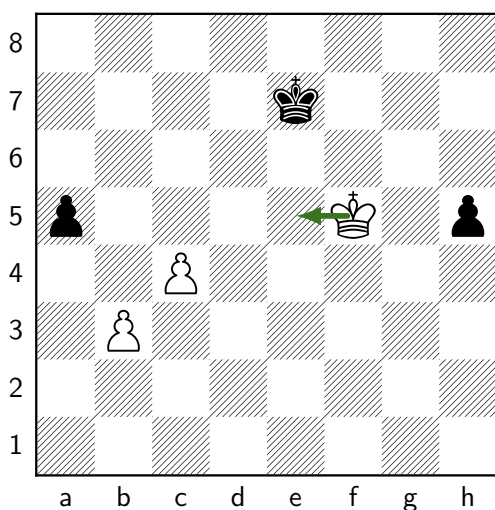
Figure 77: **Move 83, Black: Ke4.** (a) Our position; Melanie plays Ke4 (e5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1199` (ply 156) [structural distance $d=3$, QAP obj=2]: the master played Kg5; under μ our Ke4 plays the role of that game's Kf4; cross-type roles: our Qd1 as their ph5; transferred move e5e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2810` (ply 151) [structural distance $d=4$, QAP obj=2]: the master played Kd4; under μ our Ke4 plays the role of that game's Kd4; cross-type roles: our Qd1 as their pb2; transferred move e5e4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke4 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



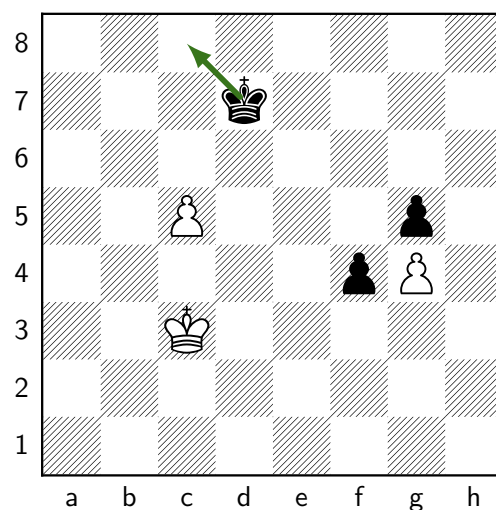
(a)



(b)

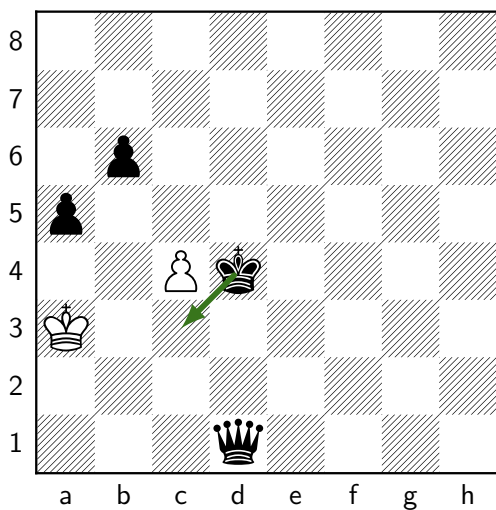


(c)

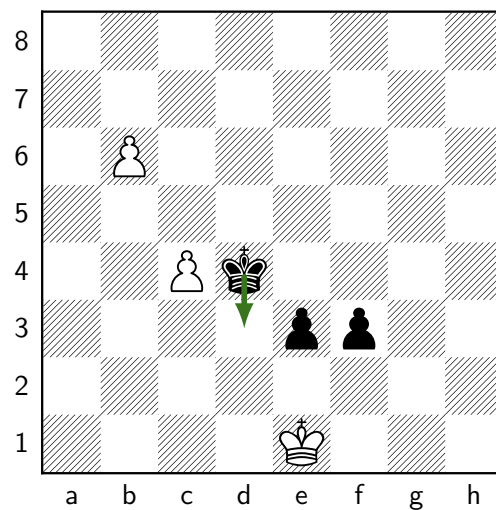


(d)

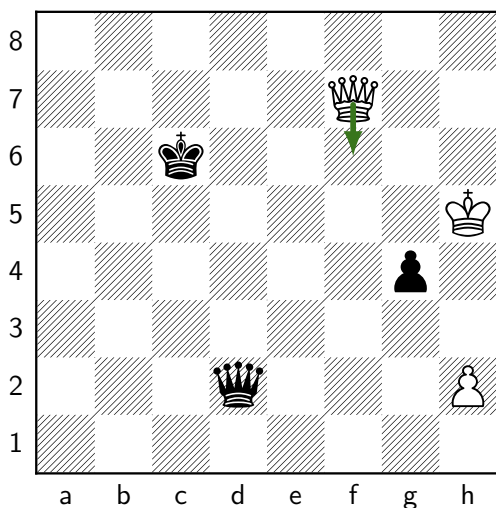
Figure 78: **Move 84, Black: Kd4.** (a) Our position; Melanie plays Kd4 (e4d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2810` (ply 171) [structural distance $d=4$, QAP obj=2]: the master played Kf6; under μ our *Ke5* plays the role of that game’s *Ke3*; cross-type roles: our *Qd1* as their *pb4*; transferred move e4d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1199` (ply 152) [structural distance $d=3$, QAP obj=2]: the master played Ke5; under μ our *Ke5* plays the role of that game’s *Kf5*; cross-type roles: our *Qd1* as their *ph5*; transferred move e4f4. (d) Retrieved from `ccr1_top_p0000#2904` (ply 183) [structural distance $d=4$, QAP obj=2]: the master played Kc8; under μ our *Ke5* plays the role of that game’s *Kd2*; cross-type roles: our *Qd1* as their *pc5*; transferred move e4f5. Synthesis: 1 retrieved analog transferred to Kd4 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



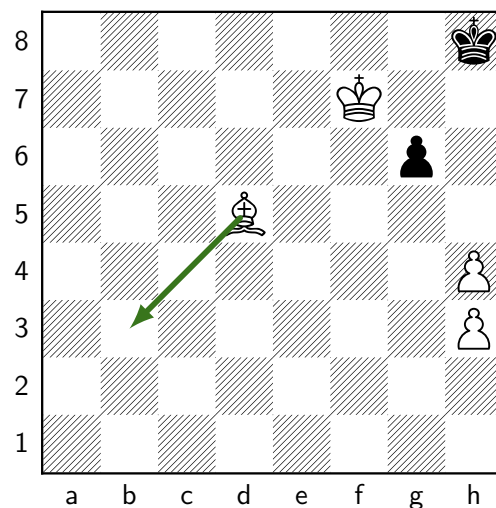
(a)



(b)

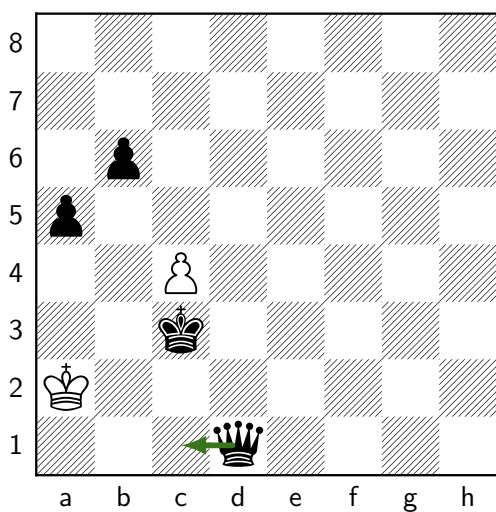


(c)

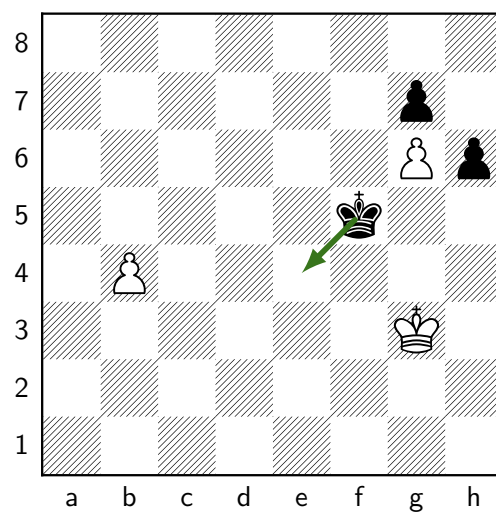


(d)

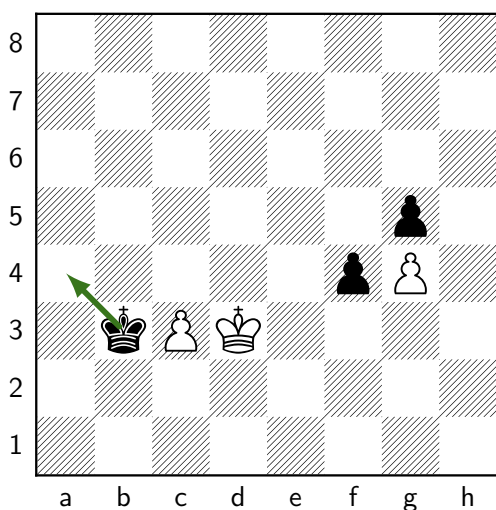
Figure 79: **Move 85, Black: Kc3**. (a) Our position; Melanie plays Kc3 (d4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2777` (ply 157) [structural distance $d=11$, QAP obj=19]: the master played Kd3; under μ our Kd5 plays the role of that game's Kd5; cross-type roles: our Qd1 as their Pe3; transferred move d4c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#998` (ply 156) [structural distance $d=11$, QAP obj=21]: the master played Qf6+; under μ our Qd8 plays the role of that game's Qf7; cross-type roles: our Pa5 as their qd2; transferred move d1f3. (d) Retrieved from `ccr1_top_p0002#3517` (ply 164) [structural distance $d=12$, QAP obj=21]: the master played Bb3; under μ our Qd8 plays the role of that game's Bd5 (a **cross-type** correspondence); transferred move d1g4. Synthesis: 1 retrieved analog transferred to Kc3 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



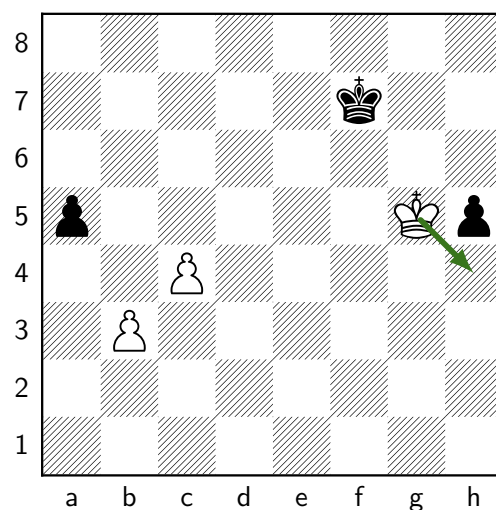
(a)



(b)

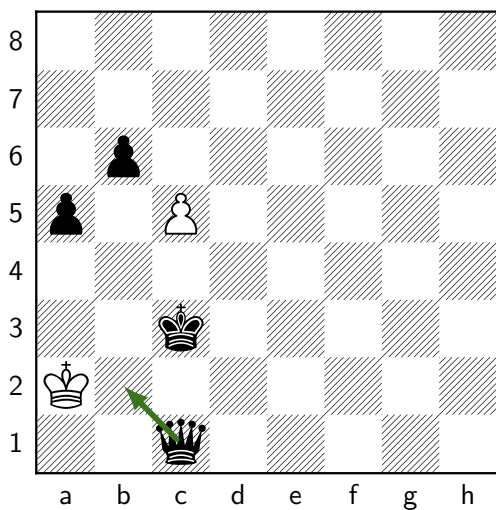


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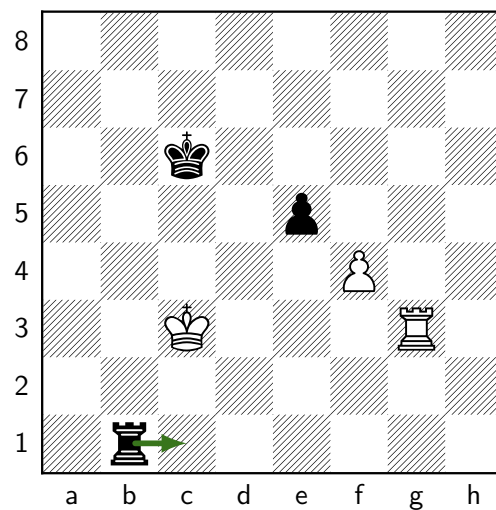


(d)

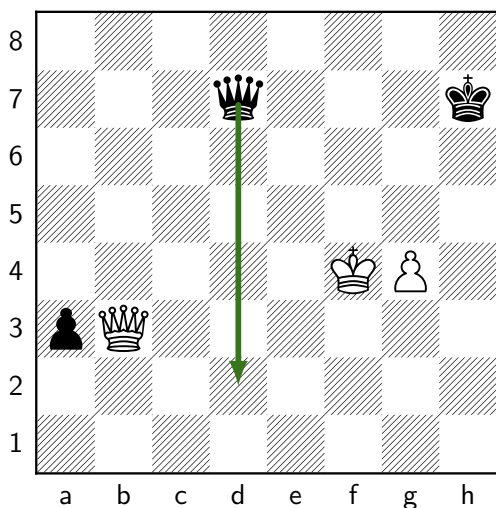
Figure 80: **Move 86, Black: Qc1.** (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#2810` (ply 177) [structural distance $d=4$, QAP obj=21]: the master played K $e4$; under μ our K $c6$ plays the role of that game’s K $f4$; cross-type roles: our Q $d1$ as their p $b4$; transferred move c3c4. (c) Retrieved from `ccr1_top.p0000#2904` (ply 165) [structural distance $d=6$, QAP obj=20]: the master played K $a4$; under μ our K $c6$ plays the role of that game’s K $b6$; cross-type roles: our Q $d1$ as their p $g4$; transferred move c3c4. (d) Retrieved from `ccr1_top.p0001#1199` (ply 158) [structural distance $d=5$, QAP obj=19]: the master played K $h4$; under μ our K $c6$ plays the role of that game’s K $g5$; cross-type roles: our Q $d1$ as their p $a5$; transferred move c3b4. Synthesis: no retrieved analog transferred to Qc1 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



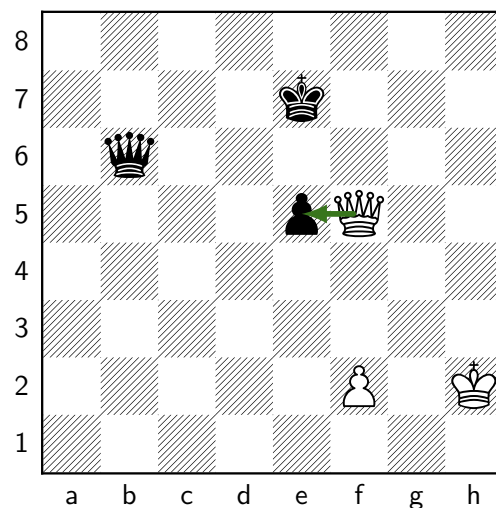
(a)



(b)



(c)



(d)

Figure 81: **Move 87, Black: Qb2#.** (a) Our position; Melanie plays Qb2# (c1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3912 (ply 189) [structural distance $d=16$, QAP obj=37]: the master played Rc1+; under μ our Qc8 plays the role of that game's Rb8 (a **cross-type** correspondence); cross-type roles: our Pa5 as their rg3; transferred move c1f1. (c) Retrieved from ccr1_top_p0000#3123 (ply 163) [structural distance $d=14$, QAP obj=19]: the master played Qd2+; under μ our Qc8 plays the role of that game's Qd2; cross-type roles: our pc5 as their qb3; transferred move c1f1. (d) Retrieved from ccr1_top_p0002#323 (ply 184) [structural distance $d=12$, QAP obj=19]: the master played Qxe5+; under μ our Pb3 plays the role of that game's Qf5 (a **cross-type** correspondence); transferred move b6c5. Synthesis: no retrieved analog transferred to Qb2# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.